

Real-Time National Land Acquisition & Management System

End-to-End Digital Monitoring and Decision Support

Master Reference: Legal Regimes, Stakeholders, Stage Workflows, RBAC, and Implementation Architecture

Field	Value
Problem statement	SIH26016 — Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support
Sponsoring ministry	Ministry of Rural Development, Department of Land Resources (DoLR)
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How to use this document

Parts I-II are the evidence base: the mandate, the existing-system landscape, and the statutory detail for every acquisition regime in active use in India. Cite these when a judge asks *“is that actually what the law says?”*

Parts III-IV are the product specification: system boundary, stakeholder registry, stage-by-stage role matrices, panel plan, and the granular RBAC and stage-gate matrices. These are the tables to build from.

Part V answers the engineering question directly — how ten-plus statutory workflows become **two bespoke state machines plus one configuration table**, not ten codebases.

Part VI is scope control: what is in, what is deliberately out, and the known research gaps that must not be presented as settled.

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PART I

Mandate and Context

What the Ministry asked for, what exists today, and what the gap actually is. This part establishes the evidence base for every design decision that follows: the twelve scope items and eight deliverables the problem statement makes mandatory, the nine platforms already operating in this space, and the precise reason none of them adds up to a national system.

In this part

- Executive summary and the one-sentence positioning
- Clause-by-clause decomposition of the official problem statement
- Diagnosis of the current fragmentation
- Deep study of Bhoomi Rashi, DILRMP, SVAMITVA, three state LAMS platforms, NGDRS and the commercial products
- International comparators and what they do not cover

1. Executive summary

Land acquisition in India is executed under at least a dozen different statutes, by at least a dozen different classes of authority, at a scale of hundreds of thousands of parcels a year — and there is no single place where the state can see it happening. SIH26016 asks for that place.

The Ministry of Rural Development's problem statement is explicit that the failure is not technological novelty but coordination: acquisition today runs on “fragmented systems, manual documentation, and state-specific processes,” producing inconsistent data, duplicated effort, delayed approvals, limited transparency and inadequate monitoring. Decision-makers, it says, lack real-time information on progress, compensation disbursement, possession status and rehabilitation. Every one of those five failures is a symptom of the same root cause: the acquisition lifecycle is recorded in dozens of unconnected ledgers, each owned by a different authority, each using a different schema, none of them obliged to reconcile with the others.

1.1 The positioning

One sentence

SIH26016 is not another Land Acquisition Management System. It is the **orchestration layer above the ones that already exist** — Bhoomi Rashi for highways, DILRMP for records, Kerala ILIMS and Tamil Nadu LAMS for their own states, PFMS for money — adding the three things none of them has: a workflow engine that is not hardcoded to one statute, a national view that crosses ministry boundaries, and decision support that is predictive rather than retrospective.

This framing matters commercially and defensively. Commercially, because it is honest: MoRTH already has a working NH Act system with 1,467 projects on it, and a submission claiming to replace it invites the obvious objection. Defensively, because it converts every existing system from a competitor into an integration point — and integration is precisely what Scope Item 7 of the problem statement demands.

1.2 The four findings that shape the design

Finding	Evidence	Design consequence
Acquisition is not one process — it is a family of processes	At minimum: LARR Act 2013 (7 statutory stages), NH Act 1956 (Sections 3A–3I), Railways Act 1989 (Sections 20A–20P), Coal Bearing Areas Act 1957 (Sections 4, 7, 9, 11), Petroleum & Minerals Pipelines Act 1962 (right-of-user, not title), and ten more.	The workflow engine must be data-driven. A hardcoded stage sequence fails at the second ministry onboarded. See Part V, Section 30.
Not all land assembly is acquisition	Urban development increasingly uses land pooling and TDR; renewable energy commonly uses government wasteland allotment or private lease; mining uses lease plus negotiated surface rights. None of these has a Section 11 notification or a compensation award.	process_type must be a first-class field with values compulsory_acquisition right_of_user land_pooling govt_allotment consent_purchase. A model that assumes every parcel has a private owner to compensate will not represent a solar park.
The gates that block acquisition are owned by other ministries	Forest clearance sits with MoEFCC's PARIVESH; Gram Sabha consent in Fifth Schedule areas sits with the village assembly under PESA and the Forest Rights Act; AERB clearance sits with the Department of Atomic Energy.	Model these as blocking gates with an evidence artefact , not as workflows to rebuild. The system asks “is the certificate uploaded?” — it does not run the clearance. See Part III, Section 19.
The statutory clock is the product	Nearly every regime hangs on a deadline: 21 days for objections under 3A, 60 days under Section 15, twelve months from Section 11 to Section 19 or the notification lapses, twelve months from Section 19 to award, 80% of compensation paid before possession under Section 38.	The timeline engine, with court-stay exclusion and lapse detection, is not a feature — it is the reason the platform has authority. See Part V, Section 36.

1.3 What is being asserted, and how confidently

This document distinguishes three tiers of claim, and marks them where it matters. **Statutory text** — section numbers, timelines, thresholds — is drawn from the Acts themselves and is stated flatly. **Platform facts** about existing systems come from official portals and press releases and are dated where the figure moves. **Design proposals** — the panel count, the two-engine strategy, the RBAC matrix — are our engineering judgement and are argued rather than asserted. Section 51 lists the handful of points where our own research is not yet settled; presenting those as certainties is the fastest way to lose a technical viva.

2. The problem statement, decomposed

The official PDF contains two normative lists: a twelve-row *Scope of Study* table and an eight-item *the system should provide* list. Together they are the marking scheme. Every module in Part V traces to at least one row below; anything that traces to none is scope creep.

2.1 The twelve scope items

#	Scope item	Literal requirement	How this document answers it
1	Existing system study	Study the current workflow across Central Ministries, States and District Authorities; identify manual processes, data duplication and operational bottlenecks.	Sections 3–6. Nine platforms studied; duplication and bottlenecks named specifically, with a migration path in Section 46.
2	Stakeholders	Land Requiring Bodies, Land Acquiring Authorities, District Collectors, State Governments, Central Ministries, Rehabilitation Authorities, Project Implementing Agencies, Policy Makers.	Sections 20–24. Thirteen primary roles across four tiers, five named sub-roles, plus the requiring-body/acquiring-authority distinction resolved explicitly.
3	Workflow digitization	End-to-end digital workflow covering proposal submission, scrutiny, approvals, notifications, awards, compensation, possession and closure.	Sections 22–23 for the stage content; Sections 30, 34 for the engine that executes it.
4	GIS & geo-tagging	Integrate GIS-based mapping and geo-tagging of acquired parcels for spatial visualization and monitoring.	Section 37. PostGIS geometry on every parcel, cadastral overlay, spatial conflict detection.
5	Data standardization	Standardized data formats, master databases, validation mechanisms and uniform reporting templates applicable across all States and UTs.	Section 44. ULPIN as universal key, per-state identifier parsers, validation rules by stage, data-quality scoring.
6	Dashboard & analytics	Real-time dashboards with project-, district-, state- and national-level analytics, KPIs and progress indicators.	Sections 40, 43. Ten KPIs with the drill-down path National → State → District → Project → Parcel → Landowner.
7	Integration	API-based integration with land record systems, cadastral maps, financial systems, GIS platforms and other government databases.	Section 38. Ports-and-adapters design, per-system reality check, three worked OpenAPI contracts.
8	Document management	Secure digital storage and management of notifications, awards, legal documents, maps and supporting records.	Section 41. Content-addressed store, version lineage, bilingual template engine, chain of custody for field evidence.
9	Monitoring & alerts	Automated alerts for pending approvals, delayed cases, statutory timelines, compensation disbursement and milestone completion.	Section 39. Nine-class alert taxonomy with trigger, recipient, channel and severity.
10	Reporting	Customizable reports, trend analysis, comparative analytics and decision-support reports for administrators and policymakers.	Section 40. Six report families, four export formats, five scheduling cadences.
11	Security & governance	Role-based access control, audit trails, secure authentication, encryption and compliance with Gol cybersecurity standards.	Sections 25–29. Granular RBAC, append-only audit ledger, STQC/CERT-In/DPDP/GIGW mapping.
12	Scalability	Nationwide deployment with multilingual support and the ability to accommodate future policy and legislative changes.	Sections 31, 45. Effective-dated rule versioning is the mechanism for “future legislative changes” — the requirement is architectural, not aspirational.

2.2 The eight system deliverables

Note the numbering quirk in the source: these appear as items 7–14 in the official PDF, a copy-paste artefact. They are items 1–8.

#	Deliverable (official wording)	What it means concretely
1	End-to-end digital workflow for land acquisition processes	A stage machine with approval gates, digital signature at each gate, automated routing and escalation rules — per regime, not per project.
2	Online submission, verification, approval and tracking of proposals	The Stage 0 proposal portal (Section 18) plus a role-scoped action inbox: each user sees only what is theirs to act on.
3	GIS-enabled geo-tagging and spatial visualization of land parcels	Polygon capture or GPS boundary walk on every parcel, cadastral overlay from DILRMP, status-coloured map with parcel-level drill-down.
4	Interactive national dashboard	Explicitly enumerated in the PS: area notified, area acquired, compensation assessed and paid, affected and displaced families, R&R status, project progress, possession status, timeline adherence. These eight are mandatory; we add two.
5	API-based integration with land records, cadastral maps and government portals	The integration hub, with a live adapter where a sandbox genuinely exists (DigiLocker) and an honest mock where it does not (PFMS write, e-Gazette publish).
6	Mobile-responsive interface for field-level data collection and verification	Note the exact words: mobile-responsive , not “offline app.” A responsive web form satisfies this literally. Offline sync is a differentiator we choose to add, not a requirement we must meet — see Section 42.
7	Secure document repository with version control and audit history	Immutable object store, version lineage per document, hash on capture, signature verification at each gate.
8	Customizable MIS reports and executive dashboards for decision-makers	Report builder with saved definitions, scheduled generation, and Parliament- and CAG-ready output formats.

The sub-theme is a scoring instruction

The PS files SIH26016 under “**AI, GIS & Data Analytics for Public Administration and Infrastructure Management.**” That is not decoration. A submission that delivers a beautiful workflow engine and treats AI as a footnote is answering a different problem statement. Section 43 specifies six AI use cases that are genuinely useful rather than ornamental — document extraction, compensation anomaly detection, duplicate-owner resolution, grievance triage, land value estimation and delay risk scoring — and is explicit about which need real training data and which can run on rules until that data exists.

2.3 The suggested technology stack, and where we diverge

Component	PS suggestion	Our choice	Reasoning
Database	PostgreSQL with PostGIS / MySQL	PostgreSQL 16 + PostGIS 3.4	Aligned. MySQL is named as an alternative but lacks the spatial predicate coverage (ST_Intersects, ST_DWithin, topology validation) that Scope Item 4 requires. Choosing PostGIS is choosing the suggestion, not diverging.
GIS platform	GeoServer, OpenLayers, Leaflet, QGIS	MapLibre GL + GeoServer WMS/WFS	MapLibre is the maintained fork in the Leaflet/OpenLayers family with vector-tile support, which matters at 1M-parcel scale. GeoServer serves the cadastral overlay. QGIS stays a desktop tool for data prep, not part of the deployment.
APIs	RESTful / GraphQL	REST for external, REST + typed client internally	REST for government integration because that is what the counterparties publish. GraphQL was considered for dashboard queries and rejected: the aggregation is better served by materialised views than by a flexible query layer over a schema that auditors need to reason about.
Cloud	NIC Cloud (MeghRaj) / AWS / Azure Government	MeghRaj target, container-portable	Data localization (Section 29) makes MeghRaj the production answer. Everything ships as OCI containers so the demo can run anywhere.
Notifications	SMS gateway, email APIs, push	NIC SMS gateway + SMTP + web push	Aligned. Section 39 adds the taxonomy the PS implies but does not specify.
Backend runtime	(not specified)	Rust (Axum + SQLx + Tokio); Python for analytics	The PS is silent, so this is ours to justify — and Section 32.3 does so honestly, including the cost.

3. What actually goes wrong today

The PS names five failures. Each has a specific mechanism, and understanding the mechanism is what distinguishes a design from a wish list.

Stated failure	Actual mechanism	What the platform must do about it
Inconsistent data collection	There is no shared schema for a parcel. Tamil Nadu records “Survey No. 45/2A,” Uttar Pradesh “Khasra No. 112,” Karnataka an RTC number, Kerala a Thandaper. Area is variously in hectares, acres, cents and bighas — and a bigha is not the same size in two states.	Normalize on ULPIN with per-state parsers and a canonical unit; keep the raw string alongside the parsed value so a revenue officer can still recognise their own record.

Stated failure	Actual mechanism	What the platform must do about it
Duplication of efforts	The same parcel is surveyed by the requiring body's consultant for the DPR, again by the revenue department for the RoR check, and again by the CALA's team for the award. Each survey produces a number, and the numbers differ.	One survey record per parcel per purpose, with lineage. Later surveys reference and supersede rather than replace, so the divergence becomes visible instead of silent.
Delays in approvals	Files move physically. A proposal that needs the requiring body, the state revenue department, and the Collector to each sign moves at the speed of the slowest despatch clerk, and nobody can see where it is stuck.	Digital routing with an explicit owner and an aging clock on every pending item; the pending-approval alert (Section 39) fires at the actor, not into a void.
Limited transparency	A landowner has no reliable way to learn the status of their own parcel. Bhoomi Rashi offers this for highways; for a state irrigation project there is usually a notice board and a queue outside the tehsil office.	A public portal keyed on survey number with OTP step-up for personal detail — the single highest-trust feature in the system and the cheapest to build.
Inadequate monitoring	Aggregation happens by asking. A ministry wanting national figures emails states, states email districts, districts fill spreadsheets. By the time it arrives the data is a month old and internally inconsistent.	Aggregation as a read of the transaction store, not a data-collection exercise. This is what “real-time” in the PS title actually means.

The bottleneck nobody lists

Across every regime the single largest cause of stalled acquisition is not process — it is **title**. A large share of Indian parcels have contested ownership, absentee owners, unrecorded succession, or pending partition suits. Both LARR (Section 77) and the NH Act (Section 3H(2)) have deposit-with-the-authority provisions precisely because payment often cannot be made to anyone.

A data model with a clean parcel-to-owner mapping cannot represent this, and a workflow that requires a payee before it can advance will deadlock on real data. ownership_status must be modelled as clear | disputed | untraceable | under_litigation | multiple_claimants with a distinct sub-flow for each. Getting this right is the difference between a demo and a system.

4. Existing system study

Scope Item 1 asks for a study, not an acknowledgement. This section examines nine platforms: one central acquisition system, one national records programme, one rural survey scheme, three state systems, one registration framework, and the commercial products. For each: what it does well, what it cannot do, and what SIH26016 should take from or give to it.

4.1 Bhoomi Rashi (MoRTH) — the closest comparator

Attribute	Detail
Ministry / operator	Ministry of Road Transport & Highways; operated with NHAI
Live since	1 April 2018; mandatory for all MoRTH land acquisition proposals
Scale	1,467 NHAI projects onboarded as of December 2023
Legal regime covered	National Highways Act 1956 only
Integrations	e-Gazette (automated 3A / 3D publication), PFMS (DBT to Aadhaar-linked accounts), built-in district/sub-district/village master data
Security	DSC plus OTP; bilingual Hindi/English interface
Reported effect	Notification processing reduced from a 3–6 month manual cycle to roughly a fortnight

What it does genuinely well

- **Gazette automation actually works.** This is the hardest integration in the whole domain — a statutory publication with legal effect, executed by API. It is the strongest single proof that the rest is possible.
- **Payment is real, not simulated.** Compensation reaches Aadhaar-linked accounts through PFMS with a traceable reference. Any claim we make about DBT should be measured against this bar.
- **Master data is curated.** The village and sub-district registry is maintained rather than imported once, which is why the notifications it generates are legally usable.

What it structurally cannot do

Limitation	Why it matters for SIH26016
Highways only	Does not cover Railways, Defence, Rural Development, Urban Development, Irrigation, or renewable energy — most of the sectors the PS background paragraph names.
NH Act only	Cannot represent SIA, consent thresholds, or the R&R award — i.e. it cannot run a LARR Act acquisition at all.
No cross-ministry view	MoRTH cannot see MoRD projects and vice versa. There is no aggregation layer above it, which is precisely the gap in the PS.
No R&R module	Rehabilitation and resettlement — an explicit PS dashboard KPI — is out of its scope entirely.
No litigation handling	Court stays are not tracked and statutory clocks do not pause, so its deadline data drifts from legal reality.
No state LAMS interoperability	Cannot exchange data with Tamil Nadu, Kerala or Odisha systems; a state highway upgraded to NH status changes systems, not records.
Closed API surface	Other ministries cannot integrate, so the replicability the government itself noted has not happened.
Responsive web, not offline-capable	No field data capture where connectivity fails, which is where surveys happen.

What to say about Bhoomi Rashi in the pitch

The government's own assessment is that the Bhoomi Rashi model is replicable and could be used by state governments and by other ministries that acquire land directly. Eight years on, it has not been replicated. That is the opening: **the gap is not that the idea is unproven — it is that nobody built the layer that makes it general.**

4.2 DILRMP and Bhulekh — records, not acquisition

The Digital India Land Records Modernization Programme is the largest land-data effort in the country and the single most important integration target. It is also, critically, not an acquisition system — a distinction worth being precise about, because conflating the two is the most common error in this problem space.

Component	Reported status
Record of Rights computerization	~99.8% of villages (early 2026 reporting); 6,25,137 of 6,57,397 villages as of Dec 2023
Cadastral map digitization	~97.4%
Text-map integration	~84.9% of villages have RoR linked to spatial maps
Sub-registrar office computerization	~95.7%
Registration-revenue integration	~88.6%
Digitally signed RoR download	Available to citizens in 19 states
Auto-triggered mutation	Implemented in 17 states — registration generates the mutation request automatically
Bank mortgage verification	Enabled in 406 districts
e-Courts linkage	Cleared in 26 states and UTs
Transliteration	Records in 22 Eighth Schedule languages across 17 states/UTs
ULPIN (“Bhu-Aadhaar”)	14-character geo-derived parcel identifier; implemented in 29 states/UTs

The consequence for our design is singular and non-negotiable: ULPIN is the join key. It is the only parcel identifier that is national, geo-derived, and already deployed almost everywhere. Every parcel record in SIH26016 carries a ULPIN foreign key, and where a legacy record has none, that absence is itself tracked as a data-quality defect rather than silently tolerated.

What DILRMP does not do: it holds no acquisition workflow, no compensation calculation, no fund tracking, no R&R, no national aggregation across ministries. It answers “who owns this parcel and where is it” — which is exactly the question our survey and verification stages need answered, and exactly why we read from it rather than rebuild it. DILRMP remains the system of record for title; we are the system of record for the acquisition process.

4.3 SVAMITVA — rural property mapping

Attribute	Detail
Launched	2020, Ministry of Panchayati Raj
Method	Drone survey of inhabited rural (abadi) areas
Reported scale	~65 lakh property cards across 50,000+ villages in 10 states (January 2025 reporting)
Output	Property cards plus village maps
Relevance	Covers inhabited rural land that DILRMP's agricultural cadastre handles poorly; property cards serve as ownership proof for rural claimants; drone imagery is directly usable for boundary verification at the survey stage
Limitation	Inhabited rural areas only — urban parcels need DILRMP/Bhunaksha, and agricultural land needs the conventional cadastre

4.4 State platforms

4.4.1 Kerala — Ente Bhoomi / ILIMS (the most advanced state system)

Kerala's Integrated Land Information Management System, built by NIC Kerala and rolling out in phases from 2024, integrates the Survey, Registration and Revenue departments behind a single citizen portal. It is worth studying closely because it has already solved several problems we will face.

- **CORS network:** 28 Continuously Operating Reference Stations covering roughly 80% of the state, with RTK GNSS and robotic total stations delivering 3–5 cm positional accuracy. This is the answer to “is phone GPS good enough for a legal boundary?” — it is not, and Kerala built the infrastructure rather than pretending otherwise.
- **Bhunaksha 5.0** with automatic ULPIN generation on cadastral map management.
- **Auto-mutation:** registration triggers the mutation in revenue records without a separate application — the pattern our Section 23 mutation track should imitate.
- **LTIN:** a Land Transaction Identification Number tracking every transaction, which is conceptually what our stage-history ledger does for acquisition events.
- **Architecture worth borrowing:** a citizen portal in front of an internal API gateway (ILM Gateway), a workflow engine and transaction broker (ILT Service), a centralized Integrated Land Bank, and authentication proxy services. That layering — public surface, gateway, workflow, repository — is close to what Section 32 proposes.
- **Single sign-on** using FIDO-compliant WebAuthn passkeys, which is ahead of most government deployments.

Its limits are jurisdictional, not technical: Kerala only, Kerala's LARR rules only, no NH Act track, no multi-ministry view, no predictive layer, no national dashboard.

4.4.2 Tamil Nadu — LAMS and TNGLMS

- Operated by the Commissionerate of Land Administration with NIC; online acquisition workflow plus e-services for Patta, Chitta, FMB sketches and TSLR extracts.
- Integrated with Tamil Nilam (urban and rural records), ColabLand, STAR registration, property tax, and the Court Case Monitoring System (CCMS).
- TNGLMS additionally maintains a government land inventory for identifying idle or underutilised holdings, and links court cases on government land.
- **What to borrow:** the CCMS linkage. Tamil Nadu already connects acquisition records to live court-case status — the exact mechanism Section 36 needs for stay-order timeline exclusion.
- **Limit:** Tamil Nadu's LARR rules only; no NH Act; no inter-state exchange.

4.4.3 Odisha — LARRMS (Bhuarjan)

District-level implementation covering Yadast preparation, award of compensation and a rent abatement module. Functionally the most basic of the three: no GIS integration, no analytics, no external connectivity. Useful mainly as evidence that the baseline in most states is far below Kerala's.

4.4.4 Other state systems worth knowing

State / UT	System	Feature of interest	Relevance
Maharashtra	I-SARITA	Centralized stamp duty, registration and e-search	Registration integration pattern
West Bengal	e-Deed	Web-based e-drafting of registered deeds	Template-driven document generation model
Punjab, HP, Goa, Jharkhand, A&N, Manipur, Mizoram	NGDRS	National Generic Document Registration System	Ready-made registration integration point
Andhra Pradesh	Blockchain pilot	Land transactions on a permissioned chain	Relevant to the companion PS SIH26019, not to us directly
19 states	Digitally signed RoR	Citizen download of legally valid records	The DILRMP endpoint our survey stage consumes

4.4.5 NGDRS — National Generic Document Registration System

Launched 2016 to give states a uniform online registration process: stamp duty calculation, online application, SMS alerts, e-KYC with UID and biometrics, and local language support. Adopted by Punjab, Himachal Pradesh, Goa, Jharkhand, Andaman & Nicobar, Manipur and Mizoram among others. Its limitation is instructive — final registration still requires a physical visit to the sub-registrar, and it is not wired into any acquisition workflow. For us it is a post-possession mutation integration point, not a dependency.

4.5 Commercial LAMS products

These matter for two reasons: they prove the feature set is buildable, and they show what private industry pays for — which is a good signal for what a district collector actually needs.

Product	Clients / focus	Capabilities	Why it is not the answer
CyberSWIFT LAMS	Tata Steel, Hindalco (Aditya Birla), Olayan Group, Expat Properties	LARR tracking, R&R, litigation management, GIS, SAP/ERP integration, Gantt charts, thematic maps, mobile field survey. Built on .NET Core and SQL Server, bilingual.	The closest technical comparator — and it demonstrates the R&R and litigation modules are tractable. But it is a private product for private acquirers: no statutory workflow engine, no Bhoomi Rashi / PFMS / DILRMP integration, no STQC path.
LAMSApp	Generic commercial	Web and mobile GIS covering acquisition, ownership, valuation, lease, litigation, encroachment and financial modules.	No government compliance posture, no STQC, no government-cloud deployment story.
Simproo LAMS	Corporate acquirers	LARR workflow, affected-community tracking, compensation and resettlement cost, rehabilitation monitoring, stakeholder communication.	Same structural gap: tracks a corporate acquirer's obligations, not a statutory authority's process.
TalkingLands REALM / Proquiro	Real estate developers, private equity	Spatial deal room with 100+ data layers including municipal plans and hydrological buffers; pipeline CRM.	Land <i>sourcing</i> , not statutory acquisition. No legal workflow engine at all.

4.6 Where the sectors are entirely undigitized

The study above covers what exists. It is equally important to be specific about what does not, because that is where the platform's marginal value is highest.

Sector / authority	Current state of digitization
Defence (DGDE / Defence Estates Officers)	Largely manual. Board of Officers convened physically, sanction issued on file, DEO liaises with the Collector by correspondence, litigation defended case by case, records entered into the Raksha Bhoomi database after the fact.
Railways	Partially digitized inside Indian Railways' internal systems; no public or inter-ministry interface for Sections 20A-20P proceedings.
Coal (Coal Controller's Office)	Manual processing of Coal Bearing Areas Act notifications.
Petroleum & gas pipelines	Manual; right-of-user compensation rates vary state by state and are revised by state notification.
Telecom right of way	Not digitized; thousands of micro-parcels per corridor, multiple licensees, disputes routine.
Irrigation and dams	PMKSY-AIBP provides an online MIS with GIS tagging at project level, but nothing acquisition-specific and nothing at parcel level.
Renewable energy	Not digitized; SECI and NTPC track internally, states apply their own land policies.
Nuclear (NPCIL)	Manual; tracked internally alongside AERB clearance.

5. International comparators

Two questions are worth pre-empting: has this been done elsewhere, and are we reinventing something. The short answer is that land *registration* digitization is common worldwide and land *acquisition* lifecycle management is not.

Programme	Scope	Acquisition lifecycle?	What is transferable
IGAD land administration work (East Africa, 2022)	Intergovernmental Authority on Development with Swedish support; technical guides for land administration across member states.	No — administration and records, not acquisition	The regional-standardization method: a shared technical guide that many jurisdictions adopt voluntarily. Close to our per-state configuration approach.
Bhutan blockchain land registry (2025)	National initiative for immutable title records and smart-contract transfer.	No — registration only	Relevant to companion PS SIH26019 rather than to us. Post-acquisition ownership, not the acquisition.
UN Innovation Network land registry proof of concept	Immutable transactional records, traceability, smart-contract sales.	No — proof of concept, registration	The audit-integrity argument: tamper-evidence as a trust mechanism. We achieve the same with hash-chained audit logs without the operational cost of a chain.
MoTA Hackathon 2.0 / National FRA platform (India, Jan 2026)	Ministry of Tribal Affairs with NIC and IIT Delhi; unifying five SIH 2025 finalist solutions into an FRA digital platform — AI archives, FRA Atlas, claimant chatbot, decision support.	No — forest rights claims, not acquisition	The most directly relevant Indian effort. If it delivers, it becomes the authoritative source for FRA claim status — which is exactly the evidence our Fifth Schedule consent gate (Section 15) needs. Treat it as a future integration, not a rival.
World Bank land governance programme	Research and policy: digitalization of registration, B-READY land indicators, Land Governance Assessment Framework (116 dimensions across 9 panels).	No — assessment framework	The LGAF dimensions are a ready-made external yardstick for arguing that the platform improves measurable governance quality, not just processing speed.
Geospatial challenges (NASA Space Apps, CopernicusLAC, GEO-OPEN-HACK)	Earth observation, land use monitoring, big geospatial processing.	No — observation, no administrative workflow	Techniques for the GIS layer; no governance content.

The honest version of the “global first” claim

It is defensible to say that **no comparable programme we could find covers the compulsory-acquisition lifecycle end to end** — notification, objection, award, compensation, possession, rehabilitation — as a single multi-statute platform. It is not defensible to say nobody has digitized land governance; a great many have, and several do parts of it better than we will at prototype stage.

State the claim narrowly and it survives questioning. State it broadly and a judge who knows about Kerala ILIMS or the Netherlands cadastre will dismantle it in one question.

6. Competitive positioning

The comparative matrix below is the single most useful artefact from the existing-system study. It should appear in the pitch deck almost unchanged. Read across a row to see which capability is genuinely unserved; read down the final column to see the claim we are making.

Capability	Bhoomi Rashi	DILRMP	Kerala ILIMS	TN LAMS	Comm. LAMS	SIH26016
National scope	Highways only	Records only	Kerala only	TN only	Private	Y — all ministries
Multi-ministry	N	N	N	N	N	Y
NH Act 1956 workflow	Y full	N	N	N	Partial	Y full
LARR Act 2013 workflow	N	N	Y (Kerala rules)	Y (TN rules)	Y	Y + configurable
Other sector regimes	N	N	N	N	N	Y via config
SIA workflow	N	N	N	Partial	N	Y
Consent threshold tracking	N/A	N	Partial	Partial	Y	Y live counter
R&R lifecycle	N	N	N	Partial	Y	Y full
Compensation engine	Y (NH)	N	Partial	Partial	Y	Y per-regime

Capability	Bhoomi Rashi	DILRMP	Kerala ILIMS	TN LAMS	Comm. LAMS	SIH26016
PFMS / DBT	Y	N	N	N	N	Y
e-Gazette publication	Y	N	N	N	N	Planned (manual fallback)
ULPIN as key	N	Y	Y	Partial	N	Y
GIS spatial queries	Basic	Maps only	Advanced	Basic	Advanced	Y PostGIS
Litigation / stay handling	N	Partial	Partial	Y (CCMS)	Y	Y + clock pause
Predictive analytics	N	N	N	N	N	Y
National dashboard	N	N	N	N	N	Y 10 KPIs
Public transparency portal	Basic	Records	Advanced	Basic	N	Y full
Field capture offline	N	N	N	N	Partial	Y PWA
Document version control	N	N	N	N	N	Y
Immutable audit trail	Basic	N	N	N	N	Y hash-chained
Open API ecosystem	Closed	Partial	Internal	Closed	N	Y
Multilingual	Hi/En	17 states	MI/En	Ta/En	N	Y i18n framework

The winning argument, stated carefully

Every existing system solves a real piece of the problem. Bhoomi Rashi solved NH Act digitization and proved gazette and payment automation work. DILRMP solved the records substrate and gave the country a universal parcel identifier. Kerala ILIMS solved cross-department integration inside one state. CyberSWIFT solved R&R and litigation tracking commercially.

None of them solves the national, multi-ministry, multi-statute acquisition lifecycle — because none of them was built to. That is a structural gap, not an oversight, and it is the gap SIH26016 is scoped to fill.

6.4 The companion problem-statement cluster

SIH26016 is not standalone. The Ministry of Rural Development released it as one of ten land-governance problem statements in the same cycle, and understanding where it sits in that cluster changes how it should be pitched: SIH26016 is positioned as the **orchestration hub**, consuming from the survey and mapping statements upstream and feeding the analytics and policy statements downstream. Building it as an isolated product, with no acknowledgement of the cluster, undersells exactly the systems-thinking argument that makes the orchestration-layer pitch (Section 1.2) credible.

PS code	Title	Relationship to SIH26016
SIH26010	Survey / Resurvey of Rural Agricultural Land in India	Upstream — supplies updated cadastral survey data that would flow into our parcel geometry and DILRMP adapter
SIH26011	3D ULPIN Generation and Vertical Property Mapping System	Upstream — the 3D data hook noted in Section 37 exists specifically so this platform's output has somewhere to land for urban, multi-storey acquisition
SIH26012	AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction using Drone Imagery	Upstream — drone-derived boundaries are a plausible future input to the survey stage (Section 22.2) alongside SVAMITVA's rural drone data
SIH26013	Automated Integration of Multi-source Geospatial Data for Urban Land Record Management	Upstream — a harmonised urban GIS layer this platform would consume rather than build
SIH26014	Integrated GIS-based Digital Public Infrastructure for Land Governance	Adjacent — the broader GIS backbone for land governance generally; SIH26016 is the acquisition-specific application that could sit on top of it
SIH26015	Geospatial Techniques for Watershed Development	Unrelated — a distinct watershed-management focus with no direct data dependency either way
SIH26016	Real-Time National Land Acquisition & Management System	This document. The orchestration layer described in Section 1.2
SIH26017	Predictive Analytics for Early Detection of Land Acquisition Delays	Downstream — this is precisely the trained-model version of the delay risk scoring in Section 43, once this platform has produced the operational history to train on

PS code	Title	Relationship to SIH26016
SIH26018	Intelligent Land Record Digitization and Validation System	Upstream — validated, digitized land records are exactly what the DILRMP adapter (Section 38) hopes to consume; today it consumes whatever DILRMP itself provides
SIH26019	National Digital Platform for Research & Policy Innovation in Land Governance	Downstream — the policy research layer above SIH26016; also the natural home for the blockchain-based audit questions raised and declined in Section 48

A caution about this positioning

None of these nine companion statements is guaranteed to be built by any team, let alone built to an interface SIH26016 could actually consume. The cluster relationship is architecturally real — the data flows described above are the correct shape for how these systems *should* relate — but it is a design statement about intended integration points, not a claim that those integrations exist today. Present it as the former.

PART II

The Legal Framework

Land in India is not acquired by one process. It is acquired under a general statute, under thirteen sector statutes that the general statute expressly exempts, under several more that sit outside that list, through right-of-user regimes that never transfer title, and through pooling and allotment mechanisms that are not acquisition at all. This part sets out each of them in the detail the workflow engine needs — sections, actors, objection windows, compensation basis and lapse rules.

In this part

- RFCTLARR Act 2013: seven statutory stages, section by section, with the schedules
- National Highways Act 1956: Sections 3A to 3J and the mutation track
- Railways Act 1989 Chapter IV-A: Sections 20A to 20P
- The remaining eleven Fourth Schedule acts
- Sector regimes outside the Fourth Schedule, including SEZ
- Right-of-user and easement regimes where title never moves
- Land pooling, TDR, government allotment and consent purchase
- Pre-condition gates: forest, environment, tribal consent, nuclear safety
- Comparative matrices and the repealed / exempt / subject distinction

7. Why one workflow is not enough

The single most consequential technical finding in this research is that “land acquisition” is not a process. It is a family of statutory processes that share a vocabulary and almost nothing else.

Two acquisitions of adjacent parcels, on the same day, for two different central projects, can differ in: which authority issues the notification, how many days the owner has to object, whether a social impact assessment happens at all, whether consent is required and at what threshold, how market value is computed, whether solatium is payable, whether rehabilitation is a statutory entitlement or a departmental policy, what triggers possession, who arbitrates a dispute, and whether the owner even loses title. A workflow engine that hardcodes any of these fails at the second ministry onboarded.

The problem statement's own background paragraph names the sectors it expects covered: “highways, railways, industrial corridors, irrigation projects, urban development, renewable energy initiatives, and other strategic infrastructure.” Mapped onto the statute book, those six categories alone reach across the LARR Act, the National Highways Act, Chapter IV-A of the Railways Act, land pooling under state town-planning law, and government wasteland allotment. A two-track system does not satisfy the background paragraph, let alone the scope table.

7.1 The regime landscape at a glance

Class	Regimes	Defining characteristic
General regime	RFCTLARR Act 2013	The default for any acquisition not covered by a special act. Only regime with mandatory SIA, consent thresholds and statutory R&R.
Fourth Schedule acts (13)	National Highways 1956; Railways 1989; Metro Railways (Construction of Works) 1978; Electricity 2003; Atomic Energy 1962; Coal Bearing Areas 1957; Petroleum & Minerals Pipelines 1962; Requisitioning & Acquisition of Immovable Property 1952; Ancient Monuments 1958; Damodar Valley Corporation 1948; Indian Tramways 1886; Land Acquisition (Mines) 1885; Resettlement of Displaced Persons (Land Acquisition) 1948	Listed in LARR's Fourth Schedule. LARR's <i>procedure</i> does not apply; each runs its own process. Since the 2015 order under Section 105(3), LARR's compensation and R&R <i>schedules</i> do apply.
Sector regimes outside the Fourth Schedule	SEZ Act 2005; MMDR Act 1957; Airports Authority of India Act 1994; Indian Telegraph Act 1885 (as read with the Right of Way Rules)	Not exempt. Where these involve compulsory acquisition of private land, LARR applies in full — a point our own research corpus got wrong and Section 17 corrects.
Right-of-user / easement	Petroleum & Minerals Pipelines 1962; Electricity Act 2003 Section 164 read with the Telegraph Act; Telegraph Act 1885 wayleave	Title never transfers. The owner keeps the land and receives compensation for the burden. There is no possession stage and no R&R.
Non-acquisition assembly	Land pooling and Town Planning Schemes; TDR; government wasteland allotment or lease; consent purchase / direct negotiation	No notification, no award, no compulsory element. The owner either contributes land for a developed plot, or the state allots land it already owns, or a willing seller sells.
Pre-condition gates	Forest (Conservation) Act 1980 as amended 2023; Environment Impact Assessment notification; PESA 1996 and Forest Rights Act 2006 in Scheduled Areas; AERB clearance for nuclear	Not acquisition regimes. They are blocking conditions that must be satisfied, by another ministry, before or alongside acquisition.

The design implication, stated once

Every regime above differs in **data** — stage names, section references, day counts, compensation formula, approving authority. Almost all of them share the same **shape**: an intention notification, an objection window with a hearing, a declaration that commits the land, a determination of money, a payment, a transfer of possession, and a dispute route.

That is the whole basis of the implementation strategy in Part V: build two bespoke engines for the two regimes whose shape genuinely differs (LARR, because of SIA and consent; the NH Act, because it is the reference implementation and the one with a live comparator), and express the rest as configuration rows against one generic Sector Acquisition Track.

8. RFCTLARR Act 2013 — the general regime

The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (Act 30 of 2013, in force from 1 January 2014) replaced the Land Acquisition Act, 1894. It is the only regime in which social assessment, owner consent and rehabilitation are statutory obligations rather than departmental policy, which makes it both the most protective and by a wide margin the slowest.

8.1 Chapter map

The chapter structure settles a question that comes up repeatedly and is worth being able to answer instantly: **does SIA come before or after the notification?** Chapter II (Sections 4–9) is Social Impact Assessment. Chapter IV (Sections 11 onward) is Notification and Acquisition. SIA comes first, and it comes first for a reason — it is meant to inform whether the project should proceed at all, which is impossible if intent has already been notified.

Chapter	Sections	Subject
I	1–3	Short title, commencement, definitions
II	4–9	Determination of social impact and public purpose
III	10	Special provision to safeguard food security
IV	11–19	Notification and acquisition
V	20–30	Compensation: enquiry, award, market value, solatium
VI	31–42	Rehabilitation and resettlement award
VII	43–50	Procedure and manner of R&R; National Monitoring Committee
VIII	51–74	Land Acquisition, Rehabilitation and Resettlement Authority
IX–XII	75–104	Apportionment, payment, temporary occupation, offences, miscellaneous
XIII	105–114	Exemptions (Fourth Schedule), rules, repeal of the 1894 Act

8.2 The seven statutory stages

The Act's own sequence. Note that the practical product workflow adds two bookend stages — a Stage 0 proposal (Section 18 of this document) and a closure/mutation step after possession — which are **not** statutory stages and must be labelled as system stages so the distinction survives a legal review.

#	Stage	Sections	What happens	Statutory clock
1	Social Impact Assessment	4–9	The appropriate government consults the local Panchayat, Municipality or Municipal Corporation and commissions an independent SIA study. A public hearing is held in the affected area. The SIA report and Social Impact Management Plan are appraised by an independent multidisciplinary Expert Group, which may recommend the project be abandoned. Government then examines the report and records whether the project serves a larger public interest and whether the minimum area is being taken.	No fixed duration for the study. Under Section 14, if the preliminary notification is not issued within twelve months of the Expert Group appraisal, the SIA lapses and must be redone (extendable by the appropriate government with recorded reasons).
2	Preliminary notification	11–15	Notification of intent to acquire, published in the Official Gazette, in two daily newspapers of which one must be in the regional language, in the local language at the Panchayat / Municipality / Municipal Corporation and at the offices of the District Collector, SDM and Tehsil, and uploaded to the website. On publication, transactions in the land are restricted. Affected persons may object.	Objections under Section 15 within 60 days of publication. The Collector must hear objectors and make a recommendation to government.
3	Declaration of acquisition	16–19	The Administrator for R&R prepares the Rehabilitation and Resettlement Scheme (Section 16) which is reviewed and approved (Sections 17–18); the declaration under Section 19 is published together with a summary of that approved R&R scheme. Land stands committed to acquisition; the requiring body must have deposited its share of the cost.	Under Section 19, the declaration must be published within twelve months of the Section 11 notification, failing which that notification is deemed rescinded (extendable, with recorded reasons).
4	Award of compensation	21–30	Notice to persons interested (Section 21); enquiry and award by the Collector (Section 23); market value determined under Section 26 and multiplied by the First Schedule factor; value of assets attached to the land added under Section 29; solatium of 100% added under Section 30.	Under Section 25 the award must be made within twelve months of the Section 19 declaration, failing which the entire proceedings lapse (extendable, with recorded reasons; where extended, compensation is recomputed).

#	Stage	Sections	What happens	Statutory clock
5	Rehabilitation and Resettlement award	31-42	The Collector passes a separate R&R award for each affected family, itemising every Second Schedule element: land for land where applicable, housing, one-time resettlement allowance, subsistence, transportation, employment or annuity or a one-time payment, and the Third Schedule infrastructure amenities at the resettlement site.	Runs alongside or immediately after the compensation award. The Third Schedule infrastructure obligations continue after possession.
6	Possession	38	The Collector takes possession only after ensuring compensation and R&R entitlements have been paid or tendered. Section 38 sets an outer limit of three months for payment of compensation and six months for the monetary components of R&R from the date of the award. The land then vests absolutely in the government free from encumbrances.	Gated on payment, not on elapsed days. See the correction in 8.6 — the widely repeated “80%” figure belongs to the urgency clause, not to Section 38.
7	Reference to the Authority	51-74 (esp. 64)	A person not accepting the award may apply to the Collector to refer the matter to the Land Acquisition, Rehabilitation and Resettlement Authority, on the measurement of land, the amount of compensation, the persons entitled, or apportionment. The Authority determines the reference and may enhance the award.	Under Section 64, the application must be made within six weeks of the Collector's award where the applicant was present or represented, otherwise within six weeks of receipt of notice or six months of the award, whichever expires first.

8.3 Consent, and when it is required

Consent under LARR is frequently over-stated. It is not required for every acquisition. Section 2(2) requires prior consent where land is acquired for a public-private partnership project or for a private company:

Acquirer / purpose	Consent threshold	Notes
Government project, land to remain vested in government	None	SIA and R&R still apply in full. Consent is not part of the process.
Public-private partnership where land continues to vest in government	70% of affected families	Counted among affected families, not merely titleholders — the denominator matters and is a live implementation question.
Private company	80% of affected families	The most demanding threshold in the Act.

Consent must be a live counter, not a rule in prose

A consent threshold that exists only as a documented rule will be missed. It must be a computed field on the project — `consent_obtained / consent_required`, recalculated on every consent record — and the Section 19 declaration gate must be **hard-blocked** while it is below threshold. The denominator (affected families as determined by the SIA, not parcels and not titleholders) must be frozen at the point the R&R scheme locks the affected-family list, or the percentage will drift as new claimants appear.

8.4 Compensation architecture

This is the part most often stated incorrectly, including in our own research corpus. The First Schedule computes compensation in four steps.

Step	Component	Basis
1	Market value	Determined under Section 26 as the higher of: the minimum land value specified in the Indian Stamp Act for registration in that area; the average sale price of the higher fifty per cent of comparable sale deeds in the nearest village or vicinity over the preceding three years; or the consented amount where land is purchased for a PPP or private company. Computed as at the date of the Section 11 notification.
2	Multiplying factor	Rural areas: a factor of 1.00 to 2.00 as notified by the appropriate government, based on distance from the nearest urban area. Urban areas: 1.00. Note the direction carefully — the higher multiplier applies to rural land, not urban. Our own act-research files state this inverted; see Section 51.
3	Value of assets attached to the land	Added under Sections 27-29: buildings and structures, standing crops, trees and plants, wells, irrigation works and any other improvement, each valued by the relevant technical authority. This is a separate line item with its own approval trail, not an adjustment to the land figure.
4	Solatium	100% of the sum of steps 1-3, under Section 30(2), in consideration of the compulsory nature of the acquisition.

First Schedule computation

```
market_value      = max(stamp_duty_minimum_value,
                        avg_top_50pct_comparable_sales_3yr,
                        consented_amount_if_ppp_or_private)

base              = market_value * factor          # factor: rural 1.00-2.00, urban 1.00
assets            = structures + crops + trees + wells + other_improvements
solatium          = (base + assets) * 1.00         # s.30(2)

award_total       = base + assets + solatium

# Net effect: up to 4x market value in rural areas, 2x in urban areas.
```

8.4.1 The two interest provisions, which are not the same

Our research corpus collapses these into a single “12% interest.” They are distinct provisions with different rates, different trigger events and different periods, and a compensation engine that implements only one will produce wrong numbers.

Provision	Rate	Period	Trigger
Section 30(3)	12% per annum	From the date of the Section 11 preliminary notification to the date of the award, or the date of taking possession, whichever is earlier	Automatic. Compensates for the delay between notifying intent and fixing the amount.
Section 80	9% per annum, rising to 15%	From the date possession is taken until the amount is paid or deposited; 15% applies to any part still unpaid one year after possession	Default interest for late payment. The step-up to 15% is the Act's penalty for prolonged non-payment and is routinely overlooked.
Section 72	As determined	On any excess awarded by the Authority over the Collector's award	Applies after a Section 64 reference succeeds.

Engine requirement

Interest accrual must be a **daily** computation over an effective-dated rate table, not a single annual multiplication — because the rate changes mid-period (9% to 15% at the one-year mark) and because court-stay exclusions remove days from the middle of the period. All monetary arithmetic uses fixed-point decimal; floating point on a crore-magnitude award produces rounding errors that an auditor will find.

8.5 The four schedules

Schedule	Subject	Contents
First	Compensation	Market value, multiplying factor, value of assets, 100% solatium — the computation in 8.4 above.
Second	R&R entitlements	The entitlement matrix for affected families: housing provision; land for land (with enhanced provision for Scheduled Caste and Scheduled Tribe families and irrigation project displacement); one-time resettlement allowance; subsistence grant for displaced families for one year; transportation cost; cattle shed or petty shop allowance; one-time grant to artisans, small traders and the self-employed; and a choice between employment, a one-time payment, or an annuity policy. Also fishing rights, and stamp duty exemption on land purchased in lieu.
Third	Infrastructure amenities	What must exist at a resettlement site: roads and drainage, safe drinking water, grazing land, fair-price shops, panchayat ghar, post office, school, sub-health centre, anganwadi, places of worship and burial or cremation ground, electric connection, village level bank, sanitation, playground, and access to public transport.
Fourth	Exempted enactments	The thirteen acts listed in Section 17 of this document, whose own acquisition procedures continue to operate. The SEZ Act 2005 is not among them.

The Third Schedule is a project management problem, not a checklist

Twenty-odd amenity obligations, per resettlement site, each with its own construction timeline, funded by the project but built by state agencies. This is why the R&R infrastructure tracker (Section 20, role 6) is a genuine module and not a status field: the obligation continues long after possession is taken and the acquisition “closed,” and it is the single most commonly unmet statutory duty in the Act.

8.6 Exceptions, exemptions and branches the engine must support

Provision	Effect	Workflow consequence
Section 10 — food security	Restricts acquisition of irrigated multi-crop land; states must specify limits, and where such land is acquired an equivalent area of culturable wasteland must be developed for agriculture, or its value credited to a fund.	A validation gate at proposal stage keyed on land classification, with a state-configured ceiling. Blocks or flags the proposal rather than the notification.
Section 40 — urgency	The appropriate government may invoke urgency for defence and national security, or for rehabilitation following natural calamities or other emergencies. The SIA requirement and much of the objection process can be dispensed with, possession can be taken on an accelerated basis after 80% of the estimated compensation is paid, and additional compensation (an additional 75% of total compensation) is payable.	A configured skip-condition on the stage sequence, not a special case in code. Every urgency invocation must be flagged in the audit ledger for oversight review — this is the most abusable provision in the Act and the platform's value here is making its use visible.
Section 41 — Scheduled Areas	In Fifth and Sixth Schedule areas, prior consent of the Gram Sabha, Panchayat or autonomous district council is required. Acquisition in Scheduled Areas is to be avoided as far as possible, a development plan is required, and land alienation restrictions apply.	A hard blocking gate: if <code>parcel.in_scheduled_area</code> then require <code>gram_sabha_consent</code> before advancing. Requires a “Gram Sabha / Tribal Consent Authority” role that our earlier stakeholder list omitted entirely.
Section 24 — transitional cases	Governs acquisitions begun under the 1894 Act and still pending. See 8.7.	
Section 46 — private purchase	Where a person purchases land through private negotiation above a limit specified by the state, the R&R provisions apply even though there is no compulsory acquisition.	A distinct <code>process_type = consent_purchase</code> track that carries R&R obligations but no notification or award stages.
State amendments	Several states — Tamil Nadu, Gujarat, Andhra Pradesh, Telangana, Jharkhand, Maharashtra, Karnataka among others — have amended LARR or made rules exempting categories such as irrigation, rural infrastructure, affordable housing, industrial corridors and PPP infrastructure from the SIA and/or consent requirements. Tamil Nadu additionally saved three of its own acquisition acts, a course that has been through the courts.	<code>sia_required</code> and <code>consent_required</code> must be functions of (state, project_category, effective_date) — never hardcoded true for the LARR track. This is the strongest single argument for the effective-dated rule table in Section 31.

8.7 Section 24 — the legacy acquisition problem

Section 24 is the most litigated provision in modern Indian land law and it is entirely absent from the current data model. Any national platform will meet these cases on its first day of operation, because thousands of pre-2014 acquisitions are still working through the courts.

Situation	Consequence under Section 24
Proceedings initiated under the 1894 Act, no award made under Section 11 of that Act as at 1 January 2014	LARR's compensation provisions apply to the pending proceeding.
Award made under the 1894 Act	The proceeding continues under the 1894 Act as if it had not been repealed.
Award made five years or more before commencement, but possession not taken and compensation not paid	The acquisition proceedings are deemed to have lapsed, and the appropriate government may initiate fresh proceedings under LARR.

The third limb generated a decade of conflicting judgments over whether the conditions were cumulative. A Constitution Bench settled it in *Indore Development Authority v. Manoharlal* (2020): the conditions are read **conjunctively** — a deemed lapse arises only where possession was not taken *and* compensation was not paid. The Court also held that compensation is “paid” for this purpose where the amount is tendered and unaccepted or deposited in the treasury; non-deposit in court attracts higher interest but does not cause a lapse. Litigation on the section continues nonetheless — the Supreme Court has since reversed High Court lapse findings on the same ground.

What the data model needs

A distinct case type — `legacy_1894_transitional` — separate from both the LARR and NH Act tracks, carrying its own status enumeration: `continuing_under_1894` | `larr_compensation_applies` | `lapsed` | `under_s24_dispute`, with the two determinative facts (possession taken date, compensation paid or deposited date) as first-class required fields rather than free text.

This is correctly deferred out of the prototype — but it must appear by name in the Phase 2 roadmap. A judge who knows land law will ask about Section 24, and “deliberately deferred, here is the data model hook” is a far better answer than a blank look.

9. National Highways Act 1956 — the fast regime

Act 48 of 1956, administered by the Ministry of Road Transport & Highways and executed through NHAI, NHIDCL and state PWDs. The acquisition machinery lives in Sections 3A to 3J, inserted by amendment in 1997. It is the only regime with a mature national digital implementation (Bhoomi Rashi), which makes it the natural reference implementation for our own engine — and the one where a judge can most easily check our claims.

Two structural features distinguish it from LARR: there is **no social impact assessment and no consent requirement** — acquisition is compulsory on the Central Government's satisfaction of public purpose — and the objection window is **21 days** rather than 60. Section 3J excludes the 1894 Act from applying to acquisitions under this Act.

9.1 Sections 3A to 3J

#	Section	Stage	What happens	Clock
1	3A	Preliminary notification	Where the Central Government is satisfied that land is required for a public purpose for the building, maintenance, management or operation of a national highway, it declares its intention to acquire by notification in the Official Gazette. Published additionally in two local newspapers, one in the vernacular. Land use is effectively frozen.	Day 0. Starts the 21-day objection clock and the one-year clock to 3D.
2	3B	Power to enter and survey	A person authorised by the Central Government may enter the land to inspect, survey, measure, value or enquire, dig or bore into the subsoil, set out boundaries and the intended line of work, and mark levels. Damage caused must be compensated.	In practice days 30–90. This is where the CALA — typically the DM, ADM or SDM — assigns the field survey.
3	3C	Hearing of objections	Any person interested may object within 21 days of publication of the 3A notification. The competent authority must give the objector an opportunity of being heard, consider the objections, and pass an order allowing or disallowing them.	21 days to file. The competent authority's order at this stage is final.
4	3D	Declaration of acquisition	On being satisfied that the land should be acquired, the Central Government declares so by notification. On publication, the land vests absolutely in the Central Government free from all encumbrances .	Must be published within one year of the 3A notification, failing which the 3A notification ceases to have effect. Automatic lapse — the single most important rule for the timeline engine to enforce.
5	3E	Power to take possession	Where land has vested, the competent authority may serve notice requiring surrender of possession, and may take possession — using police assistance if required — after the notice period expires.	60 days from the surrender notice.
6	3F	Right to utilise land	Once vested, the Central Government may use the land for the purposes of the Act.	—
7	3G	Determination of compensation	The competent authority determines the amount payable. Where the amount is not agreed, Section 3G(5) provides for determination by an arbitrator appointed by the Central Government . Section 3G(7) sets the parameters — market value, damage to standing crops and trees, damage from severance, damage to other property, and reasonable relocation expenses.	Typically days 180–365 in practice; no single statutory outer limit equivalent to LARR Section 25.
8	3H	Deposit and payment	The amount determined is deposited with the competent authority and disbursed to the entitled persons. Where there is a dispute as to apportionment or as to the persons entitled, or where entitlement cannot be established, the amount is deposited with the authority or the court rather than paid out. Interest is payable where deposit is delayed.	Deposit expected promptly after determination; Bhoomi Rashi operationalises this as a 10-day target.
9	3I	Competent authority's powers	Powers of a civil court for the purposes of the Act in respect of summoning, examination and document production.	—
10	3J	1894 Act excluded	Nothing in the Land Acquisition Act 1894 applies to an acquisition under this Act.	Structural — this is why the NH Act is a self-contained code.
11	—	Mutation (parallel track)	Post-possession, land records are mutated to record the vesting. In practice the title records the President of India / Central Government. Executed by the state revenue department, not by MoRTH.	Target of 30 days after possession; not a statutory deadline under this Act.

9.2 Compensation under the NH Act, before and after 2015

As enacted, Section 3G provides for market value plus damages, with no multiplying factor and no solatium — the source of the long-standing complaint that highway acquisition paid materially less than LARR acquisition of adjacent land. Interest at 9% is associated with delayed deposit and with arbitration excess.

The 2015 order changes the arithmetic — and our files have not caught up

Section 105(3) of LARR obliged the Central Government to notify, within one year of commencement, that the First, Second and Third Schedule provisions — compensation, R&R entitlements and infrastructure amenities — would apply to the thirteen Fourth Schedule acts. An order to that effect was issued in **2015**.

The practical consequence: acquisition under the NH Act, the Railways Act and the other Fourth Schedule acts should now attract First Schedule compensation (multiplying factor plus 100% solatium) and Second and Third Schedule R&R — not the pre-2015 bare-market-value figures.

Our act-research corpus still carries the pre-2015 figures for several of these acts. This is listed as an open gap in Section 51 and must not be presented as verified. The compensation engine's regime configuration should carry the post-2015 basis with an effective-date boundary at the 2015 order, so both the historical and current computation remain reproducible.

10. Railways Act 1989 — Sections 20A to 20P

The problem statement's background paragraph names railways explicitly, so the absence of this regime from a two-track design is a hole a judge can find in one question. Chapter IVA of the Railways Act 1989, inserted by the Railways (Amendment) Act 2008, is a self-contained acquisition code for **special railway projects**. It does not follow the NH Act's stages and it is not LARR.

Section	Stage	Detail
20A	Notification of intention	The Central Government declares its intention to acquire land for a special railway project by notification, specifying the land.
20B	Power to enter and survey	Authorised persons may enter, survey, measure, dig or bore, and set out boundaries and levels — parallel to NH Act Section 3B.
20D	Objections	Any person interested may object within 30 days of publication of the 20A notification. Note the window: 30 days, against 21 under the NH Act and 60 under LARR. This alone makes a shared hardcoded objection period impossible.
20E	Declaration of acquisition	On satisfaction, the Central Government declares the acquisition; the land vests absolutely in the Central Government free from encumbrances.
20F	Determination of compensation	The competent authority determines the amount, with provision for arbitration where the amount is not accepted. The arbitrator is an officer of gazetted rank appointed by the Central Government.
20G	Criteria for market value	Sets out how market value is to be determined, including the minimum land value under the Stamp Act and average sale prices of comparable land.
20H	Deposit and payment	Payment of the determined amount, with deposit provisions where entitlement is disputed or the payee untraceable.
20I	Power to take possession	Possession may be taken once the land has vested and the amount has been deposited.
20J-20P	Remaining provisions	Completing the chapter: powers of the competent authority, exclusion of other acquisition law, and related machinery.

A correction our own files needed

An earlier fact-check round found the act-research corpus citing the Railways Act land acquisition provisions as **Sections 120-124**. They are **Sections 20A-20P**. The correction was noted at the top of the file but the body text and every stage heading beneath it still carry the wrong numbers. That file must not be used as a source of truth until the body is reconciled — see Section 51.

11. The remaining Fourth Schedule acts

Eleven acts beyond the NH Act and the Railways Act appear in LARR's Fourth Schedule. Each is active law, each runs its own acquisition procedure, and since the 2015 order each should apply LARR's compensation and R&R schedules. None is a candidate for a bespoke engine; all are candidates for configuration rows.

11.1 Coal Bearing Areas (Acquisition and Development) Act, 1957

Attribute	Detail
Ministry / authority	Ministry of Coal; processed through the Coal Controller's organisation rather than the District Collector
Applies to	Acquisition of land or rights in land in coal-bearing areas, for government companies and coal public sector undertakings
Stage sequence	Section 4(1) prospecting notification → Section 7(1) notification of intention to acquire → Section 9(1) declaration of acquisition → vesting in the Central Government → Section 11 transfer of rights to a government company
Distinctive features	A prospecting stage before any acquisition intent; the intention notification carries a multi-year validity window; and rights ultimately transfer to a PSU rather than remaining with government
Compensation	Determined under the Act's own provisions with a tribunal route for disputes; R&R historically governed by Coal India's own policy rather than statute
Workflow consequence	Needs a pre-acquisition prospecting stage and a post-vesting transfer stage that no other regime has. Both are expressible as extra configured stages.

11.2 Requisitioning and Acquisition of Immovable Property Act, 1952

Attribute	Detail
Ministry	Principally Ministry of Defence, through the Directorate General of Defence Estates and Defence Estates Officers
Enacted	14 March 1952
Distinctive feature	Requisition, not permanent acquisition. Property is requisitioned for a public purpose for a period, with provision to subsequently acquire it, and provision for release. Recurring compensation is payable for the requisition period.
Stage sequence	Notice of intended requisition → requisition order → possession → (optionally) acquisition → or release back to the owner
Pre-acquisition process	Defence acquisition has a substantial internal workflow before any statutory notification: in-principle Ministry approval, a Board of Officers, the Defence Estates Officer's advice on cost and R&R, Ministry sanction, then a demand placed on the District Collector. Records land in the Raksha Bhoomi database and the Military Land Register.
Workflow consequence	Two things no other regime needs: a requisition period with recurring payment and a release path, and a modelled pre-statutory internal approval chain.

Note on our own data

The act-research file for this Act currently records an enacted date of 14 March 1952 and an effective date of 7 March 1952 — i.e. effective before enactment. This looks like two values swapped during a correction pass rather than a genuine finding. Flagged in Section 51.

11.3 Petroleum and Minerals Pipelines (Acquisition of Right of User in Land) Act, 1962

The clearest example of a regime that is not acquisition at all, and the one where our data was most badly wrong.

Attribute	Detail
Ministry / initiators	Ministry of Petroleum & Natural Gas; pipeline companies (GAIL, IOCL, ONGC and others) initiate, the Central Government declares
What is acquired	A right of user in the land — not the land. The owner retains title and may continue to cultivate, but loses the right to construct buildings or plant deep-rooted trees along the corridor.
Stage sequence	Section 3 notification of intention → Section 5 objections within 21 days → Section 6 declaration, on which the right of user vests in the Central Government → entry and laying → compensation determination
Compensation	A one-time payment for the right of user based on a percentage of market value — 10% under the Act — plus separate compensation for damage to crops, trees and structures and for any other damage caused. State governments have from time to time notified enhanced rates.
R&R	None. There is no displacement because there is no dispossession.
Workflow consequence	No possession stage, no mutation, no R&R award. The parcel geometry is a corridor intersecting parcels rather than whole parcels — the spatial model must support partial-parcel burdens with an area computed by intersection.

Three different numbers for the same calculation

The act-research file for this Act contains **three conflicting compensation percentages**: 33–50% in the general formula description, 10% in one worked example and the comparison table, and 40% in a second worked example. 10% is the corrected figure. If the compensation engine is seeded from that file, the number it produces depends on which paragraph the developer happened to read. This is the single highest-priority data defect in the corpus — see Section 51.

11.4 Electricity Act, 2003

Attribute	Detail
Ministry	Ministry of Power; executed by transmission licensees and distribution companies
Mechanism	Primarily not acquisition . Section 164 empowers the appropriate government to confer on a licensee the powers of the telegraph authority under the Indian Telegraph Act 1885 for placing telegraph lines and posts — i.e. a statutory wayleave for transmission lines and towers. Sections 67–68 govern works and overhead lines.
Compensation	For the tower footprint, the corridor burden and damage to crops and trees, under the Telegraph Act machinery and state-notified rates; not a full market-value award for the land.
Workflow consequence	A right-of-user track with a linear corridor geometry and a point footprint per tower. Disputes are common and the District Magistrate is the compensation authority.

11.5 Atomic Energy Act, 1962

Attribute	Detail
Ministry	Department of Atomic Energy; projects executed by NPCIL
Mechanism	Fourth Schedule act with its own acquisition provisions. Our act-research corpus records the relevant sections as Sections 9–12 , correcting an earlier claim of Sections 16–18; this has not been independently verified against the gazette text and is flagged in Section 51.
Practical route	Nuclear projects in practice combine this Act with LARR acquisition executed by the state government, with NPCIL funding an extensive R&R package implemented by the state.
Pre-condition	Atomic Energy Regulatory Board site clearance is a gate before acquisition can meaningfully proceed.
Workflow consequence	Dual-statute projects are real: the same project may carry an Atomic Energy Act basis and a LARR-executed acquisition. The data model must allow a project to declare a primary regime and a secondary regime rather than exactly one.

11.6 Metro Railways (Construction of Works) Act, 1978

Fourth Schedule act for metro construction, with its own notification, objection and compensation machinery and a metro-specific competent authority. In practice metro projects frequently proceed under LARR for compensation purposes, and urban metro acquisition is dominated by structure and business-loss valuation rather than land value — densely built parcels, partial takings, and substantial disturbance claims. Section numbers for this Act are not independently verified in this document.

11.7 The five older acts

Act	Status and relevance
Ancient Monuments and Archaeological Sites and Remains Act, 1958	Active. Acquisition of land in and around protected monuments, by the Archaeological Survey of India. Low volume, but the prohibited and regulated area rules around monuments act as a constraint layer on other projects — which is a useful GIS overlay rather than a workflow.
Damodar Valley Corporation Act, 1948	Active. Acquisition for the Damodar Valley Corporation's dams, canals and power works. Regionally confined but a live example of a statutory-corporation acquirer.
Indian Tramways Act, 1886	Active but effectively dormant for new acquisition. Retained in the Fourth Schedule; a configuration row for completeness, not a build target.
Land Acquisition (Mines) Act, 1885	Active. Concerns acquisition of land for mining purposes and the treatment of minerals underlying acquired land. Interacts with the MMDR regime.
Resettlement of Displaced Persons (Land Acquisition) Act, 1948	Active. Note the year: our summary file recorded 1954, which was corrected to 1948. Historical partition-resettlement statute; retained in the Fourth Schedule.

12. Sector regimes outside the Fourth Schedule

These regimes govern major land-consuming sectors but are **not** exempted from LARR. Where they involve compulsory acquisition of private land, LARR applies in full — SIA, consent, four-step compensation, statutory R&R.

12.1 SEZ Act, 2005 — the correction that matters most

SEZ is not a Fourth Schedule act

The Fourth Schedule contains thirteen enactments and the Special Economic Zones Act 2005 is **not** one of them. An SEZ acquisition of private land therefore runs on LARR: social impact assessment, 80% consent where the acquirer is a private developer, First Schedule compensation, Second and Third Schedule R&R.

Our own corpus had this wrong in two places and inconsistently in a third: the master comparative summary listed SEZ as exempt act number five and priced it at the old 30%-solatium rate, while the SEZ act file's own header correctly said it is not exempt and its body then described a pre-2013-style notification-and-declaration workflow anyway. The summary has been corrected; **the SEZ file body has not**. See Section 51.

Attribute	Detail
Ministry	Ministry of Commerce & Industry; Board of Approval and Development Commissioners
Sequence	The developer assembles contiguous land first , then applies for SEZ notification. Acquisition or purchase precedes the SEZ status, rather than flowing from it.
Land route	LARR acquisition by the state on the developer's behalf, or direct purchase, or state industrial-corporation land
Distinctive feature	The requiring body is frequently a private company , not a ministry. That changes both the consent threshold (80%) and the access model — a private developer account with tightly restricted visibility, which is a genuine RBAC design problem addressed in Section 25.

12.2 Mines and Minerals (Development and Regulation) Act, 1957

Non-coal minerals. The state grants a mining lease or composite licence; the mineral vests in the state, but the **surface rights** do not come with the lease. The lessee must separately acquire or obtain consent for surface access — usually by negotiated purchase or under state land revenue law, occasionally by LARR acquisition where the state acquires for a public purpose. Compensation is negotiated rather than statutory, with royalty flowing to the state and District Mineral Foundation contributions to the affected area. The system requirement is a **surface rights negotiation** track — consent records per parcel within a lease boundary — not an acquisition workflow.

12.3 Airports Authority of India Act, 1994

Greenfield airports and airport expansion. AAI may acquire directly under its own Act, but large airport acquisitions in practice run through LARR with SIA. Two features matter for the platform. First, airport acquisition is **heavily litigated** and courts have quashed declarations for procedural non-compliance, including on the ground that urgency provisions cannot be used to bypass hearing rights. Second, the surrounding constraint regime — obstacle limitation surfaces, noise zones — means the acquisition footprint is smaller than the affected footprint. The design consequence is that **strict procedural compliance evidence** — hearing dates, notice service, SIA completeness — is not bureaucratic overhead but the primary defence against a quashing order, and the audit ledger is what produces it.

12.4 Industrial corridors — DPIIT and the SPV model

The Delhi-Mumbai, Chennai-Bengaluru, Vizag-Chennai and Amritsar-Kolkata corridors operate through special purpose vehicles jointly held with state industrial development corporations. The SPV assembles land, develops trunk infrastructure, and allots serviced plots to investors. Land comes through a **mix** of LARR acquisition, land pooling and direct purchase — often within the same node. The modelling requirement is that a single project must be able to carry parcels on **different process types** simultaneously, which is a stronger requirement than one regime per project.

13. Right-of-user and easement regimes

Grouped here because they share a property that breaks most acquisition data models: **title never moves**. There is no vesting, no possession, no mutation, no displaced family and no R&R. What is created is a burden on land that someone else continues to own.

Regime	What is created	Compensation basis	Geometry
Petroleum & Minerals Pipelines Act 1962	Right of user in a corridor; construction and deep-rooted planting prohibited over it	10% of market value for the right of user, plus damage to crops, trees and structures	Linear corridor clipped against parcels; area by spatial intersection
Electricity Act 2003 s.164 with Telegraph Act 1885	Wayleave for transmission lines, towers and poles	Tower footprint, corridor burden, crop and tree damage at state-notified rates	Line geometry plus point footprints
Indian Telegraph Act 1885 (with the Right of Way Rules)	Wayleave for telecom towers, poles and underground fibre; District Magistrate as authority	One-time or recurring payment for the footprint, plus restoration and crop damage	Thousands of micro-parcels per corridor, multiple licensees
Temporary occupation (LARR Chapter on temporary occupation)	Time-limited occupation for construction, borrow pits, haul roads	Rent for the occupation period plus restoration to original condition	Whole or partial parcel, time-bounded

Why telecom right of way is the hardest of these

A single fibre route generates thousands of parcel intersections, each individually trivial and collectively unmanageable. Multiple licensees cross the same corridor. There is no central register, and disputes are routine. If the platform ever extends here it needs a **micro-parcel bulk mode** — corridor-level notification with per-parcel derivation — not the same per-parcel workflow used for a highway. Correctly out of scope for this submission; worth naming as a deliberate exclusion rather than an omission.

14. Non-acquisition land assembly

Three mechanisms in wide and growing use that produce land for public projects without acquiring anything. Each is directly relevant to project categories the problem statement names, and none fits an acquisition workflow.

14.1 Land pooling and Town Planning Schemes

Landowners contribute land to an authority, which develops trunk infrastructure and returns a smaller but serviced plot — typically 40–60% of the contributed area — plus, in some schemes, a cash or annuity component. There is no notification, no objection window, no award and no solatium, because there is no compulsory taking. Used at scale in Delhi's land pooling policy, the Amaravati capital scheme, and the long-established Town Planning Scheme mechanism under Gujarat and Maharashtra town planning law.

Transferable Development Rights are the related instrument: development potential is moved from a parcel that must be surrendered for a public purpose to another parcel elsewhere in the city, compensating the owner in buildable area rather than cash.

The workflow shape is entirely different — **contribution → scheme preparation and objection → reconstitution of plots → infrastructure development → redistribution and TDR certificate issuance** — and the money flows in the opposite direction from acquisition. Any model that assumes every parcel has compensation payable to an owner cannot represent a pooling scheme.

14.2 Government allotment and lease

The dominant route for renewable energy. Solar parks consume roughly 3.5–7.5 acres per MW and wind farms more; most projects are sited on government wasteland allotted or leased to the developer by the state revenue or energy department, or on private land taken on long-term lease at negotiated rates. Neither involves acquisition, and the entire owner-compensation workflow is skipped. What matters instead is the allotment order, the lease term and rate, and any encroachment or common-land grazing rights on the allotted parcel — which is where these projects actually generate disputes.

14.3 Consent purchase and direct negotiation

Direct purchase from willing sellers, used across sectors and available under the NH Act as a limited exception that short-circuits the 3A–3D sequence entirely. Under LARR Section 46, private purchase above a state-specified area threshold triggers the R&R obligations even though nothing was acquired — a genuinely counterintuitive rule that a compliance check should enforce automatically. Workflow: negotiation → consent record → sale deed registration → mutation, with an R&R branch above the threshold.

14.4 The process_type enumeration

First-class field, not a special case

```
process_type ENUM:
  compulsory_acquisition -- LARR, NH, Railways, Coal, Metro, Atomic, DVC, ...
  right_of_user           -- pipelines, transmission, telecom wayleave
  requisition             -- 1952 Act: time-bounded, releasable
  land_pooling            -- TP schemes, pooling policies, TDR
  govt_allotment          -- state wasteland allotment or lease
  consent_purchase        -- direct negotiation (R&R above s.46 threshold)
  legacy_1894_transitional -- Section 24 cases

-- A project may carry parcels of more than one process_type simultaneously.
-- Therefore process_type belongs on the PARCEL, not only on the PROJECT.
```

15. Pre-condition gates

These are not acquisition regimes and must not be built as workflows. They are clearances owned by other ministries which block acquisition until satisfied. The platform's job is to know the status and hold the evidence — the boundary argument in Section 19.

Gate	Statutory basis	Owner	Process	How we model it
Forest clearance	Forest (Conservation) Act 1980, renamed by the 2023 amendment as the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980	MoEFCC, through the PARIVESH portal; state forest departments	Diversion of forest land to non-forest use: Stage-I in-principle approval, compliance with conditions including compensatory afforestation and net present value payment, then Stage-II final approval	Gate with three states (not_required / in_progress / obtained), a certificate artefact, and a reference number. Blocks stage advance where forest land is involved.
Environmental clearance	EIA notification under the Environment (Protection) Act 1986	MoEFCC or the State Environment Impact Assessment Authority	Screening, scoping, EIA study, public hearing, appraisal, clearance	Same gate pattern. Note that renewable energy is generally in the exempt category, which is why the gate must be conditional on project category rather than always required.
Tribal consent in Scheduled Areas	PESA 1996; Forest Rights Act 2006; LARR Section 41	The Gram Sabha, with the Sub-Divisional, District and State Level Committees under the FRA process	Gram Sabha resolution; where forest rights are claimed, recognition of individual and community rights before diversion	Hard blocking gate plus a new stakeholder role. Requires spatial detection of Fifth and Sixth Schedule areas as a parcel attribute — this cannot be a manual tick box.
Nuclear site clearance	Atomic Energy Act 1962 machinery	Atomic Energy Regulatory Board	Site safety review and clearance before project land is committed	Gate on nuclear projects only; certificate artefact.
Coastal, wildlife and monument constraints	CRZ notification; Wild Life (Protection) Act 1972; Ancient Monuments Act 1958	MoEFCC; National Board for Wild Life; ASI	Zone-specific prohibitions and regulated areas	Better modelled as GIS constraint layers that flag a proposed alignment at Stage 0 than as gates — the value is in early warning, not in approval tracking.

The line to hold

It is tempting to build the forest clearance workflow. It is also the fastest way to lose the submission: PARIVESH exists, works, belongs to a different ministry and operates under a different Act. **The right answer to “what about environmental clearance?” is “deliberately out of scope, here is why, and here is the integration point.”** That reads as engineering judgement. Silence reads as an oversight, and a half-built clearance module reads as scope indiscipline.

16. Comparative matrices

The tables in this section are the direct input to the regime configuration table in Section 31. Each column of the first matrix becomes a configuration field.

16.1 Procedural comparison

Feature	LARR 2013	NH Act 1956	Railways 1989	Coal Bearing 1957	Pipelines 1962
Notifying authority	Appropriate govt (usually State)	Central Govt	Central Govt	Central Govt	Central Govt
Intention notification	s.11	s.3A	s.20A	s.7(1)	s.3
Pre-notification stage	SIA (s.4-9)	None	None (consultation by practice)	Prospecting (s.4)	None
Objection window	60 days (s.15)	21 days (s.3C)	30 days (s.20D)	After s.7(1) notification	21 days (s.5)
Declaration	s.19	s.3D	s.20E	s.9(1)	s.6
Lapse rule	12 months s.11→s.19; 12 months s.19→award (s.25)	1 year 3A→3D	Per chapter provisions	Validity window on s.7(1)	—
Consent required	80% private / 70% PPP	No	No	No	No
Award authority	Collector	Competent authority / CALA	Competent authority	Coal Controller machinery	Competent authority
Compensation basis (post-2015)	First Schedule: MV x factor + assets + 100% solatium	First Schedule via 2015 order	First Schedule via 2015 order	Own provisions + 2015 order	10% of MV for RoU + damages
Statutory R&R	Yes — Second and Third Schedules	Via 2015 order	Via 2015 order	Policy-based + 2015 order	None
Possession trigger	Payment made or tendered (s.38)	Vesting + 60-day notice (s.3E)	Vesting + deposit (s.20I)	Vesting then transfer	No possession — RoU only
Dispute route	LARR Authority (s.64)	Arbitrator appointed by Central Govt (s.3G(5))	Gazetted officer arbitrator	Tribunal	Competent authority / court
Title transfers?	Yes	Yes	Yes	Yes, then to PSU	No

16.2 Relative burden and duration

Indicative rather than statutory — these are the practical durations that make LARR the slowest and most expensive route and explain why sector acts are used wherever available.

Regime	Typical elapsed time	Cost intensity	Principal driver of delay
LARR Act 2013	3-5 years	Highest	SIA and Expert Group appraisal; consent collection; R&R award and site development
NH Act 1956	1-2 years	Moderate (higher post-2015)	3C objection hearings; compensation disputes to arbitration
Railways Act 1989	1-2 years	Moderate	Similar to NH; 30-day objection window
Coal Bearing Areas 1957	2-3 years	Moderate	Prospecting stage; PSU transfer
Pipelines 1962 (RoU)	6-18 months	Low	Crop-damage valuation disputes across a long corridor
Land pooling	2-4 years	Low cash cost	Owner participation rates; scheme reconstitution and objections
Government allotment	3-12 months	Lowest	Encroachment and common-land rights on the allotted parcel

17. Repealed, exempt, or subject: getting the status right

Three states of affairs are routinely conflated, and the conflation produces wrong answers in both directions — treating live statutes as dead, and treating LARR-bound acquisitions as exempt. The distinction below is worth being able to state precisely.

Status	Meaning	Enactments
Repealed	No longer in force. LARR Section 114(1) repeals it outright.	The Land Acquisition Act, 1894 — and this one only. No other acquisition statute was repealed by LARR.
Active but procedurally exempt	Still in force and in active use. LARR's <i>procedure</i> does not apply because the Act is listed in LARR's Fourth Schedule; each runs its own acquisition process. Since the 2015 order under Section 105(3), LARR's First, Second and Third Schedules — compensation and R&R — do apply.	The thirteen: National Highways Act 1956; Railways Act 1989; Metro Railways (Construction of Works) Act 1978; Electricity Act 2003; Atomic Energy Act 1962; Coal Bearing Areas (Acquisition and Development) Act 1957; Petroleum and Minerals Pipelines (Acquisition of Right of User in Land) Act 1962; Requisitioning and Acquisition of Immovable Property Act 1952; Ancient Monuments and Archaeological Sites and Remains Act 1958; Damodar Valley Corporation Act 1948; Indian Tramways Act 1886; Land Acquisition (Mines) Act 1885; Resettlement of Displaced Persons (Land Acquisition) Act 1948.
Active and fully subject to LARR	In force, not in the Fourth Schedule. Where compulsory acquisition of private land is involved, the full LARR process applies — SIA, consent, First Schedule compensation, statutory R&R.	SEZ Act 2005 ; MMDR Act 1957 (where the state acquires); Airports Authority of India Act 1994 (in practice); and every sector not otherwise exempted.

Two errors this table prevents

“Exempt” is not “repealed.” NHAI still acquires highway land under the NH Act every day; Railways still uses Sections 20A–20P. Describing the Fourth Schedule acts as superseded by LARR is simply wrong, and it is the sort of wrong that ends a technical viva early.

“Sector act” is not “exempt.” The Fourth Schedule is a closed list of thirteen. An SEZ, a non-coal mine or an airport does not get to skip SIA and consent because it has its own statute — and a compensation engine that treats it as exempt will under-pay by a factor of roughly two to four.

PART III

System Boundary and Stakeholders

Who uses the system, what each of them does at each stage, and — the question that decides whether the project is buildable — where the system stops. This part answers the specific question raised in the earlier sessions: when the requiring body asks for land, which stage is that, and is it inside our boundary or outside it.

In this part

- Stage 0: the proposal, and why it is not a statutory stage
- The in-bound / out-of-bound table, with the reasoning for each line
- Master stakeholder registry: thirteen roles, four tiers, five sub-roles
- How many login panels are actually needed, and which to merge
- LARR Act: all seven stages with per-stakeholder actions
- NH Act: all eight stages plus the mutation track
- Cross-cutting roles that span every stage

18. Stage 0: where the requiring body enters

The problem statement calls it the **Land Requiring Body**. It is the actor that needs land: NHAI or NHIDCL for a highway, Indian Railways for a rail corridor, a state PWD or CPWD for a building, an irrigation department for a canal, a renewable energy PSU for a solar park, an industrial corridor SPV for a node, a private developer for an SEZ.

18.1 Requiring body and acquiring authority are different actors

The problem statement lists “Land Requiring Bodies” and “Land Acquiring Authorities” as separate stakeholders, and they are: the requiring body needs the land and pays for it; the acquiring authority — the District Collector, or the competent authority / CALA under the sector acts — executes the statutory process on its behalf and holds the legal power. NHAI cannot issue a Section 3C order; the CALA cannot decide a highway alignment.

Our source material sometimes treats the requiring body and the Project Implementing Agency as separate stakeholders and sometimes as the same one. In most real projects **the requiring body and the PIA are the same entity** — NHAI both requires the land and builds the road. The recommendation is to collapse them into a single role in v1 and carry a distinguishing attribute rather than maintain two nearly identical RBAC rows. Where they genuinely differ — an SPV that assembles land and allots it to investors who build — the difference is one field, not one role.

18.2 What Stage 0 actually consists of

Stage 0 sits **before any statutory stage** — before Section 4 or Section 11 on the LARR track, before 3A on the NH Act track. It is the trigger event for everything downstream, and it is entirely a system stage rather than a legal one. The content below is drawn from the project's own LARR process map, which sets it out in more detail than most secondary sources.

Component	What the requiring body submits
Project justification	Project proposal and covering letters; the public purpose for which land is required; justification for the requirement; applicable prior approvals and reports
Land requirement	Extent of land required; location of the required land; project design, alignment or layout information where applicable
Land and revenue information	Village details; survey or khasra numbers; parcel areas; land classification; revenue records; ownership and tenancy information available in the records; maps and land plans
Commercials	Estimated cost including compensation and R&R; budget head and fund availability; confirmation of the requiring body's obligation to deposit its share
Regime selection	The statutory basis proposed — which determines the entire downstream stage sequence, and therefore the single most consequential field in the system
Clearance status	Whether forest, environmental and other clearances are obtained, in progress or not required, with reference numbers and certificates where obtained

On acceptance, Stage 0 also produces the first public-facing acts recorded in the process map: publication of the requirement in the **Official Gazette**, and publication in the **local language** at the Panchayat, Municipality or Municipal Corporation as applicable and at the offices of the District Collector, the Sub-Divisional Magistrate and the Tehsil. Note that the requirement is the *local* vernacular, not a generic Hindi-English toggle — a national system needs per-state language packs for notice templates, which is a concrete requirement rather than a nice-to-have.

18.3 Scrutiny and routing at Stage 0

The problem statement asks specifically for “digital scrutiny” and “automated workflow routing.” Stage 0 is where both live.

Check	Type	Behaviour on failure
Completeness	Automated	Reject back to submitter with the specific missing fields — never a generic “incomplete”
Regime validity	Rule	Block if the selected regime is not available to that requiring body class (e.g. a private developer selecting the NH Act)

Check	Type	Behaviour on failure
Spatial conflict	Automated (PostGIS)	Flag where the proposed footprint intersects an existing project's parcels — two requiring bodies proposing the same survey numbers is a real and currently undetected failure
Constraint intersection	Automated (PostGIS)	Warn where the footprint touches forest land, a Scheduled Area, a CRZ zone, a monument regulated area or multi-crop irrigated land
Food security ceiling	Rule (LARR s.10)	Flag where irrigated multi-crop area exceeds the state-configured limit
Budget adequacy	Rule	Warn where the estimate is materially below a modelled expectation for that area and land class — an early signal of later payment failure
Document classification	AI-assisted	Route uploaded documents to the right slots and flag apparent mismatches; the human decides

Routing rules are configuration, not code. Worked examples: a proposal above a configured cost threshold requires central rather than state sanction; where objections at the hearing stage exceed a configured count, the case escalates to the state government; where a project spans more than one state, a sub-project is created per state with its own approver chain.

Multi-state projects are a data model requirement, not a reporting feature

A highway or rail corridor routinely spans several states — different circle rates, different state government approvers, different vernacular languages, different state amendments to LARR — for the **same** project. A model that carries state only at parcel level cannot produce a per-state approval chain or a per-state rollup. The fix is a project → state_subproject → parcel hierarchy, with the state subproject as the unit of approval, language selection and rule resolution. Retrofitting this later means rewriting every aggregation query in the system.

19. System boundary: in-bound and out-of-bound

This is the table to put on a slide. Judges reliably ask what happens about environmental clearance, or about the actual construction, and “deliberately out of scope, here is why, here is the integration point” is a far stronger answer than either silence or an over-claim.

Domain	In the system — we build this	Outside the system — we integrate or attach	Why the line is here
Proposal	Requiring body submits purpose, extent, location, village and survey list or alignment corridor, budget, regime selection; scrutiny and routing; approval record	The DPR and the technical alignment engineering itself — accepted as an uploaded artefact, not authored in-app	We are a process system, not a civil engineering tool. Authoring a DPR is a different product.
Clearances	A gate per clearance: status, reference number, certificate artefact, blocking behaviour	The forest, environmental, wildlife and coastal clearance workflows — MoEFCC's PARIVESH, under different Acts	Different ministry, different statute, working system already exists. Rebuilding it is scope indiscipline.
Land records	Cache and display of survey numbers, ULPIN, area, classification, RoR extract, circle rate; the parcel's acquisition state	DILRMP, Bhulekh and the state revenue records — which remain the system of record for title	We are the system of record for the acquisition <i>process</i> . Becoming a second land registry would create exactly the duplication the PS complains about.
Identity	Consuming an identity assertion; storing a reference, never the raw identifier in the clear	UIDAI's authentication service itself; DigiLocker for document-backed identity	Direct Aadhaar authentication requires licensed AUA/KUA status. Using DigiLocker's sign-in is a compliance requirement, not a convenience.
Money	Award computation, payment instruction, disbursement status, UTR capture, reconciliation view, interest accrual	PFMS and treasury accounting — which remain the system of record for the ledger	Duplicating a treasury ledger is both out of scope and a governance risk. We instruct and track; PFMS holds the money.
Publication	Generation of the notification and award documents in the correct languages, with the statutory content, ready for publication	The e-Gazette publication act itself where no write API is available; local newspaper insertion	Bhoomi Rashi proves gazette automation is possible for MoRTH. Until that access is extended, generating a publication-ready artefact for manual submission is the honest current state.
Consent (Scheduled Areas)	The gate, the resolution artefact, the blocking rule, the audit record	The Gram Sabha's deliberation and the FRA committee process	A village assembly's decision is not a workflow we get to implement. We record that it happened and hold the evidence.

Domain	In the system — we build this	Outside the system — we integrate or attach	Why the line is here
Disputes	Case record, linkage to affected parcels, stay order upload, timeline exclusion, outcome recording, enhanced award processing	The judicial or arbitral decision itself; e-Courts and NJDG as the case-status source	Courts are not users of our workflow engine. We consume status and react to it.
R&R delivery	Affected family registry, entitlement matrix, award, disbursement tracking, resettlement site infrastructure checklist and progress	Actual construction of the resettlement site amenities	We track the Third Schedule obligation to completion; the state agencies build.
After possession	Possession record, handover certificate, mutation trigger and status, closure	Project construction and execution	The acquisition lifecycle ends at clean possession and mutated title. Construction monitoring is a different system — and a different problem statement.

The boundary is the strongest defensive asset in the submission

Every line above converts a potential criticism (“you didn’t handle X”) into evidence of judgement (“X belongs to another authority; here is where we meet it”). A team that can draw this line crisply is demonstrably a team that has understood the domain rather than the feature list.

20. Master stakeholder registry

Thirteen primary roles across four tiers. Tier 1 creates and processes acquisition data; Tier 2 enables compliance, finance and rehabilitation; Tier 3 oversees policy and audit; Tier 4 is the public and external actors. For each: what they do, what is broken for them today, what the system must give them, and what access that implies.

20.1 Tier 1 — core operational

#	Stakeholder	Role	What is broken today	System need	Access level
1	Central Ministry official (MoRD, MoRTH, MoD, Railways, MoUD, MoPNG, Power)	Project sanctioning authority, fund allocator, policy overseer. Issues or approves central notifications where the regime places the power centrally (3A, 3D, 20A, 20E).	No national view. Relies on spreadsheets and PDF returns from states; cannot compare state performance in real time or see where a project is actually stuck.	National dashboard with drill-down from India to parcel; fund utilisation view; statutory deadline breach alerts; export in Parliament-ready format.	Super-admin: read all projects, write policy and regime configuration, sanction. No parcel-level edit.
2	State Government / Revenue Department	Coordinates between Centre and district; appoints the CALA; issues the LARR notification and declaration where the appropriate government is the state; releases the state share.	District collectors report in different formats with no common schema; fund release approvals move on paper.	State dashboard aggregating all districts; standardized ingestion; e-signed fund release approval; inter-district dispute view.	State-admin: all projects in state, write district-level configuration, appoint CALA.
3	District Collector / CALA / competent authority	The ground-level executing authority and the busiest user in the system. Issues local notices, hears objections, passes orders, determines compensation, declares the award, takes possession.	Manual file movement; hearings scheduled by hand; compensation computed on paper; hundreds of parcels per project tracked in registers.	A workflow inbox with stage gates; auto-generated notices and orders in the correct language; the compensation calculator; a hearing calendar; DSC signing at each gate; a landowner registry.	District-admin: full create-read-update on parcels and stages within the district; stage advance authority; document signing.
4	Project Implementing Agency / Land Requiring Body (NHAI, NHIDCL, PWD, CPWD, Railways, irrigation dept, SPV, SEZ developer)	Requisitions the land, provides the technical alignment, funds the acquisition, receives the right of way and commences construction.	No visibility of acquisition progress against the construction timeline; cost overruns driven by land delays it cannot see coming.	Project timeline showing acquisition stages against construction milestones; delay impact alerts; GIS overlay of alignment against acquired parcels; the document repository.	Project-manager: own projects only; write project metadata and upload alignment; read-only on the parcel workflow. It funds the process; it does not run it.
5	Field Surveyor / Lekhpal / Patwari / Amin	Conducts the ground survey, verifies boundaries, enumerates structures, crops and trees, captures GPS, obtains the owner's acknowledgement.	No connectivity where the work happens; paper forms lost or re-keyed at the office; no GPS linkage between the measurement and the record.	A mobile-first interface with offline capability; automatic GPS capture; camera evidence with hash and timestamp; digital acknowledgement; visible sync status.	Field-user: assigned parcels only; create survey data; no approval rights.

20.2 Tier 2 — support and monitoring

#	Stakeholder	Role	What is broken today	System need	Access level
6	Rehabilitation & Resettlement Administrator	Prepares the R&R scheme and award; delivers Second Schedule entitlements; develops and monitors the resettlement site against the Third Schedule.	Affected-family lists go stale; site infrastructure progress tracked in separate spreadsheets with no linkage to the acquisition stage.	Affected family registry with the entitlement matrix; per-family entitlement cards; site infrastructure checklist with progress; displacement timeline from pre-displacement through transit to resettlement.	R&R-admin: full access to the R&R module; read-only on the acquisition workflow.
7	Finance / PFMS Controller / Treasury Officer	Manages budget allocation, deposit with the authority, DBT disbursement, reconciliation and the handling of unclaimed amounts.	No real-time view of fund flow; manual reconciliation between the project agency and treasury; delayed payments generate interest liability and litigation.	Fund flow tracker from budget through sanction, deposit, disbursement to reconciliation; PFMS integration; interest accrual calculator; pending payment alerts; utilisation certificates.	Finance-admin: read all financial data, write payment status. No workflow approval authority — separation of duties.
8	Legal / Litigation Officer	Handles arbitration and references, court cases, stay orders, and the government's defence of awards.	Court stays are not reflected in the acquisition record, so statutory clocks do not pause and deadline data drifts from legal reality; no central case repository.	Case management with parcel linkage; automatic timeline pause on stay upload and resume on vacation; arbitration and reference tracking; assembled document bundle per case.	Legal-user: read-write on litigation data; read-only workflow for context; authority to apply a timeline exclusion.

20.3 Tier 3 — policy and oversight

#	Stakeholder	Role	What is broken today	System need	Access level
9	Policy maker / NITI Aayog	National policy review, interstate benchmarking, infrastructure corridor planning.	Data siloed by ministry; no models for acquisition bottlenecks; cannot correlate land delay with infrastructure delivery.	Delay probability models with feature attribution; comparative state rankings; scenario analysis; export to briefing formats.	Policy-read: national aggregated data and analytics; no operational write; no personal data.
10	CAG / audit / Parliamentary Committee	Financial and compliance audit; review of fund utilisation and timeline adherence.	Inconsistent formats across states; manual document verification; no immutable trail to verify against.	Append-only audit log with before and after values; document version lineage and integrity verification; compliance scorecards; log export.	Auditor: read-only across everything, including the audit log itself; export; no operational write of any kind.
11	National Monitoring Committee (LARR Sections 48-50)	The statutory committee overseeing R&R implementation.	Reporting is periodic, self-reported and retrospective.	R&R implementation status by project and state; Third Schedule amenity completion; displaced family outcomes.	Oversight-read, R&R scoped. Merges naturally with role 9 and 10 — see Section 21.

20.4 Tier 4 — public and external

#	Stakeholder	Role	What is broken today	System need	Access level
12	Landowner / affected family	The claimant. Entitled to compensation, R&R, information, objection and grievance redressal.	No transparency at all. Does not know the status of their own land; cannot estimate what they are owed; grievances disappear into the system.	Public portal with no login for general information and OTP step-up for personal detail; parcel status tracker; compensation estimator; objection and grievance filing with a ticket number; download of notices and awards; hearing dates.	Public: read own parcel only; write objections and grievances.
13	Arbitrator / LARR Authority / court	Adjudicates compensation disputes; passes enhanced awards.	No digital case file; physical award copies; delayed communication of orders.	Case-specific dossier with all documents, maps and award history; award recording with signature; e-Courts linkage.	Judicial: case-scoped read-write; no access to unrelated projects. Best implemented as a scoped guest token, not a standing portal.
14	NGO / civil society / RTI applicant / researcher	Monitors transparency, files RTI applications, advocates for affected communities.	Data not available in analysable form; no interface for researchers.	Open data endpoints with aggregation and no personal data; project-level public reports; RTI response assistance.	Public-read: aggregated, non-personal data only.

20.5 The five sub-roles the problem statement implies

These are treated as distinct stakeholders in the problem statement's framing but function as sub-roles beneath the District Collector. They should exist as assignable roles inside the district panel rather than as separate panels — the alternative is five more near-duplicate portals to build and maintain.

Sub-role	Function	Why it must exist as a distinct role
Sub-Divisional Magistrate	Frequently appointed as the CALA or competent authority in large districts	Needs the full CALA workflow inbox, scoped to a sub-division rather than the district — a jurisdiction scoping problem, not a new permission set
Tehsildar / Revenue Inspector	Revenue record verification; coordinates the field survey; local notice publication	Needs read access to revenue data and write access to verification outcomes, without award or possession authority
Bank / Treasury Officer	Holds the deposited compensation; verifies disbursement	Needs payment reconciliation access without any view of the workflow — the narrowest role in the system
Police / law and order	Provides assistance where possession must be taken without surrender	Needs a possession-execution task and nothing else. Auto-alerted where surrender does not occur by the notice deadline
SIA agency and Expert Group	Conducts the social impact assessment; the Expert Group appraises it	Needs an SIA report upload and review surface. Critically, these are supposed to be empanelled entities, not merely users who register — see below

Empanelment is not the same as an account

SIA Expert Groups and independent valuers are meant to be certified or empanelled bodies with recorded credentials, not simply whoever logs in. A lightweight **vendor empanelment sub-module** — register once with credentials and validity period, then get assigned across projects — closes a real integrity gap: it makes it possible to ask later who appraised this SIA and were they qualified to.

21. Panel plan

The question asked directly in the earlier session was how many panels are needed. Fourteen roles do not mean fourteen panels. The recommendation is **eight login-based panels plus one no-login public portal**, with the remaining roles implemented as scoped roles or tokens inside those panels.

#	Panel	Primary users	Defining surface
1	Central Ministry / Sanctioning Authority	Role 1	National dashboard, project sanction, regime configuration, policy export
2	State Government / Revenue	Role 2, Tehsildar sub-role	State dashboard, district comparison, CALA appointment, fund release queue
3	District Collector / CALA	Role 3; SDM, Tehsildar, Police sub-roles	The workflow board — the operational heart of the system. Stage inbox, document generator, hearing scheduler, compensation calculator, landowner registry
4	Land Requiring Body / PIA	Role 4	Proposal submission, alignment upload, acquisition-versus-construction timeline, delay alerts
5	Field Surveyor (mobile-first)	Role 5	A deliberately minimal offline-capable surface: assigned parcel list, survey form, GPS capture, camera, sync status
6	R&R Administrator	Role 6; National Monitoring Committee read view	Affected family registry, entitlement cards, R&R award, site infrastructure tracker
7	Finance / PFMS Controller	Role 7; Bank/Treasury sub-role	Fund flow pipeline, per-landowner payment status, interest calculator, reconciliation
8	Legal / Litigation Officer	Role 8	Case list, stay order upload with timeline impact, document bundle assembly, arbitration and reference tracking
9	Public portal (<i>no login</i>)	Roles 12 and 14	Survey-number lookup, parcel status, compensation estimator, notice download, grievance filing with ticket, open aggregate data. OTP step-up for personal detail only

21.1 The merges, and the reasoning

Decision	Reasoning
Merge policy maker, CAG/audit and the Parliamentary Committee into one read-only Policy & Audit role	Their permission sets are nearly identical — aggregate read, log read, export, no operational write. Building three near-duplicate dashboards spends scarce development time on differences that are cosmetic. Differentiate by saved views within one panel, not by three panels.
Implement the arbitrator as a case-scoped guest token rather than a standing portal	An arbitrator engages with a handful of cases, briefly, and must be unable to see anything else. A time-limited scoped token is both less to build and a tighter security posture than a full account with a jurisdiction filter.
Landowner as OTP-based public access, not an administrative panel	Requiring millions of citizens to hold managed accounts is an onboarding problem with no upside. Public information needs no login; personal detail needs a one-time step-up against the parcel record.
Collapse requiring body and PIA into one role with an attribute	In most projects they are the same entity. Two RBAC rows that differ in nothing measurable is maintenance cost without benefit — see Section 18.1.
SDM, Tehsildar, Revenue Inspector and Police as roles inside the district panel	Each needs a slice of the CALA surface, differing in jurisdiction scope and a small permission subset. Roles with jurisdiction scoping handle this; separate panels would quadruple the district build.

This is a scope-reduction recommendation driven by team bandwidth, not a claim that the merged roles matter less. Nine surfaces built well will demonstrate multi-stakeholder coordination more convincingly than fourteen built thinly — and multi-stakeholder coordination is the core requirement of the problem statement.

The demonstration that proves the point

Show **one project, five views**. Central Ministry: an India map with two states flagged as delayed. District Collector: a board with twelve parcels at the notification stage, five at declaration, two awaiting award. Field Surveyor: an offline form with three records pending sync. Landowner: enter a survey number, see status and an estimated entitlement. Finance: the fund pipeline from allocation through deposit and disbursement to pending.

The same underlying data, five roles, five different truths — that is the requirement made visible in about ninety seconds.

22. LARR Act 2013 — stage-by-stage stakeholder matrix

For each statutory stage: who does what, and what the system does about it. These tables are the direct specification for the stage definitions in the workflow engine.

	SIA (s.4-9)	Prelim. Notification (s.11)	Declaration (s.19)	Award (s.21-30)	R&R Award (s.31-42)	Possession (s.38)	Reference (s.64)
Appropriate Govt (Centre / State)	●	●	●		○	○	
District Collector	○	○	○	●	○	●	○
SIA Agency / Expert Group	○			○			
R&R Administrator	○		○		●	○	
Landowner / Affected Family	○	○	○	○	○	○	○
Finance		○	○	○	○		○
Legal Officer				○			○
LARR Authority							●

● lead actor for that stage ○ materially involved, not the lead

Figure 3. LARR Act 2013 stage flow and stakeholder involvement — filled circle is the lead actor for that transition (matches the gate matrix in Section 27.1); open circle is materially involved but not the one who advances the stage.

22.1 Stage 1 — Social Impact Assessment (Sections 4-9)

Stakeholder	Role in this stage	System action
Appropriate Government (State, or Central for central projects)	Mandates the SIA; consults the local Panchayat, Municipality or Municipal Corporation; defines the affected area; appoints the SIA agency; examines the report and records its decision on public purpose and minimum area	Initiates the SIA workflow; records the affected-area definition as a geometry; assigns the empanelled agency; captures the recorded decision as a signed artefact
SIA agency / Expert Group (external, empanelled)	Conducts the independent multidisciplinary assessment; prepares the Social Impact Management Plan; the Expert Group appraises the report and may recommend abandonment	Report and SIMP upload against a structured schema; public hearing minutes upload; Expert Group appraisal recorded with the group's empanelment reference
District Collector	Facilitates the public hearing in the affected area; provides local revenue and village data	Schedules the hearing; generates and serves hearing notices in the local language; uploads the hearing record; verifies village-level data
Landowner / affected family	Participates in the public hearing; provides livelihood and dependency information	Receives the hearing notice; views the schedule on the public portal; submits livelihood documentation and representations
R&R Administrator	Begins affected-family profiling from the SIA findings; identifies vulnerable groups	Creates the initial affected-family registry from SIA data; tags Scheduled Caste, Scheduled Tribe and other vulnerable categories
Gram Sabha (Scheduled Areas only)	Consultation and, under Section 41, prior consent	The Scheduled Area gate becomes active here; resolution artefact required before the stage can close
NGO / civil society	Monitors the fairness of the process; represents vulnerable groups	Views hearing schedules on the transparency portal; submits third-party observations to the record

The assessment content the system must structure — taken from the project's own process map, which is more specific than most secondary sources: number of affected families; number of displaced families; land and property affected; livelihoods dependent on the land; agricultural and non-agricultural employment; public and community assets; common property resources; infrastructure and facilities; impact on vulnerable groups; impact on the local economy; resettlement requirements; and whether people will in fact have to relocate.

And the questions the report must answer, which make good section headings for the upload schema: why is the land being acquired; who will be affected; how will they be affected; how many people will be displaced; what social and economic effects will occur; are there alternatives that would cause less displacement or damage; and what mitigation and rehabilitation measures are required.

22.2 Stage 2 — Preliminary notification (Sections 11-15)

Stakeholder	Role in this stage	System action
Appropriate Government	Issues the Section 11 notification; defines the affected area	Generates the notification with statutory content; publishes to Gazette, two newspapers including one in the regional language, local offices and website; starts the 60-day objection clock and the 12-month clock to declaration
District Collector	Publishes locally; receives and hears objections; makes a recommendation to government	Manages objection intake with acknowledgement; schedules hearings; records outcomes; generates the recommendation
Landowner / public	Files objections within 60 days; attends the hearing	Objection filing on the public portal with document upload and ticket number; hearing status tracking; view of the order
Legal Officer	Reviews objection orders for legal soundness	Reviews draft orders; flags legal risk; validates order language against templates
Requiring body	Deposits its share of the acquisition cost; presents the project justification at hearings	Deposit confirmation recorded as a gate condition; uploads technical justification; views hearing outcomes
Central Ministry (central projects)	Monitors notification compliance	Views status; receives the lapse warning alert as the 12-month deadline approaches

The survey and verification work in practice runs across Stages 2 and 3 and is where the field role is active. The process map breaks it into two distinct exercises which the data model should keep separate: the **physical survey** — where exactly is the land, what are its boundaries, what area is involved — and **land record verification** — what does the official revenue record say about this parcel, who is recorded as having an interest, what is the recorded area and classification. A third exercise, **verification for**

compensation, asks who is legally interested in the land, what structures and assets are affected, and what compensation is payable. Three different questions, three different record types, one parcel.

22.3 Stage 3 — Declaration of acquisition (Sections 16-19)

Stakeholder	Role in this stage	System action
R&R Administrator	Prepares the R&R Scheme under Section 16; it is reviewed and approved under Sections 17-18 and its summary published with the declaration	R&R scheme authoring against the Second Schedule; review and approval workflow; locks the affected-family list; generates the published summary
Appropriate Government	Issues the Section 19 declaration once the SIA, hearing and R&R scheme are complete and the requiring body's deposit is made	Hard gate: blocks generation unless SIA valid, objections disposed, R&R scheme approved, consent threshold met where applicable, and deposit confirmed. Confirms compliance with the 12-month rule and stops the clock
District Collector	Publishes locally; initiates the award process	Records the declaration date; transitions parcel state; opens the award stage and starts the 12-month award clock
Landowner / public	Loses the right to transfer the land; becomes entitled to the award	Alert on publication; status update; download of the declaration
Finance	Ensures compensation and R&R budget is allocated	Views total liability estimate; records fund release

The publication channels the process map records for the declaration are worth encoding as a checklist the stage cannot close without: the Official Gazette; two daily newspapers circulating in the locality, of which at least one must be in the regional language; and local government offices — the Panchayat, Municipality or Municipal Corporation, the Collector's office, the SDM's office and the Tehsil — in the local language. The declaration itself must state the district or other territorial division in which the land is situated, the purpose for which it is needed, the approximate area, and where a plan has been prepared, the place at which that plan can be inspected free of cost.

22.4 Stage 4 — Award of compensation (Sections 21-30)

Stakeholder	Role in this stage	System action
District Collector	Serves notice to persons interested; holds the enquiry; determines market value; applies the First Schedule factor; values attached assets; adds solatium; passes the award	Compensation calculator with the audited formula per regime; award document generation with the full computation breakdown; DSC signature; per-parcel and per-owner apportionment
State Revenue Department	Provides land classification and circle rates; confirms recorded interests	Circle rate and classification fetched from the records integration; recorded against the parcel with a source and retrieval timestamp
Landowner / person interested	Receives the award; accepts it or seeks a reference under Section 64	Award breakdown visible on the portal in plain language; acceptance recorded; reference application filed with valuation evidence
Finance	Verifies funds; deposits the amount	Liability roll-up; deposit instruction; interest accrual begins where deadlines are crossed
Legal Officer	Reviews the award for legal compliance; prepares for references	Award review queue; reference case creation with the award as the primary document
Requiring body	Bears the cost; verifies reasonableness against its estimate	Views total compensation against budget; variance alert where the award exceeds the Stage 0 estimate materially

The award record content the process map enumerates: the true area of land acquired; the compensation payable; how that compensation is apportioned among the persons interested; and other relevant particulars required by the Act. Where a person does not agree with the award and raises a dispute, the Authority may consider the Collector's award, the land records, valuation evidence, evidence of sale deeds and comparable transactions, documents submitted by the parties, expert valuation evidence, evidence regarding ownership and interests, and any other relevant evidence — which is precisely the document bundle the legal module must be able to assemble on demand.

22.5 Stage 5 — Rehabilitation and Resettlement award (Sections 31-42)

Stakeholder	Role in this stage	System action
R&R Administrator	Passes the R&R award for each affected family; identifies the resettlement site; plans the Third Schedule infrastructure	Per-family R&R award generation from the entitlement matrix; resettlement site geometry; infrastructure checklist created with target dates
District Collector	Approves the R&R award; ensures convergence with the compensation award	Approval gate; consistency check that every family in the locked registry has an award
Affected family	Chooses among available options — employment, one-time payment or annuity; land for land where applicable; submits preferences	Entitlement card showing what has been awarded and what remains; option selection recorded; site progress visible
State Government	Provides resettlement land; approves site infrastructure and fund release	Site allocation recorded; infrastructure fund release approval; progress monitoring
Requiring body	Funds R&R as part of project cost	Views R&R cost; approves against project budget
National Monitoring Committee	Statutory oversight of R&R implementation	Aggregate implementation view; Third Schedule completion by site

The process map records the R&R award components as: monetary R&R assistance; housing-related entitlements; transportation assistance; subsistence allowance; employment or annuity options where applicable; and other statutory benefits. The resettlement site obligations are recorded as roads, drinking water, electricity, sanitation, healthcare facilities and community facilities — the Third Schedule list. Both are entitlement **lines** with independent delivery status, not a single “R&R complete” flag.

One definitional point the map captures well: depending on the circumstances defined in the Act, an affected family can include people who have lost land, been displaced, are dependent on the acquired land for livelihood, have lost residence, or have certain recognised livelihood or occupation relationships with the affected land. The registry's eligibility model must therefore be broader than titleholders — this is exactly where affected-family counts diverge from parcel counts, and where the consent denominator comes from.

22.6 Stage 6 — Possession (Section 38)

Stakeholder	Role in this stage	System action
District Collector	Takes possession after ensuring compensation and R&R entitlements are paid or tendered, within the Section 38 outer limits	Payment verification gate against every award line; possession recording; possession certificate generation; land vests and the record is updated
Landowner / affected family	Vacates; receives any remaining entitlements and transitional assistance	Acknowledgement of vacation; final payment status; transitional assistance tracking
R&R Administrator	Ensures transit accommodation; coordinates the physical displacement	Tracks each family through pre-displacement, transit and post-resettlement states
Requiring body / PIA	Receives clean possession; commences the project	Handover event fired; possession certificate available for download; right of way marked available on the alignment view
Police / law and order	Assistance where possession is not surrendered	Auto-alert on non-surrender past the notice deadline; possession mode recorded as surrendered, forcible or disputed
Field Surveyor	Verifies physical possession; documents site condition	Post-possession photographs with hash and geotag; boundary integrity confirmation

22.7 Stage 7 — Reference to the Authority (Section 64)

Stakeholder	Role in this stage	System action
Landowner / person interested	Applies for a reference within the Section 64 window	Reference application with the deadline enforced and evidence upload; case created and linked to the parcel and award
LARR Authority	Adjudicates; may enhance the award	Case dossier assembled automatically; enhanced award recorded with the differential computed
Legal Officer	Defends the original award	Counter-evidence upload; case status and hearing calendar
Finance	Processes the enhanced amount with interest	Supplementary payment instruction; interest on the excess computed from the correct base date
District Collector	Implements the Authority's determination	Parcel and award state updated; enhanced disbursement tracked

23. NH Act 1956 — stage-by-stage stakeholder matrix

The same treatment for the second bespoke engine. Note how much thinner the early stages are: no SIA, no consent, no R&R scheme before declaration. That structural difference is why this warrants its own engine rather than a configuration row.



Figure 4. NH Act 1956 stage flow and stakeholder involvement, including the 3G(5)/3I arbitration branch and the parallel mutation track.

23.1 Stage 1 — Section 3A, preliminary notification

Stakeholder	Role in this stage	System action
Central Ministry (MoRTH)	Satisfied of public purpose; issues the 3A notification in the Official Gazette; approves the land requirement proposal	Creates the project; selects the NH Act regime; uploads the DPR and alignment; triggers 3A generation; starts the 21-day and one-year clocks
Requiring body (NHAI / NHIDCL / PWD)	Submits the land requirement, technical alignment and estimated cost	Uploads alignment geometry, village lists and survey numbers; specifies area required
State Government	Receives intimation; coordinates district readiness and CALA availability	Notification appears in the state inbox; assigns a monitoring officer
District Collector / CALA	Acknowledges receipt; prepares for the field survey; publishes local notices	Acknowledges 3A; initiates village notice-board publication; applies the land-use freeze flag
Landowner / public	Receives notification; may object within 21 days	Views and downloads the 3A notice on the public portal; files an objection, which creates the 3C proceeding
Finance	No direct role yet	Read-only view of projected compensation liability

23.2 Stage 2 — Section 3B, survey and earmarking

Stakeholder	Role in this stage	System action
District Collector / CALA	Authorises the survey; appoints the team; reviews the survey report	Assigns surveyors to parcels; approves survey completion; reviews boundary geometry
Field Surveyor	Conducts the ground survey; measures boundaries; enumerates structures, crops and trees; captures GPS	Offline-capable capture; parcel geometry; photographic evidence with hash and geotag; sync on connectivity
State Revenue Department	Provides revenue records, cadastral maps and ownership data	Records fetched through the DILRMP / Bhulekh adapter and cached against the parcel with provenance
Landowner / public	Present during survey; points out boundaries; acknowledges the measurement	Receives the survey schedule; views the survey record; files a discrepancy where the measurement is contested
Requiring body	Provides the technical alignment for verification	Views the GIS overlay of surveyed parcels against alignment; flags deviations

23.3 Stage 3 — Section 3C, hearing of objections

Stakeholder	Role in this stage	System action
Landowner / public	Files a written objection within 21 days; appears at the hearing	Objection filed with supporting documents; hearing schedule received; outcome tracked
District Collector / CALA	Schedules the hearing; hears the objector; passes an order allowing or disallowing; records minutes	Hearing scheduler; minutes upload; order generation and signature; stage status update. The order is final at this stage — which makes the completeness of the record the only protection against a later challenge
Legal Officer	Advises on the legal validity of objections; reviews draft orders	Reviews objection documents; flags legal risk; validates order language
State Government	Monitors hearing compliance	Views the hearing calendar; alerted where a hearing slips past the statutory window
Requiring body	Presents the project justification at the hearing	Uploads technical justification; views the outcome

23.4 Stage 4 — Section 3D, declaration of acquisition

Stakeholder	Role in this stage	System action
Central Ministry (MoRTH)	Approves and publishes the declaration; land vests absolutely in the Central Government	Hard gate on the one-year rule from 3A; declaration generation and approval; vesting recorded; mutation track opened
District Collector / CALA	Publishes locally; posts notices; initiates title vesting in the records	Confirms publication; sets land status to vested; triggers the parallel mutation workflow
State Revenue Department	Updates land records to reflect vesting	Receives the mutation event through the records adapter; RoR update tracked to completion
Landowner / public	Informed that the land has vested; can no longer transfer or encumber	Alert issued; declaration downloadable; status reflects vesting
Finance	Prepares for the compensation deposit	Views total award liability; confirms budget allocation
Legal Officer	Anticipates writ petitions; prepares the defence	Flags high-risk projects; uploads preventive legal opinions

23.5 Stage 5 — Section 3E, possession

Stakeholder	Role in this stage	System action
District Collector / CALA	Issues the 60-day surrender notice; takes possession, with police assistance where necessary	Possession notice generation; 60-day countdown; possession mode recorded as surrendered, forcible or disputed
Landowner / public	Surrenders possession within 60 days, or does not	Receives the notice; acknowledges surrender digitally; or files an urgent grievance
State Government / Police	Provides assistance where possession is not surrendered	Auto-alert on day 60 where surrender has not been recorded; law-and-order coordination task
Requiring body / PIA	Receives handover; commences construction	Possession status update; handover certificate download; right of way marked available
Field Surveyor	Verifies physical possession; documents site condition	Post-possession evidence upload; boundary integrity confirmation

23.6 Stage 6 — Section 3G, determination of compensation

Stakeholder	Role in this stage	System action
District Collector / CALA	Determines the amount payable under the Section 3G(7) parameters — market value, damage to standing crops and trees, severance damage, damage to other property, reasonable relocation expenses	Compensation calculator configured for the NH Act basis, including the post-2015 First Schedule application; award order with computation breakdown; signature
State Revenue Department	Provides land classification and circle rates	Classification and rate fetched and recorded with provenance
Finance	Ensures funds are available	Views liability; issues the deposit instruction
Landowner / public	Receives the determination; accepts or disputes	Views the breakdown; accepts, which moves to 3H; or disputes, which opens the 3G(5) arbitration
Legal Officer	Reviews for compliance; handles disputes	Review queue; arbitration case creation
Requiring body	Bears the cost; verifies reasonableness	Views total against budget; variance alert

23.7 Stage 7 — Section 3H, deposit and payment

Stakeholder	Role in this stage	System action
Finance / PFMS Controller	Deposits the amount with the competent authority; processes disbursement to entitled persons	Payment instruction with idempotency key; account verification; UTR capture; failure handling
District Collector / CALA	Notifies entitled persons to collect; holds undisbursed and disputed amounts	Per-owner payment status; unclaimed and deposited amount tracking; apportionment dispute handling
Landowner / public	Provides account details; receives payment; acknowledges	Account detail submission; payment status view; receipt download
Central Ministry / requiring body	Monitors fund utilisation	Fund flow view: allocated, deposited, disbursed, pending
Auditor	Tracks financial compliance	Append-only payment log; reconciliation evidence

23.8 Stage 8 — Section 3G(5) / 3I, arbitration

Stakeholder	Role in this stage	System action
Landowner / public	Applies for arbitration disputing the amount	Application with supporting valuation; case created and linked to parcel and award
Central Government	Appoints the arbitrator	Appointment order processed; arbitrator credentials recorded; scoped access token issued
Arbitrator	Conducts the arbitration; passes the award	Case dossier view; hearing record; award upload; enhanced amount computed
Legal Officer	Represents the government; defends the original determination	Defence document upload; case status tracking
Finance	Deposits the enhanced amount with interest	Interest computed from the correct base date at the applicable rate; supplementary payment
District Collector / CALA	Implements the arbitral award	Parcel and award state updated; enhanced disbursement tracked

23.9 Parallel track — mutation

Stakeholder	Role in this stage	System action
State Revenue Department	Mutates the title to record the vesting in the Central Government	Mutation request processed through the records adapter; RoR update confirmed
District Collector / CALA	Submits the mutation request after possession	Auto-triggered on possession recording; 30-day target tracked
Landowner / public	No direct role — title has vested	Informed through the portal that mutation is complete
Requiring body / PIA	Receives mutation-certified land	Mutation certificate download; project record updated; acquisition closed

24. Cross-cutting roles

Three processes run alongside every stage of every regime and cannot be modelled as stages.

24.1 Grievance management (all stages)

Stakeholder	Role
Landowner / public	Files a grievance at any stage; tracks resolution against a ticket number
District Collector	Adjudicates or assigns to the relevant officer; accountable for the service level
State Government	Monitors resolution timelines; escalation where unresolved beyond the configured period
Central Ministry	Views the national grievance pattern; intervenes on systemic issues
NGO / civil society	Monitors resolution rates; files RTI applications

Ticket lifecycle: filed → acknowledged → assigned → under enquiry or hearing → resolved → closed, with an escalation branch on breach of the service level. A grievance is **not** the same object as a statutory objection under Section 15 or 3C — the objection has legal consequences and a statutory deadline, the grievance does not. Conflating them in one table is a common and consequential modelling error.

24.2 Litigation and stay management (all stages)

Stakeholder	Role
Legal Officer	Uploads stay orders; requests the timeline exclusion; tracks case status
District Collector	Implements the stay; workflow pauses; resumes on vacation
Landowner / public	Informed of the stay's effect on their case
Court / arbitrator	Issues stay and vacation orders — external to the system, consumed through the case-status integration
System (automatic)	Pauses countdown timers on the affected parcels only; recalculates deadlines on vacation; records the exclusion period in the audit ledger

Stay scope is a precision requirement

A stay rarely covers a whole project. It usually covers named parcels, sometimes a village, occasionally a stage. A system that pauses the entire project's clocks on any stay upload will produce deadline data that is wrong in the permissive direction — the most dangerous kind. The exclusion must be scoped to the parcels the order actually names, and the scope must be recorded as a deliberate, attributable decision by the Legal Officer.

24.3 Monitoring and audit (continuous)

Stakeholder	Role
Policy maker	Views analytics; compares state performance; formulates policy
CAG / auditor	Queries the audit trail; verifies compliance and fund utilisation; checks document integrity
Parliamentary Committee	Receives periodic MIS output; reviews sectoral progress
National Monitoring Committee	Statutory oversight of R&R implementation under LARR Sections 48–50
Central Ministry	Publishes reports; responds to audit observations

PART IV

Access Control and Governance

Scope Item 11 asks for role-based access control, audit trails, secure authentication, encryption and compliance with government cybersecurity standards. In a system that generates legally operative documents and moves public money, these are not a hardening pass at the end — they are the reason anyone would trust the output. This part specifies the permission model, the stage-gate rules that make the workflow enforceable, the audit ledger that makes it auditable, and the statutory compliance surface.

In this part

- Role definitions and how jurisdiction scoping actually works
- The granular permission matrix — verbs, not read/write
- Stage-gate enforcement: who advances, who returns, who is notified
- Audit trail schema and evidence chain of custody
- STQC, CERT-In, DPDP, IT Act, GIGW and WCAG obligations

25. Role definitions and jurisdiction scoping

Most government RBAC designs fail not on the permission list but on scope. Two District Collectors have identical permissions and must see entirely different data. Getting that right is more important than getting the verb list right.

25.1 The three-dimensional model

Every access decision in the system resolves three independent questions. Collapsing them into one role string is the mistake that forces a rewrite.

Dimension	Question it answers	Values
Role	What kind of action may this user take?	central_admin, state_admin, district_admin, sub_division_admin, requiring_body, field_user, rr_admin, finance_admin, legal_user, policy_audit, public, judicial_guest, system_admin
Jurisdiction	Over which subset of the data?	A scope object, not a string: {level: national state district sub_division project parcel_set case, ids: [...]}. Resolved into a predicate applied to every query.
Regime and stage context	Under which statute and at which stage?	Some permissions exist only for some regimes (there is no consent record to approve under the NH Act) and only at some stages (compensation cannot be computed before declaration).

Assignment and scoping

-- Access is the conjunction of all three, evaluated at query time.

```
CREATE TABLE user_assignment (  
  user_id      UUID      NOT NULL REFERENCES app_user,  
  role_code    TEXT      NOT NULL REFERENCES role,  
  scope_level  TEXT      NOT NULL,    -- national|state|district|sub_division|project|case  
  scope_ids    TEXT[]    NOT NULL,    -- LGD codes, project ids, or case ids  
  valid_from   DATE      NOT NULL,  
  valid_to     DATE,      -- NULL = current; officers transfer, roles expire  
  appointed_by UUID      REFERENCES app_user,  
  appointment_ref TEXT,    -- the CALA appointment order reference  
  PRIMARY KEY (user_id, role_code, scope_level, valid_from)  
);
```

-- Enforced in the database, not only in the application:

```
ALTER TABLE parcel ENABLE ROW LEVEL SECURITY;  
CREATE POLICY parcel_scope ON parcel USING (  
  district_code = ANY (current_setting('app.scope_districts')::text[])  
  OR current_setting('app.scope_level') IN ('national', 'state_matched')  
);
```

Why role assignment must be time-bounded

District Collectors transfer. CALAs are appointed by order for specific projects and replaced. An award signed by an officer who had ceased to hold the office is a defect a court can act on.

The consequence: role assignments carry `valid_from` and `valid_to`, the appointment order reference is stored, and **the audit ledger records the assignment that was in force at the moment of the action** — not the assignment that is in force when someone later reads the log. Reconstructing “who was authorised on that date” after the fact is impossible unless it was captured at the time.

25.2 Separation of duties

Four separations are load-bearing and should be stated explicitly, because each prevents a specific and realistic abuse:

- **The officer who computes the award cannot disburse it.** District admin computes and declares; finance admin disburses. One role doing both is the classic compensation fraud pattern.
- **The requiring body that pays cannot advance the workflow.** It funds and receives; it does not decide that an objection has been disposed of. This is why role 4 is read-only on the parcel workflow.
- **Nobody can edit the audit log.** Not the system administrator, not the central ministry. The auditor role can read it and export it; no role can write to it other than the system itself, append-only.

- **Policy and audit roles cannot see personal data.** Aggregate analytics do not require identified individuals, and giving national-level read access to identified landowner records creates a data-protection exposure with no operational benefit.

26. Granular permission matrix

“Read” and “write” are too coarse to build from. The permission vocabulary below is verb-and-object, and each cell states either an unqualified grant or the scope qualifier that limits it.

Legend: **Y** = permitted within the user's assigned jurisdiction; a word such as *own*, *state*, *district*, *assigned*, *case*, *agg* (aggregate only) = permitted but narrowed to that scope; **—** = not permitted.

Permission	Centr.	State	Distr.	Req. Body	Field	R&R	Fin.	Legal	Pol./ Audit	Publ.	Judic.
project.create	Y	—	—	—	—	—	—	—	—	—	—
project.read	Y	state	district	own	—	linked	fin	case	agg	—	case
project.sanction	Y	—	—	—	—	—	—	—	—	—	—
proposal.submit	—	—	—	Y	—	—	—	—	—	—	—
proposal.scrutinise	Y	state	district	—	—	—	—	—	—	—	—
regime.config.edit	Y	—	—	—	—	—	—	—	—	—	—
regime.config.read	Y	Y	Y	Y	—	Y	Y	Y	Y	—	—
parcel.create	—	—	Y	—	—	—	—	—	—	—	—
parcel.read	Y	state	district	own	assigned	linked	—	case	agg	own	case
parcel.edit	—	—	Y	—	survey	—	—	—	—	—	—
parcel.geometry.edit	—	—	Y	align	Y	—	—	—	—	—	—
survey.create	—	—	Y	—	Y	—	—	—	—	—	—
survey.approve	—	—	Y	—	—	—	—	—	—	—	—
sia.commission	Y	Y	—	—	—	—	—	—	—	—	—
sia.upload	—	—	Y	—	—	—	—	—	—	—	—
sia.appraise	—	—	—	—	—	—	—	—	—	—	—
consent.record	—	—	Y	—	Y	—	—	—	—	—	—
consent.read	Y	state	district	own	—	linked	—	case	agg	own	—
notification.generate	Y	Y	Y	—	—	—	—	—	—	—	—
objection.file	—	—	—	—	—	—	—	—	—	Y	—
objection.dispose	—	—	Y	—	—	—	—	—	—	—	—
hearing.schedule	—	—	Y	—	—	—	—	—	—	—	—
declaration.issue	Y	Y	—	—	—	—	—	—	—	—	—
workflow.advance	—	—	Y	—	—	—	—	—	—	—	—
workflow.return	Y	state	Y	—	—	—	—	—	—	—	—
compensation.calculate	—	—	Y	—	—	—	—	—	—	—	—
compensation.estimate	Y	Y	Y	Y	—	Y	Y	—	—	Y	—
award.declare	—	—	Y	—	—	—	—	—	—	—	—
award.enhance	—	—	—	—	—	—	—	—	—	—	Y
rr.scheme.author	—	—	—	—	—	Y	—	—	—	—	—
rr.award.approve	—	—	Y	—	—	—	—	—	—	—	—
rr.entitlement.edit	—	—	—	—	—	Y	—	—	—	—	—
rr.site.progress.edit	—	state	—	—	—	Y	—	—	—	—	—
payment.instruct	—	—	—	—	—	—	Y	—	—	—	—
payment.read	Y	state	district	own	—	—	Y	—	agg	own	—
possession.record	—	—	Y	—	—	—	—	—	—	—	—
mutation.trigger	—	—	Y	—	—	—	—	—	—	—	—
grievance.file	—	—	—	—	—	—	—	—	—	Y	—

Permission	Centr.	State	Distr.	Req. Body	Field	R&R	Fin.	Legal	Pol./ Audit	Publ.	Judic.
grievance.resolve	—	esc	Y	—	—	Y	—	—	—	—	—
litigation.create	—	—	Y	—	—	—	—	Y	—	—	—
litigation.stay.upload	—	—	—	—	—	—	—	Y	—	—	—
timeline.exclusion.apply	—	—	—	—	—	—	—	Y	—	—	—
document.upload	—	Y	Y	Y	Y	Y	Y	Y	—	—	Y
document.sign	Y	Y	Y	—	—	Y	Y	—	—	—	Y
document.read	Y	state	district	own	assigned	linked	fin	case	agg	own	case
report.export	Y	state	district	own	—	linked	fin	case	Y	—	—
opendata.read	Y	Y	Y	Y	—	Y	Y	Y	Y	Y	—
audit.log.read	—	—	—	—	—	—	—	—	Y	—	—
user.manage	Y	state	district	—	—	—	—	—	—	—	—
urgency.invoke	Y	Y	—	—	—	—	—	—	—	—	—

Cell value	Meaning
own	Limited to projects where the user's organisation is the requiring body
state / district	Limited to the user's assigned state or district
assigned	Limited to parcels explicitly assigned to that field user
linked	Limited to projects with an active R&R obligation
case	Limited to parcels and documents linked to a case the user is assigned to
fin	Financial fields only; no workflow or personal detail beyond payee identification
agg	Aggregate figures only; no record-level personal data
align	Alignment and corridor geometry only; not parcel boundaries
esc	Only where the grievance has escalated to that level

Two cells worth explaining

sia.appraise is denied to every role. That is deliberate. The Expert Group is an external empanelled body, not a system role — it receives a scoped, time-limited token against a specific SIA, like the judicial guest. Granting appraisal authority to any standing role would misrepresent the statutory position.

urgency.invoke exists as a distinct permission rather than being implied by central or state admin. LARR Section 40 is the most consequential and most abusable provision in the Act. It gets its own permission, its own justification field, and a mandatory audit flag that surfaces on the oversight dashboard.

27. Stage-gate enforcement

The rule that makes the workflow more than a status field: **a stage cannot advance unless the designated actor for that transition performs the approval action, and every gate condition is satisfied.** Not a UI convention — a database-enforced invariant.

27.1 Gate matrix

Transition	Who may advance	Who may return or reject	Gate conditions checked	Notified on success
Stage 0 → regime start	Central Ministry (central regimes) or State Government (LARR)	Scrutinising authority, back to the requiring body	Proposal complete; regime valid for the requiring body class; no unresolved spatial conflict; clearance gates recorded; requiring body deposit committed	State, District, requiring body
SIA → Section 11	Appropriate Government, on the Expert Group appraisal	Central Ministry; Expert Group may recommend abandonment	SIA report and SIMP uploaded; public hearing recorded; Expert Group appraisal on file; government decision recorded; SIA within the 12-month validity	Public, District Collector, R&R Administrator
Section 11 → Section 19	Appropriate Government	Collector's recommendation may return it	60-day objection window closed; all objections disposed with orders ; R&R scheme approved; consent threshold met where applicable ; within 12 months of Section 11; requiring body deposit received	All stakeholders; landowners by SMS
Section 19 → award	District Collector	—	Declaration published in all required channels; notice to persons interested served; enquiry completed; survey and valuation records approved	R&R Administrator, Finance, requiring body
Award → possession (LARR)	District Collector	—	Compensation paid or tendered on every award line ; R&R award passed for every family in the locked registry; monetary R&R components paid or tendered; no subsisting stay on the parcel	Requiring body, R&R Administrator, landowners
3A → 3B	District Collector / CALA	Central Ministry, on irregularity	3A published in Gazette and two newspapers including one vernacular; land-use freeze applied; CALA appointed and in force	Landowners by SMS, requiring body
3B → 3C	District Collector / CALA	—	Survey complete and approved on all listed parcels; geometry validated; evidence uploaded	Landowners, State Government
3C → 3D	District Collector / CALA (recommendation); Central Ministry (declaration)	—	21-day window closed; every objection heard with a recorded order; minutes on file	All stakeholders; objection outcome to each objector
3D → 3E	Central Ministry (Gazette)	—	Within one year of 3A — hard block, no override; vesting recorded; mutation track opened	All stakeholders, Finance, State Revenue
3E → possession	District Collector / CALA	—	60-day surrender notice served and expired; amount deposited under 3H; no subsisting stay	Requiring body, Police where forcible
3G → 3H	District Collector / CALA on determination	Landowner dispute opens 3G(5) arbitration in parallel	Determination signed; apportionment resolved or amount deposited with the authority	Finance, requiring body, landowners
Any stage → closed	District Collector	—	Possession recorded; mutation confirmed; all payments settled or deposited; no open litigation blocking closure; R&R obligations either complete or explicitly carried forward with an owner	Requiring body, Central Ministry, R&R Administrator

27.2 Overrides and how to make them safe

Some gates can be extended by law; some cannot be overridden at all. The distinction must be in configuration, not in a developer's judgement.

Gate	Overridable?	Mechanism
Twelve-month Section 11 to Section 19 window	Yes, by law	The Act permits extension by the appropriate government with recorded reasons. The system requires the reasons as a mandatory text field, the extension is signed, and it appears on the oversight dashboard.
Twelve-month Section 19 to award window	Yes, by law	Same mechanism, with the additional consequence that where extended, compensation is recomputed — which the engine must apply automatically rather than leave to the officer.
One-year 3A to 3D window	No	The NH Act provides no extension. A hard block with no override path. Where it lapses, the only route is a fresh 3A.

Gate	Overridable?	Mechanism
Consent threshold	No	Statutory. Blocked absolutely. Where a state amendment exempts the project category, the exemption is resolved from the rule table and the gate is not raised in the first place — it is never overridden after being raised.
Payment before possession	No	The core protective provision of LARR. Not overridable outside the Section 40 urgency path, which is itself a separate permission with its own audit flag.
Clearance gates	By exception, recorded	Where a clearance is genuinely not required for that project category, the not_required state must carry the basis for that conclusion — not simply be selected.

The one design rule for overrides

Every override is **a permission, a mandatory reason, a signature, and a dashboard entry**. An override that leaves no trace is indistinguishable from a bug, and an override that any operator can apply silently destroys the platform's value as an audit instrument. Making the exception visible is more useful than preventing it — because the exceptions are lawful and the invisibility is what causes harm.

28. Audit trail and evidence integrity

The audit trail is the system's most valuable output. It is what lets a court reconstruct whether a hearing was actually held, what lets the CAG verify a disbursement, and what lets a Collector demonstrate that the process was followed. It has to be designed for that purpose rather than emerging from application logging.

28.1 Schema

Append-only, hash-chained audit ledger

```
CREATE TABLE audit_log (
  log_id          BIGSERIAL PRIMARY KEY,
  occurred_at     TIMESTAMPTZ NOT NULL DEFAULT now(),
  actor_user_id   UUID        NOT NULL,
  actor_role      TEXT         NOT NULL,    -- role IN FORCE at the time of the action
  actor_scope     JSONB        NOT NULL,    -- jurisdiction in force at the time
  actor_ip        INET,
  action          TEXT         NOT NULL,    -- the permission verb exercised
  entity_type     TEXT         NOT NULL,
  entity_id       UUID        NOT NULL,
  old_value       JSONB,        -- NULL on create
  new_value       JSONB,        -- NULL on delete
  reason          TEXT,        -- mandatory for overrides and returns
  document_refs   UUID[],
  signature       BYTEA,        -- DSC/eSign where the action is a signed act
  prev_hash       BYTEA        NOT NULL,
  row_hash        BYTEA        NOT NULL    -- H(prev_hash || canonical(this row))
);

REVOKE UPDATE, DELETE ON audit_log FROM ALL; -- append-only, enforced by grant
CREATE INDEX ON audit_log (entity_type, entity_id, occurred_at DESC);
CREATE INDEX ON audit_log (actor_user_id, occurred_at DESC);
```

The hash chain is what converts a log into evidence. Each row commits to its predecessor, so any retrospective alteration or deletion breaks the chain at a detectable point. Periodically publishing the chain head — to a separate append-only store, or simply recording it in a signed daily report — gives an external verifier a fixed point to check against, which delivers the tamper-evidence benefit that is usually the stated reason for reaching for a blockchain, at a fraction of the operational cost.

28.2 What must be logged

Category	Events
Workflow	Every stage advance, return, block and override; every gate evaluation that failed and why
Money	Award computation with all inputs and the formula version; payment instruction; disbursement outcome; interest accrual runs; deposit and unclaimed handling
Documents	Upload, version supersession, signature application and verification, publication, download by whom
Records	Every external fetch — which adapter, which endpoint, what was returned, when — because a compensation figure derived from a circle rate must be traceable to the moment that rate was retrieved
Access	Authentication, authorisation failure, scope resolution, role assignment change, guest token issue and expiry
Personal data	Every read of identified landowner data, by whom and under which permission — a DPDP requirement, not merely good practice
Configuration	Every change to a regime definition, rule version or template, with the effective-date range affected

28.3 Chain of custody for field evidence

Photographs taken by a surveyor may end up as exhibits. Treating them as ordinary file uploads throws away their evidentiary value.

- **Hash at capture**, on the device, before any upload — so the hash covers the original bytes rather than whatever the server received.
- **Bind the metadata**: capture timestamp, GPS fix with reported accuracy, device identifier, capturing user, and the parcel it was captured against — recorded together and included in the hashed payload.
- **Write-once storage**. Evidence objects are never overwritten; a correction is a new object superseding the old, with both retained.
- **Record the custody events**: captured by, synced at, verified by, included in bundle. “Who could have altered this between capture and hearing” must have an answer.
- **Bundle export** that assembles every evidence object for a parcel or case with its hashes and custody record in one verifiable package — this is what the Legal Officer actually needs and what no existing system provides.

A note on GPS accuracy honesty

A consumer phone GPS fix is accurate to several metres in good conditions and far worse under tree cover. That is adequate for locating a parcel and inadequate for a legal boundary. Kerala's answer was a 28-station CORS network delivering centimetre accuracy; ours should be to **record the reported accuracy with every fix** and to state plainly that phone-captured geometry is indicative, with survey-grade instruments required where a boundary is contested. Claiming otherwise is the kind of overreach a technical judge will test.

29. Security, privacy and statutory compliance

The problem statement asks for compliance with Government of India cybersecurity standards. That phrase covers several distinct obligation sets, and a submission that names them specifically reads very differently from one that says “encryption and RBAC.”

29.1 The obligation surface

Obligation	Source	What it requires of this system
Data protection	Digital Personal Data Protection Act 2023, with Rules notified 14 November 2025 and enforcement staggered across 14 November 2025, 14 November 2026 and 14 May 2027	The system stores identifiers, bank details and family composition — squarely personal data. Requires notice and purpose limitation at collection, a lawful basis, data minimisation, retention limits, breach notification, and a grievance and correction route for data principals. Full enforcement lands while this project would still be live , so it must be designed in, not retrofitted.
Breach reporting	CERT-In directions	Reportable cyber incidents notified within six hours of noticing. Requires detection capability and a named responder, not just a policy document — six hours is not achievable by escalation through a committee.
Electronic signatures	Information Technology Act 2000, Section 3A and the eSign framework	Approval at stage gates must use a legally recognised electronic signature — DSC or eSign — for the signed act to have the effect the statute requires. Signature verification and long-term validity are part of the document store, not the auth layer.
Accessibility	Guidelines for Indian Government Websites (GIGW); WCAG 2.1 Level AA	Mandatory for a citizen-facing government portal. ARIA labelling on every form control, full keyboard operability of the workflow, contrast compliance on dashboards and map legends, text scaling to 200% without loss of function, and a screen-reader-usable status tracker.
Security certification	STQC assessment for government applications	Application security testing, secure configuration baseline, audit logging adequacy, and documented cryptographic controls. Worth designing to the checklist even at prototype stage, because retrofitting it is expensive.
Data localization	Government cloud policy; MeghRaj	Primary hosting in India — and, critically, backups and disaster recovery replicas too . This is the part most commonly overlooked in a compliance claim.
Right to information	RTI Act 2005	Much of what the system holds is disclosable. Building the public portal and open aggregate data properly reduces RTI load rather than adding to it — a genuine administrative benefit worth stating.
Interoperability	India Stack and government API guidelines	Published API contracts, versioning discipline, and standard identifiers (ULPIN, LGD codes) so other systems can consume this one.

29.2 Cryptographic and data handling controls

Control	Specification
In transit	TLS 1.3, HSTS, certificate pinning on the field client where feasible
At rest	AES-256 full-volume encryption on database and object storage
Field-level	Separate encryption for identifiers and bank account details, with keys held outside the database. A database backup that leaks must not leak these fields.
Identifier handling	Store a hash or a reference token, never the identifier in the clear; display masked; log every read of the unmasked value
Key management	A dedicated secret store for signing keys and adapter credentials; rotation schedule; no credential in source control or environment variables in a shared namespace
Money arithmetic	Fixed-point decimal throughout. Floating point on a crore-magnitude award produces reconciliation differences that an auditor will find and cannot be explained away.
Session	Short-lived access tokens with refresh; step-up authentication for signing actions; device binding on the field client
Document integrity	Content hash on every stored object; signature verification at each gate; version lineage with supersession rather than overwriting
Network posture	Public portal in a segregated zone; internal services on private subnets; egress to external government APIs through a controlled gateway with per-adapter credentials
Retention	Audit logs seven years; legal documents ten years or longer per records policy; personal data beyond that period minimised or removed under the DPDP retention obligation — which requires a defined retention schedule per data category, not a blanket policy

The consent and grievance obligation is a feature, not a policy page

DPDP requires that a data principal can find out what is held about them, have it corrected, and complain if it is misused. For this system the data principals are landowners and affected families — the same people the public portal already serves.

That is a fortunate alignment: the portal that provides transparency also discharges the data-protection obligation. Building it well means the correction route and the grievance route are the same mechanism, which is both cheaper and more likely to actually be used than a separate compliance workflow nobody visits.

Implementation Architecture

The engineering question raised at the close of the working sessions was concrete: “I don't know how to implement all the workflows — 7 or 8 of them — into a platform.” This part is the direct answer. It is not ten workflow engines. It is two bespoke state machines, one configuration table that the remaining regimes instantiate, and the supporting services — data model, compensation and timeline engines, GIS, integration, alerting, reporting, documents, field capture and analytics — that every regime shares.

In this part

- The core strategy: why two engines and a configuration table, not ten codebases
- The regime configuration schema that makes a new act an admin task, not a deployment
- System architecture: services, data flow, and where Rust actually earns its scalability claim
- The data model, including the ownership, process-type and Section 24 cases the earlier design missed
- Compensation, timeline, GIS, integration, alerting, reporting, document, field-capture and AI modules
- Non-functional targets and the migration path from Bhoomi Rashi and the state systems

30. The core strategy: two engines and a configuration table

Part II documented at least a dozen active acquisition regimes. Building a bespoke workflow engine for each is not a 12-times-harder version of the same problem — it is a different, much larger problem, and one the team's timeline cannot absorb. The answer is not to reduce ambition. It is to notice that the regimes differ in **data**, not in **shape**.

30.1 The shared shape

Read across every regime in Part II and a common skeleton appears: an intention notification, an objection window with a hearing, a declaration that commits the land, a determination of money (or, for right-of-user regimes, a determination of a smaller figure), a payment or deposit, a transfer of possession (or, for right-of-user regimes, an entry right with no transfer), and a dispute route. The differences are entirely in the *parameters*: which authority acts at which step, how many days the objection window runs, what the compensation formula computes, whether R&R attaches, and what triggers the next step.

Regime	Notify	Object	Declare	Determine money	Transfer	Dispute
LARR 2013	s.11	60d	s.19	First Sched. formula	Possession, s.38	Authority, s.64
NH Act	3A	21d	3D	3G(7) params	Vesting + possession, 3E	Arbitrator, 3G(5)
Railways	20A	30d	20E	20G params	Vesting + possession, 20I	Gazetted officer
Coal Bearing	4(1)/7(1)	—	9(1)	Act provisions	Vesting, then PSU transfer	Tribunal
Pipelines (RoU)	s.3	21d	s.6	10% of MV	No transfer — entry only	Competent authority

30.2 Two engines, thirteen-plus configuration rows

Build	What it covers	Why it is bespoke rather than configured
Engine A — LARR Act 2013	The general regime: all central and most state civil, urban and rural infrastructure projects not covered by a sector act	The only regime with a mandatory pre-notification assessment stage, a consent threshold that gates the declaration, and a statutory R&R award running in parallel with the compensation award. These are structural differences in the state machine, not parameter differences — SIA and consent are extra states and extra gate conditions that no other regime has.
Engine B — National Highways Act 1956	MoRTH, NHAI, NHIDCL and state PWD highway acquisition	The regime with a live national comparator (Bhoomi Rashi) that every design decision can be checked against, and structurally the leanest regime — no SIA, no consent, a shorter objection window. Building it bespoke, rather than folding it into the generic track, keeps the reference implementation faithful to the system every judge and every integration partner already knows.
Sector Acquisition Track — one configurable engine, many rows	Railways, Coal Bearing Areas, Metro Railways, Atomic Energy, Damodar Valley Corporation, Ancient Monuments, Indian Tramways, Land Acquisition (Mines), Resettlement of Displaced Persons, and any future sector act	Each of these differs from the NH Act pattern only in parameters: authority names, day counts, section labels on generated documents, and which post-2015 compensation basis applies. None needs a materially different state shape. A single parameterised state machine, driven by a configuration row per act, covers all of them — see 30.3.
Right-of-User Track — one configurable engine	Petroleum & Minerals Pipelines Act, Electricity Act wayleave, Telegraph Act wayleave	Shares the notify-object-declare shape but never transfers title and never triggers possession or R&R. One additional track type with a shorter terminal state, not a third bespoke engine.
Non-Acquisition Track — one configurable engine	Land pooling and TDR, government allotment or lease, consent purchase	A genuinely different shape — contribution or negotiation rather than compulsory taking — but simple enough, and used across few enough regimes in this platform's initial scope, to configure rather than build bespoke.
Legacy / Transitional Track	Section 24 cases under the repealed 1894 Act	Deliberately deferred to Phase 2 (Section 49). The data model hook — the legacy_1894_transitional process type from Section 14.4 — is designed now so the extension is additive rather than a migration.

What this claim actually licenses saying to judges

“New legal regime? An administrator adds a configuration row in the admin console — no code deployment, no schema change.” This is true for any regime that fits the Sector Acquisition Track shape, which is every Fourth Schedule act beyond the NH Act, and it is the load-bearing claim of the whole submission. It is **not** true for a regime with a genuinely different shape — that requires a third bespoke engine, and overclaiming this is the fastest way to lose credibility under questioning. State the boundary of the claim exactly as it is stated here.

30.3 What “configuration, not code” actually means

For the Sector Acquisition Track to be genuinely configurable rather than configurable in name only, four things must be true of the engine, and all four are concrete engineering commitments rather than aspirations.

- 1 **The stage sequence is data, read at runtime** — an ordered list of stage definitions per regime, each with its section reference, its actor role, its document template, and its gate conditions, none of it in application code.
- 2 **Gate conditions are composed from a fixed library of predicates** — `objection_window_closed`, `deposit_confirmed`, `consent_threshold_met`, `clearance_obtained(kind)` — combined per regime rather than written bespoke per regime. A new regime selects which predicates apply; it does not need a new predicate written for it unless it genuinely introduces a new legal concept.
- 3 **Document templates are parameterised, not per-regime files** — one notification template with fields for the regime name, section reference, day count and issuing authority, rendered per regime and per state language.
- 4 **The compensation basis is a formula selector plus effective-dated parameters**, not a per-regime code branch — see Section 35.

Where any of these four is not true — where a “configurable” regime secretly needs an application-code change — the honest statement is that the regime needed a new predicate or a new document field, which is still a small change relative to a new engine, but is not the zero-code claim above. Track this distinction in the regime onboarding checklist so the claim made to a judge matches what actually happened for each regime.

31. Regime configuration schema

This is the schema that operationalises Section 30. It is the single most important table in the data model, because every other stage, gate and document in the system is derived from a row in it.

Core configuration tables

```
CREATE TABLE regime (
  regime_code      TEXT PRIMARY KEY,      -- 'LARR_2013','NH_1956','RAIL_1989', ...
  display_name     TEXT NOT NULL,
  statute_ref      TEXT NOT NULL,         -- 'RFCTLARR Act 2013' etc.
  process_type     TEXT NOT NULL,         -- see Section 14.4 enumeration
  engine           TEXT NOT NULL,         -- 'larr' | 'nh' | 'sector' | 'row' | 'non_acq'
  requires_sia     BOOLEAN NOT NULL DEFAULT FALSE,
  consent_rule     JSONB,                 -- {threshold: 0.80, denominator: 'affected_family'}
  objection_days   INT NOT NULL,
  declare_deadline_days INT,              -- NULL = no statutory outer limit
  award_deadline_days INT,
  possession_gate  TEXT NOT NULL,         -- 'deposit' | 'full_payment' | 'pct_80' | 'none'
  rr_applies       BOOLEAN NOT NULL DEFAULT FALSE,
  compensation_formula TEXT NOT NULL,     -- selector, see Section 35
  interest_schedule TEXT,                 -- selector into the rate table
  dispute_route    TEXT NOT NULL,         -- 'larr_authority' | 'arbitrator_central' |
                                          -- 'gazetted_officer' | 'tribunal' | 'court'

  effective_from   DATE NOT NULL,
  effective_to     DATE,                  -- e.g. pre/post the 2015 s.105(3) order
  superseded_by    TEXT REFERENCES regime(regime_code)
);

CREATE TABLE regime_stage (
  regime_code      TEXT REFERENCES regime,
  stage_order      INT,
  stage_code       TEXT,                  -- 'SIA','S11','S19','AWARD','RR','POSSESSION','REFERENCE'
  section_ref      TEXT,                  -- for the generated document
  actor_role       TEXT,
  document_template_id UUID,
  gate_predicates TEXT[],                 -- composed from the fixed predicate library
  PRIMARY KEY (regime_code, stage_order)
);
```

31.1 Worked example: onboarding the Coal Bearing Areas Act

To make the claim in 30.3 concrete, here is what onboarding an eleventh regime actually involves once Engines A and B and the Sector Acquisition Track exist.

Step	Action	New code required?
1	Insert a regime row for COAL_1957: engine sector, no SIA, no consent, objection days not applicable (uses a validity-window model instead — see below), possession gate vesting_then_transfer, compensation formula selector coal_act_plus_2015_order	No
2	Insert four regime_stage rows: prospecting (s.4(1)), intention (s.7(1)), declaration (s.9(1)), transfer to PSU (s.11)	No
3	Author the four document templates with Coal Act section references and the Coal Controller as issuing authority	No — template authoring, not code
4	The prospecting stage's multi-year validity window does not exist as a predicate in the library yet	Yes — one new predicate , validity_window_not_expired(years), generically useful for any future regime with the same pattern
5	Everything downstream — the workflow inbox, the document generator, the audit ledger, the dashboard — consumes the new regime automatically because they all read from regime and regime_stage	No

One new predicate for one genuinely new legal concept (a multi-year prospecting validity window), reusable for the next regime that needs it. That is what the “configuration, not code” claim looks like when it is kept honest.

31.2 Effective-dated rules solve two different problems at once

Legislative change and litigated interpretation both need a date range

Scope Item 12 explicitly requires the platform to accommodate future policy and legislative change. Every rule table above carries `effective_from` and `effective_to` for exactly this reason — not as a compliance checkbox, but because it is the only way two facts established in Part II can both be represented correctly at once: the 2015 order changed Fourth Schedule compensation, and the *Indore Development Authority* ruling changed how Section 24 lapse is read, without either change silently rewriting the history of a project that started before it. State amendments compound this: `sia_required` and `consent_required` resolve as a function of (state, project_category, effective_date) against this table, precisely because Tamil Nadu, Gujarat and others have exempted categories at different times. A project keeps the rule that was in force when it started unless the law itself says otherwise — which is what a court would actually ask.

32. System architecture

32.1 Component view

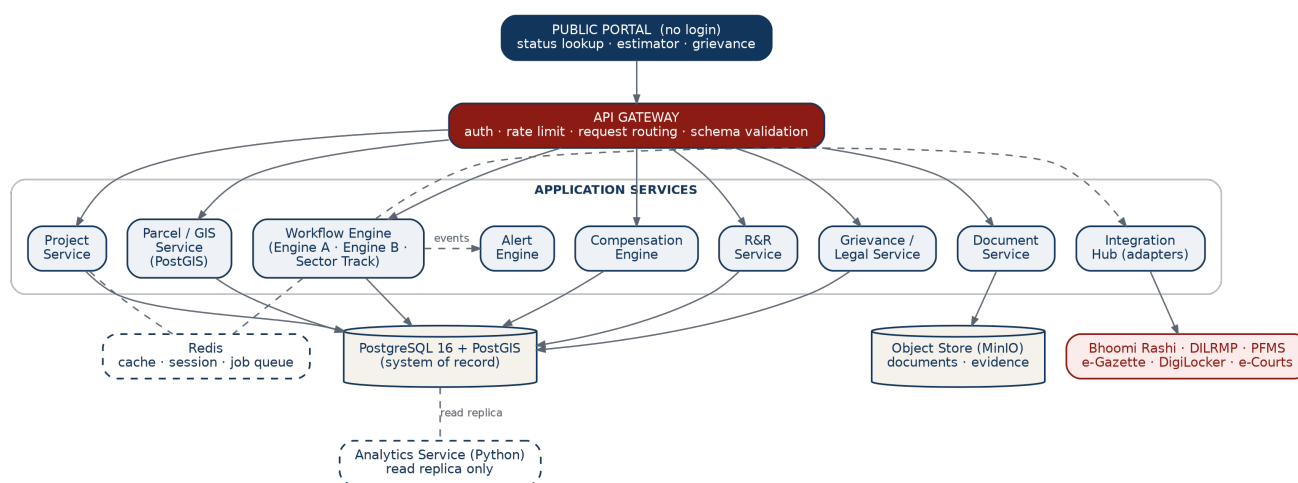


Figure 1. Service topology — the API Gateway fronts one workflow engine parameterised across Engine A, Engine B and the Sector Acquisition Track (Section 30), with the Analytics Service isolated behind a read replica so predictive queries never contend with an officer advancing a stage.

32.2 Why services are split this way

Boundary	Reasoning
Workflow Engine is one service, parameterised by regime.engine	Keeps the state-machine logic in one place so Engine A, Engine B and the Sector Acquisition Track share the same execution, audit and gate-evaluation code, differing only in the configuration they read. Splitting these into separate services would duplicate the gate-evaluation and audit-writing logic three times.
Parcel / GIS is separate from Workflow	Spatial queries (conflict detection, constraint intersection, corridor clipping) have different scaling characteristics from workflow state transitions and are called by several other services (proposal scrutiny, compensation valuation, field capture) independently of workflow state.
Compensation Engine is separate from Workflow	The formula selector and interest accrual are pure computation over a documented input set, called both synchronously (an officer requests a calculation) and as a scheduled job (nightly interest accrual across every open award). Isolating it makes the calculation independently testable against the worked examples in Section 35 — which matters given how many compensation figures in the source research were simply wrong.
Integration Hub is one service with an adapter per external system	Each external system fails independently, authenticates differently, and may or may not have a live sandbox. Isolating failure and retry logic per adapter, behind one hub interface, is what makes Section 38's mock-versus-live distinction operationally real rather than a documentation note.
Analytics Service reads a replica, never the primary	Predictive queries and dashboard aggregation must never contend with, or be blocked by, an officer trying to advance a workflow stage. This single decision is most of what makes “real-time” in the product title actually true under load.

32.3 The Rust decision, argued honestly

React for the frontend needs little defence — mature GIS map-library support through MapLibre wrappers, and mature internationalisation tooling for the vernacular requirement. Rust for the backend is a more debatable choice and deserves the same scrutiny here that it would get from a judge.

Consideration	Assessment
Where scalability is a genuinely true argument	The Integration Hub , fanning concurrent requests out to many slow or unreliable government APIs without blocking, and the audit-log writer , which must be fast, correct and never lose a write under load. Both benefit concretely from Tokio's async model and the absence of GC pauses.
Where it is not the honest argument	A few thousand government officials authenticating and filling forms is not, by itself, an extreme-scale problem, and a well-built Node.js or Go service handles it without difficulty. Framing the whole system's technology choice around raw concurrent-user scale overstates the case.
The framing that survives scrutiny	Correctness and resource efficiency on constrained government cloud infrastructure — compile-time-checked SQL through SQLx catching a wrong compensation-field type before it reaches production, and predictable memory behaviour on modestly provisioned MeghRaj instances — rather than “scale.”
The real cost, stated plainly	This is a document-generation-heavy, business-rule-heavy CRUD system. Django or NestJS give substantial boilerplate for free that Rust does not. If no one on the team has shipped in Rust before, budget real ramp-up time in the first month, not an afterthought.
What is deliberately not forced into Rust	The predictive-analytics scope (Section 43) needs Python's ecosystem — scikit-learn and XGBoost are far ahead of Rust's linfa or smartcore for this kind of tabular modelling. Analytics runs as a separate Python service the core calls internally, which also matches the microservice boundary in 32.2.

Layer	Choice	Note
Web framework	Axum (Tokio-based)	Prefer Axum over Actix-web unless the team already knows Actix
Database driver	SQLx	Compile-time-checked SQL, pairs naturally with hand-written PostGIS spatial functions
Money type	rust_decimal or bigdecimal	Never floating point — see Section 29.2
Auth	jsonwebtoken + argon2	Short-lived JWT, argon2 for any password path, DSC/eSign at signing gates
API documentation	utoipa	Generates OpenAPI from Rust types, keeping the published contract and the implementation from drifting apart
GIS	the geo crate plus hand-written PostGIS SQL	The heavy spatial predicates run as SQL functions regardless of backend language — this is the real strength of the PostgreSQL/PostGIS pairing
Analytics bridge	Python microservice behind an internal HTTP or gRPC boundary	Isolates the ML dependency surface from the Rust core entirely

32.4 Multi-tenancy architecture

A “national” platform means MoRTH, MoRD, MoD, Railways, Ministry of Power and every state government share one deployment while each sees only its own data. This is a distinct architectural concern from the jurisdiction scoping in Section 25 — jurisdiction scoping controls *which rows* a District Collector sees within their own ministry's data; multi-tenancy controls whether *MoRTH can see MoRD's data at all*, and it needs its own mechanism because the failure mode is worse: a jurisdiction-scoping bug leaks one extra district; a multi-tenancy bug leaks one ministry's entire portfolio to another.

Design question	Answer
Isolation model	Shared infrastructure, isolated data. One PostgreSQL cluster, one Kubernetes deployment — not a database per ministry, which would defeat the national-aggregation purpose of the platform (Scope Item 6) before it started
Tenant key	ministry_id as the tenant discriminator, present on every tenant-scoped table (PROJECT, and everything that hangs off it) — not inferred from a join, stored directly so the isolation check never depends on query correctness elsewhere
Enforcement layer	PostgreSQL Row-Level Security, in addition to the application-layer checks in Section 25 — not instead of them. RLS is the layer that holds even if an application bug forgets to filter by tenant
Cross-tenant aggregation	The Policy & Audit role (Section 21.1) and the national dashboard (Section 40) are the two deliberate exceptions, reading pre-aggregated materialised views that never expose record-level cross-tenant data — aggregation, not access
A parcel touched by two ministries	Genuinely occurs — the spatial conflict check in Section 18.3 exists specifically to surface two requiring bodies proposing overlapping survey numbers. The conflicted parcel is visible to both tenants for the specific purpose of resolving the conflict, via a narrow, audited exception rather than a general RLS bypass

Design question	Answer
State government as a tenant	States are tenants in their own right for LARR acquisitions they run directly, and simultaneously counterparties on central-ministry projects within their territory — modelled as a state role scoped to the relevant state_subproject rows (Section 18.3) rather than as tenant ownership of those rows

Row-level security as the tenancy backstop

```
ALTER TABLE project ENABLE ROW LEVEL SECURITY;
```

```
CREATE POLICY project_tenant_isolation ON project
  USING (ministry_id = current_setting('app.tenant_ministry_id')::uuid
        OR current_setting('app.role') = 'policy_audit_aggregate');
```

```
-- Every tenant-scoped table cascades the same predicate through its FK to project,
-- so a single session variable set at connection time – not per query – governs
-- isolation for the entire request. The application never constructs this filter;
-- it cannot forget to.
```

33. Data model

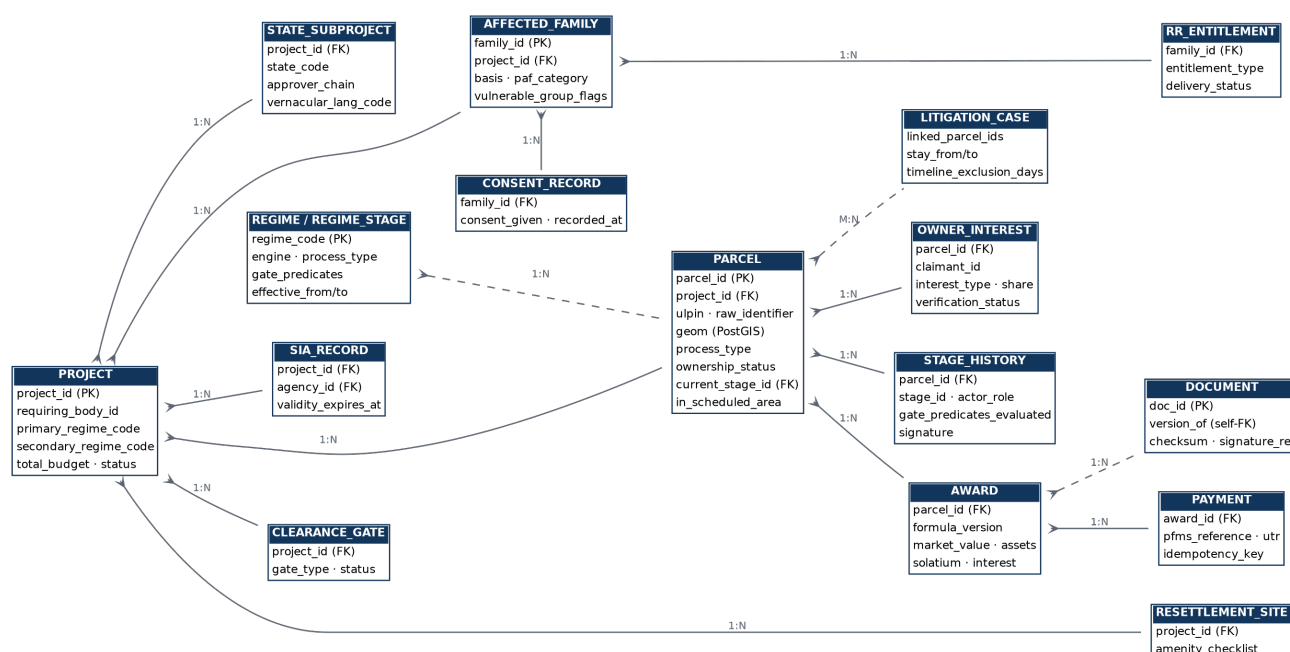


Figure 2. Simplified entity-relationship diagram — the fourteen core entities and their primary relationships. Several supporting tables (EVIDENCE_ITEM, ALERT, EMPANELLED_AGENCY, REGIME_STAGE) are omitted here for legibility and are specified in full in the table below.

The entities below correct three specific gaps identified across the research sessions: the assumption of a clean parcel-to-owner mapping, the absence of a process-type distinction, and the absence of a hook for Section 24 legacy cases. Every monetary field is a fixed-point decimal type; every geometry field is a native PostGIS type, never a serialised blob.

Entity	Purpose	Fields worth calling out
PROJECT	The master record for one acquisition effort	project_id, requiring_body_id, primary_regime_code, secondary_regime_code (nullable — see the dual-statute nuclear case, Section 11.5), total_budget, status, created_at
STATE_SUBPROJECT	One state's slice of a multi-state project	project_id, state_code, approver_chain, vernacular_language_code — the hierarchy Section 18.3 requires
PARCEL	One land unit under acquisition or another process	parcel_id, project_id, ulpin (nullable with a data-quality flag where absent), raw_identifier, raw_identifier_state_format, geom (PostGIS), area_recorded, area_computed, land_class, process_type (Section 14.4 enum, per-parcel not only per-project), ownership_status (clear / disputed / untraceable / under_litigation / multiple_claimants), current_stage_id, in_scheduled_area (boolean, computed spatially)
OWNER_INTEREST	A person's or entity's claimed or recorded interest in a parcel	parcel_id, claimant_id, interest_type, share_fraction, claim_basis, verification_status — deliberately many-to-many against PARCEL, because co-ownership and disputed multiple claims are the normal case, not the exception

Entity	Purpose	Fields worth calling out
AFFECTED_FAMILY	An entitlement-bearing unit, broader than titleholders	family_id, project_id, basis (land loss / displacement / livelihood dependency / other recognised relationship — per Section 22.5), paf_category, vulnerable_group_flags — linked to parcels and owners but not defined by title alone
REGIME / REGIME_STAGE	Configuration, as specified in Section 31	See Section 31 for the full schema
STAGE_HISTORY	Immutable record of every stage transition	parcel_id or case_id, stage_id, actor_user_id, actor_role, action, occurred_at, gate_predicates_evaluated (JSON, recording which passed and which were overridden), signature — feeds the audit ledger, does not replace it
SIA_RECORD	Social impact assessment for a project or affected area	project_id, affected_area (PostGIS), agency_id (references EMPANELLED_AGENCY), report_doc_id, hearing_doc_id, expert_group_appraisal, validity_expires_at — the 12-month SIA validity clock from Section 8.2
CONSENT_RECORD	One consented family's decision, rolling up to the live counter	project_id, family_id, consent_given, recorded_at, recorded_by — the aggregation is a materialised view, not a stored percentage, so it is always derivable and auditable
EMPANELLED_AGENCY	SIA agencies, Expert Group members, independent valuers	agency_id, credential_ref, empanelment_valid_from, empanelment_valid_to — closes the gap identified in Section 20.5
AWARD	A compensation determination	parcel_id, regime_code, formula_version, market_value, factor_applied, asset_values (JSON line items), solatium, interest_accrued_to_date, total, status, signed_by, signed_at
RR_AWARD / RR_ENTITLEMENT	Second Schedule entitlements per family	family_id, entitlement_type, entitlement_detail (JSON per Section 8.5's Second Schedule list), delivery_status — one row per entitlement line, not a single completion flag
RESETTLEMENT_SITE	A Third Schedule site and its amenity checklist	site_id, project_id, geom, amenity_checklist (JSON, the twenty-item Section 8.5 list), per-amenity target_date and status
PAYMENT	A disbursement instruction and its outcome	award_id or rr_entitlement_id, payee_id, amount, pfms_reference, utr_number, status, idempotency_key (never retry a disbursement without one — see Section 38.3)
LITIGATION_CASE	A dispute at any level	case_type (objection / reference / arbitration / writ / Section 24), linked_parcel_ids, stay_from, stay_to, timeline_exclusion_applied_days, outcome
DOCUMENT	A stored artefact with lineage	doc_id, doc_type, storage_ref, version_of (self-reference for supersession), checksum, signature_ref, language_code
EVIDENCE_ITEM	Field-captured evidence with chain of custody	captured_by, captured_at, geom, accuracy_metres, device_id, hash, custody_events (JSON array) — per Section 28.3
ALERT	A generated alert instance	alert_type (Section 39's nine classes), severity, entity_ref, recipient_roles, channel, status
CLEARANCE_GATE	A pre-condition gate instance on a project	project_id, gate_type (forest / environmental / tribal_consent / nuclear_site / other), status (not_required / in_progress / obtained), basis_for_not_required, certificate_doc_id

Three corrections this model makes to the earlier design

Ownership is not assumed clean. ownership_status and the many-to-many OWNER_INTEREST table replace a single owner field, so a disputed or untraceable parcel is representable rather than a special case the application code has to guess at.

Entitlement follows affected families, not titleholders. AFFECTED_FAMILY is a first-class entity with its own eligibility basis, because Section 22.5 establishes that displacement, livelihood dependency and other recognised relationships create entitlement independent of ownership.

Process type lives on the parcel, not only the project. A single industrial corridor project can carry acquired parcels, pooled parcels and allotted government land simultaneously — Section 12.4 documents exactly this pattern — and the schema must not force a single process choice at the project level.

34. Workflow engine design

The engine that executes the configuration from Section 31 against the data model from Section 33, enforcing the gate matrix from Section 27.

34.1 Execution model

Transition evaluation

```
fn attempt_transition(parcel_or_case, target_stage, actor) -> Result<Transition, GateFailure> {
  let regime = load_regime(parcel_or_case.regime_code);
  let stage_def = regime.stage(target_stage);

  // 1. Role check – is this actor's role the designated advancer for this transition?
  require(actor.role == stage_def.actor_role, RoleMismatch);

  // 2. Jurisdiction check – does the actor's scope cover this parcel/case?
  require(actor.scope.covers(parcel_or_case), OutOfJurisdiction);

  // 3. Gate predicates – evaluate every predicate the regime attaches to this stage,
  //   in the fixed predicate library. ALL must pass, or the officer sees exactly
  //   which one failed and why – never a bare 'cannot advance'.
  for predicate in stage_def.gate_predicates {
    let result = evaluate(predicate, parcel_or_case);
    if !result.passed { return Err(GateFailure(predicate, result.reason)); }
  }

  // 4. Effective-dated rule resolution – sia_required, consent_required, compensation
  //   formula version, all resolved against (state, category, as_of = today).
  let resolved_rules = resolve_effective_rules(parcel_or_case, today());

  // 5. Commit: state change, stage_history row, audit_log row (hash-chained),
  //   document generation trigger, alert trigger – as one transaction.
  commit_transition(parcel_or_case, target_stage, actor, resolved_rules)
}
```

34.2 What must be visible on failure

A gate that fails silently or with a generic message is worse than no gate — it teaches officers to route around the system. Every gate failure returns the specific predicate that failed, in language an officer can act on: *“Cannot issue Section 19 declaration — 3 of 47 objections remain undisposed (Sections 15A, 15B); consent stands at 76% against an 80% requirement.”* That specificity is what makes the workflow inbox in Section 21's district panel genuinely useful rather than a queue of unexplained blockages.

34.3 The predicate library

Predicate	Checks
objection_window_closed	Days elapsed since notification exceed the regime's objection_days
objections_disposed	Every filed objection has a recorded order
within_declare_deadline	Days since notification are within declare_deadline_days, or an extension with recorded reasons is on file
consent_threshold_met	Live consent ratio meets or exceeds the regime's configured threshold
sia_valid	An SIA record exists, is appraised, and is within its 12-month validity
rr_scheme_approved	The R&R scheme has passed its approval gate
deposit_confirmed	The requiring body's cost-share deposit is recorded
compensation_paid_or_tendered	Every award line for the parcel or family set is paid, tendered, or deposited with the authority
rr_monetary_paid_or_tendered	Every monetary R&R entitlement line is settled
no_subsisting_stay	No active litigation stay covers this parcel
clearance_obtained(kind)	The named clearance gate is in the obtained state, or not_required with a recorded basis
gram_sabha_consent_obtained	Required only where in_scheduled_area is true
validity_window_not_expired(years)	For regimes with a multi-year intention validity, e.g. Coal Bearing Areas s.7(1)

35. Compensation and interest engine

Given how many compensation figures in the source research were internally inconsistent (Section 8.4, Section 11.3), this engine is specified with worked, checkable arithmetic rather than a formula description alone.

35.1 Formula selector

Selector	Applies to	Computation
larr_first_schedule	LARR Act; any Fourth Schedule act after its 2015 compensation basis applies	market_value × factor(1.00–2.00 rural, 1.00 urban) + assets + 100% solatium, per Section 8.4
nh_pre_2015	NH Act awards with a notification date before the 2015 order	market_value + damage to crops/trees + severance damage + other damage + relocation expense — no factor, no solatium
coal_act_plus_2015_order	Coal Bearing Areas Act	The Act's own valuation basis, with the 2015 order's compensation and R&R schedules layered on top where applicable — unverified against the gazette text; flagged in Section 51
pipeline_right_of_user	Petroleum & Minerals Pipelines Act	10% of market value for the right of user, plus itemised crop, tree and structure damage — see the correction in Section 11.3
wayleave_footprint	Electricity Act and Telegraph Act wayleave	Tower or pole footprint value plus corridor burden plus crop and tree damage, at state-notified rates

35.2 Interest accrual — the two-provision engine

LARR interest, Sections 30(3) and 80 (see Section 8.4.1)

```
fn accrue_interest(award) -> Decimal {
  let s30_3 = day_count(award.notification_date, min(award.award_date,
                                                    award.possession_date))
    * award.base_plus_solatium * dec!(0.12) / dec!(365); // s.30(3), automatic

  let days_since_possession = day_count(award.possession_date, today());
  let s80_rate = if days_since_possession > 365 { dec!(0.15) } else { dec!(0.09) };
  let s80 = day_count(award.possession_date, award.payment_date.unwrap_or(today()))
    * award.total_before_interest * s80_rate / dec!(365);

  // Subtract any court-stay-excluded days from BOTH periods before this call –
  // exclusion is applied to the day_count inputs, never to the accrued total.
  s30_3 + s80
}
```

The rate step-up at the one-year mark and the exclusion of stayed days from the day count are both easy to omit and both material to the amount owed — an award that is a year overdue and includes ninety days of a since-vacated stay must exclude exactly those ninety days from both provisions' day counts, not from the total.

35.3 Structure, crop and tree valuation as line items

Section 8.4 already establishes these as a separate schedule component; the data model consequence is that each is its own row with its own approval trail, not a lump adjustment to the market-value figure. A structure valuation disputed at reference stage should be traceable and re-computable independent of the land value dispute.

36. Timeline engine

The statutory clock is, as Section 1.2 argues, the reason the platform has authority. Three behaviours make it real rather than decorative.

36.1 Lapse detection

Two lapse regimes, deliberately different

```
-- NH Act: hard lapse, no extension available
IF today() - notification_3A_date > 365 days AND declaration_3D_date IS NULL THEN
    project.status := 'LAPSED';
    ALERT Central Ministry, State Government, requiring body;
    BLOCK all further workflow actions on this project;
    -- the only route forward is a fresh 3A notification

-- LARR: soft lapse, extendable with recorded reasons
IF today() - notification_s11_date > 365 days AND declaration_s19_date IS NULL
AND NOT extension_recorded THEN
    notification.status := 'DEEMED_RESCINDED';
    ALERT appropriate government;
-- an extension record, with mandatory reasons, resets the deadline and is itself audited
```

36.2 Timeline exclusion for court stays

As argued in Section 24.2, exclusion must be scoped to the parcels the stay order actually names. The engine walks litigation events per parcel, accumulating excluded day ranges, and recomputes the effective deadline by advancing only through days not covered by an active exclusion and not a declared public holiday. Every recomputation is written to the audit ledger with the litigation case reference that caused it.

36.3 The alert cadence a deadline generates

Days before deadline	Alert
90	Advisory — informational, to the responsible officer only
30	Warning — to the officer and their immediate supervisor
7	Critical — to the officer, supervisor and the next tier up
0 (breach)	Breach — statutory timeline breach alert per Section 39, to all stakeholders in the gate matrix's notification list for that transition, and a permanent flag on the project dashboard

37. GIS and spatial module

Capability	Implementation
Parcel geometry capture	Polygon drawing against a cadastral basemap, or a field-walked GPS boundary with per-vertex accuracy recorded (Section 28.3)
Cadastral overlay	DILRMP and ULPIN layers served through GeoServer WMS/WFS, rendered in MapLibre GL on the frontend
Corridor geometry	Linear features for pipelines, transmission and telecom right of way, with per-parcel burden computed by intersection (ST_Intersection) rather than assigned whole-parcel
Spatial conflict detection	ST_Intersects at proposal scrutiny (Section 18.3) between the proposed footprint and any existing project's parcels
Constraint intersection	Pre-loaded layers for forest boundaries, Scheduled Areas (Fifth and Sixth Schedule), CRZ zones, monument regulated areas and irrigated multi-crop land, checked at Stage 0 and again before any gate that depends on them
Status-coloured dashboard map	Parcel colour keyed to current_stage, with drill-down from national to parcel per Section 40
Spatial queries for operations	“All parcels within 500 metres of this alignment,” “total area acquired in this district,” “parcels with an unresolved spatial conflict” — all standard PostGIS predicates over an indexed geometry column
3D hook	A schema placeholder for vertical property data from the companion PS SIH26011, so urban acquisition involving multi-storey structures is not a later migration

38. Integration architecture

The realistic version of “how do we connect to other departments' APIs,” argued honestly rather than assumed.

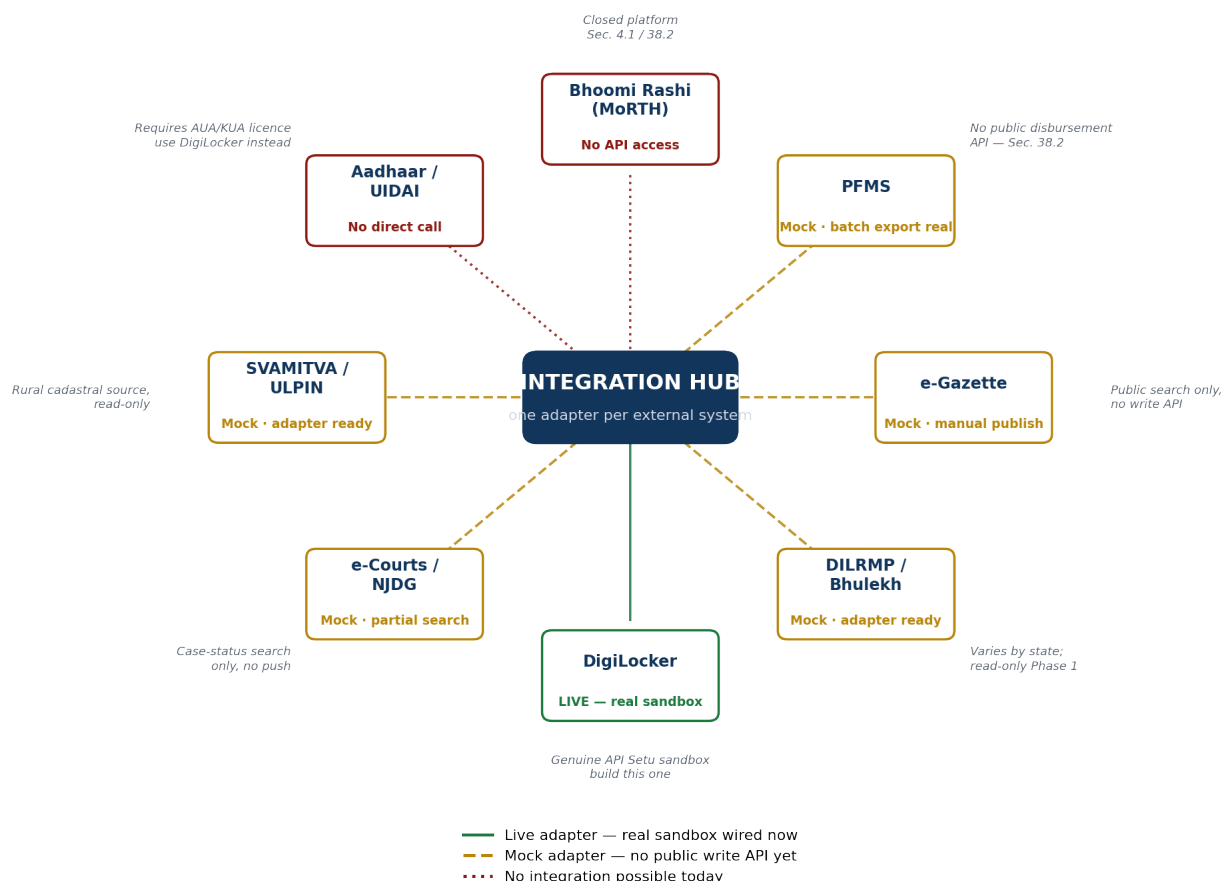


Figure 5. The Integration Hub and its eight external systems, colour-coded by what is actually available today rather than what would be ideal — the honest version of this diagram is more useful than the aspirational one.

38.1 The formal channel

The production path for inter-department integration is **API Setu** (MeitY/NeGD) and **NAPIX** (NIC's parallel exchange). API Setu is positioned as an open, interoperable platform for cross-silo government service delivery, hosting well over a thousand partners and several thousand published APIs, at meaningful monthly transaction volume — the real route for eventual production onboarding, through published guidelines rather than point-to-point agreements with each ministry.

38.2 Per-system reality check

System	What is realistically available now	Design response
DigiLocker	A genuine public sandbox via API Setu	Wire it for real. This is the one integration worth building live rather than mocked, and it is also the compliant route to Aadhaar-backed identity — see below.
PFMS	No broad public disbursement API for third parties; DBT in practice flows through the sponsoring department's own integration or a batch file	Design the Finance module to export a batch-ready disbursement file matching PFMS's format, and provide a mock live-API adapter for demonstration. State this distinction plainly rather than implying a live payment integration exists.
e-Gazette	Public search; no public write API	Generate a publication-ready notification document with the correct statutory content, for manual submission today; treat automated publication as a stated future integration, not a current capability.
e-Courts / NJDG	Some case-status search APIs; stay orders in practice still arrive as documents	Consume case-status search where available; the Legal Officer's stay-order upload (Section 24.2) remains the primary and reliable input regardless.
Aadhaar / UIDAI	Direct core authentication requires licensed AUA/KUA status	Never call the core authentication API directly without that licence. Use DigiLocker's login-with-Aadhaar for landowner OTP sign-in instead — a compliance requirement, not a convenience.
DILRMP / Bhulekh / state land records	Varies by state; several states publish digitally signed RoR download	Adapter per state format, normalising into the ULPIN-keyed schema from Section 44. Read-only in Phase 1 regardless of what a given state's API permits.
Bhoomi Rashi	Closed API surface (Section 4.1)	No integration is currently possible; note this as a limitation and a target for the replication argument in Section 6.

38.3 Adapter architecture

One interface per external system — LandRecordsPort, PaymentPort, GazettePort, CourtStatusPort, IdentityPort — each with a MockAdapter built against the published schema for demonstration and a LiveAdapter wired for the systems in 38.2 where a real sandbox exists. This is not a shortcut dressed up as architecture: ports-and-adapters is the correct pattern regardless of mock availability, because every external system in this list fails independently and on its own schedule.

Resilience mechanism	Behaviour
Circuit breaker	Open after five consecutive failures within sixty seconds; half-open retry after a cooldown
Stale-cache serving	Redis holds the last-known-good response for read operations; served with an explicit “as of” timestamp, never silently as current
Queued retry	Write operations queue for retry with exponential backoff rather than failing the officer's request outright
Idempotency keys	Mandatory on the disbursement trigger specifically — a retried PFMS call must never cause a duplicate payment
Manual reconciliation trigger	An admin-facing forced-sync action for when automated recovery is insufficient

38.4 A worked contract, kept intentionally minimal

One normalized adapter contract, not per-source contracts

```
openapi: 3.0.3
info:
  title: Land Records Adapter (normalized)
  version: 1.0.0
paths:
  /records/{ulpin}:
    get:
      parameters:
        - {name: ulpin, in: path, required: true, schema: {type: string}}
      responses:
        '200':
          content:
            application/json:
              schema:
                type: object
                properties:
                  ulpin: {type: string}
                  source_system: {type: string, enum: [dilrmp, state_lams, svamitva]}
                  retrieved_at: {type: string, format: date-time}
                  raw_identifier: {type: string}
                  land_class: {type: string}
                  owners: {type: array, items: {$ref: '#/components/schemas/OwnerInterest'}}
```

38.5 Full contracts: Bhoomi Rashi, PFMS, DILRMP

Section 38.4 makes the architectural point — one normalized internal contract rather than leaking each external system's schema into the application. That does not remove the need to know exactly what each external system's **own** contract looks like, since the adapter has to translate against it. These three are the contracts referenced throughout Sections 4, 9 and 38.2, written out in full rather than summarised, precisely because the detail is what a developer actually implementing the MockAdapter needs — and what an integration partner reviewing the pitch would expect to see rather than take on faith.

Read these as mock contracts against published schemas, not live specifications

None of the three services below currently publishes these exact endpoints for third-party consumption at the access level implied — Section 38.2's per-system reality check stands over all of it. Bhoomi Rashi's API is closed (Section 4.1); PFMS has no broad public disbursement API (Section 38.2); DILRMP's schema varies by state implementation. These contracts are written to the **shape** such an integration would plausibly take, based on each system's publicly documented capability, so the mock adapters produce realistic sample data and the live adapters have a concrete target to implement against once access is granted through API Setu.

38.5.1 Bhoomi Rashi Integration API (mock — NH Act acquisition status)

bhoomi_rashi_adapter.yaml (1 of 2)

```
openapi: 3.0.3
info:
  title: Bhoomi Rashi Integration API
  version: 1.0.0
  description: >
    Interface with MoRTH's Bhoomi Rashi portal for NH Act 1956 land acquisition data.
    Adapter status: MOCK – Bhoomi Rashi's API surface is closed to external consumers
    today (Section 4.1); this contract targets what a granted integration would expose.
servers:
  - url: https://bhoomirashi.gov.in/api/v1
    description: Production (not currently accessible to third parties)

paths:
  /projects/{projectId}/acquisition-status:
    get:
      summary: Get acquisition status for an NH Act project
      parameters:
        - {name: projectId, in: path, required: true, schema: {type: string}}
        - {name: X-API-Key, in: header, required: true, schema: {type: string}}
      responses:
        '200':
          content:
            application/json:
              schema:
                type: object
                properties:
                  projectId: {type: string}
                  notification3A: {type: string, format: date}
                  notification3D: {type: string, format: date, nullable: true}
                  awardDeclared: {type: boolean}
                  totalParcels: {type: integer}
                  acquiredParcels: {type: integer}
                  compensationDisbursed: {type: number, format: decimal}
                  status: {type: string, enum: [active, completed, lapsed]}

  /parcels/{parcelId}/details:
    get:
      summary: Get parcel-level NH Act acquisition detail
      parameters:
        - {name: parcelId, in: path, required: true, schema: {type: string}}
      responses:
        '200':
          content:
            application/json:
              schema:
                type: object
                properties:
                  parcelId: {type: string}
                  surveyNumber: {type: string}
                  village: {type: string}
                  district: {type: string}
                  state: {type: string}
                  areaHectares: {type: number}
                  compensationAmount: {type: number, format: decimal}
                  paymentStatus: {type: string, enum: [pending, paid, disputed]}
                  possessionStatus: {type: string,
                    enum: [not_taken, surrendered, forcible]}
                  awardOrderUrl: {type: string, format: uri}
```

bhoomi_rashi_adapter.yaml (2 of 2)

```
/acquisition/notify:
  post:
    summary: Push an acquisition-stage update to Bhoomi Rashi
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              projectId: {type: string}
              stage: {type: string,
                enum: [3A, 3B, 3C, 3D, 3E, 3G, 3H]}
              parcelIds: {type: array, items: {type: string}}
              documents: {type: array, items: {type: string, format: uri}}
              timestamp: {type: string, format: date-time}
              digitalSignature: {type: string}
    responses:
      '201': {description: Notification accepted}
      '409': {description: Conflict – stale data on the receiving side}
```

38.5.2 PFMS Payment Integration API (mock – DBT and disbursement)

pfms_adapter.yaml (1 of 2)

```
openapi: 3.0.3
info:
  title: PFMS Payment Integration API
  version: 2.1.0
  description: >
    Direct Benefit Transfer for land acquisition compensation.
    Adapter status: MOCK for the live endpoints below – PFMS does not expose a broad
    public disbursement API to third parties today (Section 38.2). The Finance module
    additionally exports a PFMS-format batch file, which is the real current path for
    DBT via the sponsoring department's own integration.
  servers:
    - url: https://pfms.nic.in/api/v2
      description: PFMS production (batch file is the current real integration path)

paths:
  /beneficiary/verify:
    post:
      summary: Verify an Aadhaar-linked bank account before disbursement
      requestBody:
        content:
          application/json:
            schema:
              type: object
              properties:
                aadhaarHash: {type: string, description: hashed, never the raw number}
                accountNumber: {type: string}
                ifscCode: {type: string}
      responses:
        '200':
          content:
            application/json:
              schema:
                type: object
                properties:
                  status: {type: string, enum: [verified, mismatch, inactive]}
                  npciStatus: {type: string}
                  bankName: {type: string}
                  accountHolderName: {type: string}
```

pfms_adapter.yaml (2 of 2)

```
/payment/initiate:
  post:
    summary: Initiate DBT for compensation or R&R entitlement
    requestBody:
      content:
        application/json:
          schema:
            type: object
            properties:
              schemeCode: {type: string, example: LA-NH-2026}
              beneficiaryId: {type: string}
              accountNumber: {type: string}
              ifscCode: {type: string}
              amount: {type: number, format: decimal}
              purpose: {type: string,
                example: 'Land Acquisition Compensation - Award 3G'}
              referenceId: {type: string}
              idempotencyKey: {type: string, description: mandatory, see Section 38.3}
              dscSignature: {type: string}
    responses:
      '202':
        content:
          application/json:
            schema:
              type: object
              properties:
                pfmsReference: {type: string}
                status: {type: string, enum: [queued, failed, rejected]}
                estimatedCreditDate: {type: string, format: date}

/payment/status/{pfmsReference}:
  get:
    parameters:
      - {name: pfmsReference, in: path, required: true, schema: {type: string}}
    responses:
      '200':
        content:
          application/json:
            schema:
              type: object
              properties:
                pfmsReference: {type: string}
                status: {type: string,
                  enum: [pending, processed, credited, failed, returned]}
                utrNumber: {type: string}
                creditedDate: {type: string, format: date-time}
                failureReason: {type: string}
```

38.5.3 DILRMP / Bhulekh Land Records API (mock — RoR, cadastral, mutation)

dilrmp_adapter.yaml (1 of 3)

```
openapi: 3.0.3
info:
  title: DILRMP Land Records API
  version: 3.0.0
  description: >
    Access to Record of Rights, cadastral maps and mutation status.
    Adapter status: MOCK, adapter-ready – actual availability and schema vary by state
    (Section 38.2); several states publish digitally signed RoR download today, which is
    the nearer-term real integration than a uniform national API.
servers:
  - url: https://bhulekh.nic.in/api/v3
    description: DILRMP national portal (schema varies by state in practice)
paths:
  /land/record/{ulpin}:
    get:
      summary: Fetch Record of Rights by ULPIN
      parameters:
        - {name: ulpin, in: path, required: true, schema: {type: string}}
        - {name: stateCode, in: query, required: true, schema: {type: string}}
      responses:
        '200':
          content:
            application/json:
              schema:
                type: object
                properties:
                  ulpin: {type: string}
                  state: {type: string}
                  district: {type: string}
                  tehsil: {type: string}
                  village: {type: string}
                  surveyNumber: {type: string}
                  rawIdentifier: {type: string, description: the state's own format}
                  area: {type: number}
                  landClass: {type: string,
                    enum: [agricultural, residential, commercial, industrial]}
                  owners:
                    type: array
                    items:
                      type: object
                      properties:
                        name: {type: string}
                        share: {type: number}
                        aadhaarLinked: {type: boolean}
                  mutationStatus: {type: string,
                    enum: [active, pending, disputed]}
                  encumbrances: {type: array, items: {type: string}}
                  circleRatePerSqM: {type: number}
                  retrievedAt: {type: string, format: date-time}
```

dilrmp_adapter.yaml (2 of 3)

```
/cadastral/map:
  get:
    summary: Get a cadastral map tile or GeoJSON for a village
    parameters:
      - {name: villageCode, in: query, required: true, schema: {type: string}}
      - {name: format, in: query,
          schema: {type: string, enum: [geojson, wms, kml], default: geojson}}
    responses:
      '200':
        content:
          application/json:
            schema:
              type: object
              properties:
                villageCode: {type: string}
                boundary: {type: object, description: GeoJSON Polygon}
                parcels:
                  type: array
                  items:
                    type: object
                    properties:
                      ulpin: {type: string}
                      surveyNumber: {type: string}
                      boundary: {type: object}
                      area: {type: number}
```

dilrmp_adapter.yaml (3 of 3)

```
/mutation/status:
  post:
    summary: Batch mutation status check, used after possession (Section 22.6/23.9)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            properties:
              ulpins: {type: array, items: {type: string}, maxItems: 100}
    responses:
      '200':
        content:
          application/json:
            schema:
              type: object
              properties:
                results:
                  type: array
                  items:
                    type: object
                    properties:
                      ulpin: {type: string}
                      mutationStatus: {type: string}
                      mutatedTo: {type: string}
                      mutationDate: {type: string, format: date}
```

39. Alert engine

Scope Item 9 names five alert categories explicitly. Building a taxonomy rather than an email-on-event mechanism is what turns this into monitoring rather than noise.

Alert type	Trigger	Recipient	Channel	Severity
Pending approval	A stage-gate item unactioned for more than 7 days	Next-level approver	SMS + email + dashboard	High
Statutory deadline approaching	90 / 30 / 7 days before a hard deadline (Section 36.3)	Responsible officer, escalating tier by tier	Email + dashboard, SMS at 7 days	Escalating
Statutory deadline breached	Deadline crossed with no action	Central Ministry, State, District, requiring body	SMS + email + dashboard	Critical
Compensation disbursement delayed	Award declared but not deposited within 10 days	Finance, District Collector	SMS + email	Critical

Alert type	Trigger	Recipient	Channel	Severity
Payment failure	PFMS DBT failed or returned	Finance, District Collector	SMS	Critical
Milestone miss	Project progress behind the configured schedule	PIA, Central Ministry	Email + dashboard	Medium
Stay order recorded	A litigation stay uploaded against a parcel	All stakeholders on that project	Email	Critical
R&R infrastructure delay	A Third Schedule amenity behind its target date	R&R Administrator, State	Email	Medium
Grievance escalation	Unresolved beyond the configured service level	State Government, Central Ministry	Email	High
Urgency invocation	Section 40 or an equivalent fast-track provision is invoked	Policy & Audit, Central Ministry	Dashboard flag, permanent	Oversight

40. Reporting and MIS

40.1 National dashboard KPIs

The eight KPIs the problem statement enumerates explicitly, plus two the system adds.

#	KPI	Source
1	Area notified (hectares)	Official PS text
2	Area acquired (hectares)	Official PS text
3	Compensation assessed (₹ crore)	Official PS text
4	Compensation paid (₹ crore)	Official PS text
5	Number of affected families	Official PS text
6	Number of displaced families	Official PS text
7	R&R status (% complete)	Official PS text
8	Possession status (% handed over)	Official PS text
9	Project progress (on time vs delayed)	Added — the operational counterpart to timeline adherence
10	Timeline adherence (% meeting statutory deadlines)	Added — the single clearest measure of whether the platform is achieving its purpose

Drill-down: national → state → district → project → parcel → landowner, on every KPI. Comparative view: state against state on the same metric set. Both are read paths against the operational data through a materialised view refreshed on a short interval, never a separate reporting-data pipeline that could drift from the operational truth — drift between the dashboard and the workflow is precisely the failure the PS complains about today.

40.2 Report families

Family	Content	Formats	Cadence
Executive	Parliament-ready, CAG-ready summaries; NITI Aayog briefs	PDF, PowerPoint-ready export	On demand, monthly
Operational	District-wise pending actions, hearing schedules, award status	PDF, Excel	Daily, weekly
Financial	Fund utilisation, interest accrual, reconciliation statements	Excel, CSV	Weekly, monthly
Compliance	Timeline adherence, statutory deadline compliance by district	PDF	Monthly, quarterly
Comparative	State-versus-state benchmarking on the KPI set	PDF, dashboard	Monthly
Trend	Year-over-year acquisition speed and cost trends	PDF, dashboard	Annual

41. Document management

Capability	Detail
Version lineage	Every document supersedes rather than overwrites; the full chain is retrievable, matching the audit ledger's expectations
Bilingual and vernacular templates	One parameterised template per document type (Section 30.3), rendered per state's configured language, not a fixed Hindi/English toggle
Content-addressed storage	SHA-256 checksum on every object; a document's identity is tied to its content
Signature binding	DSC/eSign application and verification recorded against the document and the stage gate it satisfied
Evidence bundle assembly	On-demand assembly of every document and evidence item linked to a case, with hashes and custody records, for the Legal Officer — see Section 28.3
Publication readiness	A notification or award document flagged publication-ready carries every statutory content element the relevant section requires (Section 22.3's declaration checklist is the model), checked before the flag is set

42. Field data capture

The problem statement's exact wording is worth being precise about: deliverable 6 asks for a “mobile-responsive interface for field-level data collection and verification.” It does not say offline sync, a native app, or auto-sync on connectivity restoration. A responsive web form satisfies the literal requirement.

Build the offline layer as a differentiator, correctly sequenced

Field surveyors in rural India routinely work without connectivity, so an offline-capable Progressive Web App is a genuine value-add and a reasonable use of development time — but only **after** the workflow engine and dashboard are solid, not instead of them. If time runs short, ship the responsive form and present the offline architecture as a designed and ready extension rather than as a missing core feature.

Layer	Technology	Purpose
Service Worker	Cache-first for the app shell and map tiles; network-first with an offline fallback for API calls	
Client storage	IndexedDB via Dexie.js, holding parcel drafts, photos and a sync queue	
Background sync	Registers a sync task on offline save; retries automatically on reconnection	
Safari fallback	A manual “Sync Now” button, since iOS Safari does not support the Background Sync API	
Conflict resolution	Server-wins with a 409 response carrying the server's current version, surfaced to the surveyor rather than silently discarded	
Evidence capture	Photo hashed and geotagged at capture per Section 28.3, stored as base64 in IndexedDB for the prototype scale, uploaded on sync	

43. AI and analytics

The sub-theme makes this a scoring dimension, not an optional extra (Section 2.2). Six use cases, each assessed honestly for what it needs to actually work versus what can run on rules today.

Use case	What it does	Data requirement	Prototype-stage honesty
Delay risk scoring	Predicts probability of missing the next statutory deadline, with feature attribution	Historical acquisition timelines across project type, state, land class, season, litigation rate — the companion PS SIH26017's stated purpose	Genuinely needs training data this platform does not yet have. A transparent rule-based risk score (days remaining against historical median for this regime and state) is an honest interim, clearly labelled as such, with the ML model as the stated evolution once data accumulates.
Document extraction and classification	Structures data from uploaded RoR extracts, survey reports and old case files	Works reasonably now with OCR plus a classifier on document layout	The most immediately deliverable AI capability — genuinely useful for legacy record migration (Section 46), not merely a demo feature.

Use case	What it does	Data requirement	Prototype-stage honesty
Compensation anomaly detection	Flags awards whose computed figure deviates materially from comparable awards for similar parcels	Needs a baseline of awards within the system, which grows over time	Start as a rule-based outlier check (standard deviation from the district's recent average per land class); genuinely valuable for the CAG/audit role from day one.
Duplicate and fraudulent claimant detection	Fuzzy name and Aadhaar-hash matching across parcels to catch the same person claiming multiple entitlements	Works with standard fuzzy-matching techniques immediately	A genuinely deliverable feature with clear governance value, buildable now.
Land value estimation	Estimates market value from location, land class and recent comparable transactions, to sanity-check officer-entered figures	Needs a comparable-sales dataset, which DILRMP integration can eventually supply	Reasonable as a bounds-check against the officer's figure rather than a replacement for it — the Act specifies who determines market value, and it is not a model.
Grievance triage	Prioritises and routes grievance text by urgency and category	Works with a modest labelled dataset or even a rule-based keyword classifier initially	A low-effort, genuinely useful addition to the grievance module in Section 20.5

The honest framing for the pitch

“Rule-based today, model-ready by design” is a stronger and more defensible position than claiming a working ML pipeline with no training data behind it. The architecture — features logged from day one, a Python analytics service already isolated behind its own boundary (Section 32.2) — is what makes the eventual model a natural evolution rather than a rewrite. That architectural readiness is itself the answer to the sub-theme requirement, even before a single model is trained.

44. Data standardization and master data

Scope Item 5, and the mechanism behind the Section 3 diagnosis that data inconsistency is the platform's central problem to solve.

State example	Local identifier	Normalized to
Tamil Nadu	Survey No. 45/2A, Village Thiruvallur, Taluk Ambattur	ULPIN 21-01-001-01-00045, with the raw string retained alongside
Uttar Pradesh	Khasra No. 112, Village Rampur, Tehsil Sadar	ULPIN lookup against the state's DILRMP feed
Karnataka	RTC Number	ULPIN lookup where Bhunaksha coverage exists
Kerala	Thandaper (via LTIN)	ULPIN lookup against Ente Bhoomi's own identifier

Pipeline: raw input → a per-state parser (regex or lookup table by state format) → normalised fields → ULPIN resolution → the unified parcel schema, with the raw string and source retained rather than discarded, so a revenue officer can still recognise their own record and a mismatch is auditable rather than silently overwritten.

Standardization concern	Mechanism
Validation rules per stage	A mandatory-field set defined per regime_stage — a survey number is required before a 3B survey record can be approved, an Aadhaar-linked account before a payment instruction can issue
Duplicate detection	Fuzzy name matching plus hashed-identifier matching across OWNER_INTEREST rows, surfaced as a review queue rather than an automatic merge
Data quality scoring	A per-parcel and per-district completeness score — ULPIN present, geometry present and validated, ownership status resolved — visible on the operational dashboard, not only to a data team
Unit normalization	Area always stored and displayed in hectares centrally, with the state's local unit (acre, cent, bigha, with its state-specific size) retained as a display alternative
Reporting templates	One schema per report family (Section 40.2), populated from the normalized model regardless of source state — this is what makes “uniform reporting templates applicable across all States and UTs” (Scope Item 5's literal wording) actually true

45. Non-functional requirements

Category	Target
Availability	99.9% uptime — at most 8.76 hours downtime per year
API response time	Under 200ms at the 95th percentile for standard reads
Dashboard load	Under 3 seconds for the national view
Map render	Under 2 seconds for a 1,000-parcel overlay
Scale target	10,000 concurrent users, 1,000,000 parcels, sustained document and photo upload traffic — stated conservatively rather than as a headline figure, per the honest framing in Section 32.3
Recovery point objective	Under 15 minutes
Recovery time objective	Under 4 hours
Audit log retention	7 years
Legal document retention	10 years or longer per applicable records policy
Backup strategy	Daily full plus hourly incremental, geo-replicated within India per the data-localization obligation in Section 29.1
Multilingual coverage	Hindi and English at minimum; an i18n framework structured for all Eighth Schedule languages, with per-state notice-template packs as the near-term deliverable

46. Migration and onboarding

The problem statement's Scope Item 1 asks specifically for a study of the existing system as input to design — not merely an awareness of it. This section is the migration path that study implies.

46.1 Phased integration, not a rip-and-replace

Phase	Scope
Phase 1 — read-only sync	Fetch project and parcel status from Bhoomi Rashi where its API allows; fetch RoR and cadastral data from DILRMP and ULPIN; fetch litigation status from e-Courts where available; verify beneficiary identity through DigiLocker. No write-back.
Phase 2 — bi-directional sync	Push SIH26016 acquisition updates back to source systems where a write API exists; trigger mutation requests through the state records adapter; sync payment status with PFMS through the batch mechanism; publish notifications to e-Gazette once a write path opens.
Phase 3 — legacy retirement	Gradual migration of state LAMS holdings into the unified schema; Bhoomi Rashi becomes a read-only historical archive for pre-migration NH Act projects; the platform becomes the single source of truth for new acquisitions.

46.2 Handling legacy data honestly

Challenge	Mechanism
Inconsistent identifiers across systems	The per-state parser and ULPIN resolution from Section 44, applied at ingestion
Missing historical data	Pre-2018 acquisitions predating Bhoomi Rashi exist largely on paper. A bulk-import module with document OCR (Section 43) plus a manual data-entry workflow for legacy projects, with an explicit data-confidence score per imported record rather than presenting imported and system-native records as equally reliable
Duplicate records across systems	The same parcel may exist in both Bhoomi Rashi and DILRMP under different identifiers. Fuzzy matching plus ULPIN as the deduplication key, with conflicts surfaced for human resolution rather than auto-merged
Section 24 legacy cases encountered during migration	Tagged as legacy_1894_transitional (Section 33) on ingestion rather than forced into the LARR or NH Act track, preserving the distinct status the law actually requires

PART VI

Scope Control

Ambition without a stated boundary reads as naivety to a technical judge. This part draws the boundary explicitly: what the prototype builds, what is deliberately deferred and why, what could go wrong, and — the section a team most wants to skip and most needs to keep — the specific points in this document's own evidence base that are not yet settled and must not be presented as certainties.

In this part

- In-scope requirements, each traced to the scope item or deliverable it answers
- Out of scope, with the specific reason for each exclusion
- The Phase 2 roadmap that turns exclusions into stated future work
- Risk register
- Open research gaps in this document's own sources
- Sources

47. In-scope requirements

Every row traces to a scope item (Section 2.1) or a deliverable (Section 2.2). A build checklist with no traceability is a wish list; this one is not.

Priorit y	Requirement	Traces to
P0	Engine A: full LARR Act 2013 stage machine, Sections 4–64 as specified in Section 22	Scope 2, 3; Deliverable 1
P0	Engine B: full NH Act 1956 stage machine, Sections 3A–3I plus mutation, Section 23	Scope 2, 3; Deliverable 1
P0	Sector Acquisition Track supporting at minimum Railways Act 1989 (explicitly named in the PS background) as the first configured regime beyond the two bespoke engines	Scope 3, 12; Section 30.2
P0	Regime configuration schema (Section 31) — the mechanism, not just the two hardcoded engines	Scope 12; Deliverable 1
P0	Master stakeholder registry and RBAC as specified in Sections 20 and 26	Scope 2, 11
P0	Stage-gate enforcement per Section 27, including the hard NH Act one-year lapse and the LARR consent threshold	Scope 3, 11
P0	PostGIS parcel geometry, geo-tagging and the status-coloured map	Scope 4; Deliverable 3
P0	Compensation engine for both bespoke regimes with the corrected First Schedule arithmetic (Section 35)	Scope 3
P0	National dashboard with the ten KPIs and full drill-down (Section 40.1)	Scope 6; Deliverable 4
P0	Public portal: status lookup, compensation estimator, grievance filing	Scope 2; Deliverable 8
P0	Append-only, hash-chained audit ledger (Section 28)	Scope 11
P1	Interest accrual engine, both Section 30(3) and Section 80 provisions	Scope 3; Section 35.2
P1	Alert engine with the nine-class taxonomy (Section 39)	Scope 9
P1	Document generation with version lineage (Section 41)	Scope 8
P1	Integration Hub with a live DigiLocker adapter and honest mocks elsewhere (Section 38)	Scope 7; Deliverable 5
P1	R&R module: affected family registry, entitlement matrix, resettlement site tracker (Section 20.2, role 6)	Scope 2
P1	Mobile-responsive field data collection form, satisfying Deliverable 6 literally	Scope 4; Deliverable 6
P2	Offline-capable PWA layer over the field form (Section 42)	Differentiator, not required by Deliverable 6's literal text
P2	Rule-based delay risk scoring and compensation anomaly detection (Section 43)	Sub-theme (AI, GIS & Data Analytics)
P2	Data standardization pipeline with per-state parsers (Section 44)	Scope 5
P3	Litigation and stay management with scoped timeline exclusion (Section 24.2, 36.2)	Scope 3, 9
P3	Gram Sabha consent gate for Scheduled Areas (Section 15)	LARR s.41 compliance

48. Out of scope, with reasons

Each line converts a potential gap into a stated decision. This table should be read alongside Section 19's boundary table, which explains the reasoning at greater length for the system-boundary items.

Excluded	Reason
Forest and environmental clearance workflows	PARIVESH exists, is MoEFCC's system, and operates under a different Act entirely. We consume clearance status as a gate; we do not rebuild the clearance process. See Section 19.
Land pooling, TDR, government allotment as fully built workflows	Modelled in the data schema as process types (Section 14.4) so the architecture does not preclude them, but not built as complete workflows in the prototype — the team's two bespoke engines are LARR and the NH Act.
The remaining nine sector regimes beyond Railways as fully tested configuration rows	The Sector Acquisition Track's genericity is demonstrated with one regime beyond the two bespoke engines (Railways, since it is named in the PS background). The other regimes are documented in Part II with enough precision to configure, but configuring and testing all of them is Phase 2 work, not prototype work.
Section 24 legacy 1894 case handling as a working feature	Deliberately deferred; the data model hook exists (legacy_1894_transitional, Section 33) so the extension is additive. See Section 8.7.
Live PFMS disbursement API integration	No broad public API exists for third parties today (Section 38.2). We build to the batch file format and mock the live-API path.
Live e-Gazette publication	No public write API exists today. We generate a publication-ready document for manual submission.
Direct Aadhaar core authentication	Requires licensed AUA/KUA status. We use DigiLocker's login-with-Aadhaar instead, which is both compliant and available.

Excluded	Reason
Blockchain-based audit ledger	A hash-chained append-only log delivers the same tamper-evidence property at a fraction of the operational cost (Section 28.1). Blockchain integration is the stated territory of the companion PS SIH26019, not ours to duplicate.
Predictive analytics as a trained ML model	No training dataset exists yet within this platform. Built as rule-based scoring with the architecture ready for a trained model once operational data accumulates (Section 43).
Full multi-state sub-project approval chains for every possible state combination	The data model supports the state_subproject hierarchy (Section 33); full per-state rule configuration for all 28 states and 8 union territories is a data-entry exercise for Phase 2, not an architectural gap.
Actual construction monitoring after possession	The acquisition lifecycle's system boundary ends at clean possession and mutated title, per Section 19. Construction monitoring is a different problem and a different problem statement.

49. Phase 2 roadmap

Named explicitly, so a judge sees engineering judgement rather than a team that ran out of time and stopped.

- **Section 24 legacy acquisition handling** — the transitional case type, the post-*Indore Development Authority* lapse test, and a bulk-classification tool for the large stock of pending pre-2014 cases a national rollout will immediately encounter.
- **Fifth and Sixth Schedule consent workflow** — the Gram Sabha resolution process end to end, spatial detection of Scheduled Areas as a standard layer, and the Sub-Divisional, District and State Level Committee chain under the Forest Rights Act.
- **The remaining sector regimes as tested configuration** — Coal Bearing Areas, Metro Railways, Atomic Energy, the older Fourth Schedule acts, and validation that the “configuration, not code” claim holds for each without a new predicate.
- **Right-of-user and non-acquisition tracks as complete workflows** — pipelines, wayleave, land pooling and TDR, government allotment, each currently modelled in the schema but not built end to end.
- **Live write integrations** as API Setu onboarding matures — PFMS disbursement, e-Gazette publication, and DILRMP bi-directional sync.
- **A trained delay-prediction model**, once the platform has accumulated enough operational history across enough projects to train on rather than assume.
- **Full DPDP operational compliance** ahead of the 14 May 2027 full-enforcement date — consent capture UX, a working erasure and correction path, and a completed data retention schedule per data category.
- **GIGW and WCAG 2.1 AA certification** of the public portal, formally assessed rather than designed-to.

50. Risk register

Risk	Likelihood	Impact	Mitigation
Team has no prior Rust experience; ramp-up consumes the early sprint budget	Medium	High	State the cost honestly (Section 32.3); front-load Rust ramp-up in week one rather than discovering the gap mid-build; keep the Analytics service in Python from day one so at least one service is in a language the team already knows well
Compensation arithmetic errors reach the demo	Medium	High	Every formula in Section 35 is checkable against the worked examples in Sections 8.4 and 11.3; unit-test the compensation engine against those worked examples specifically, not just against arbitrary inputs
Live integrations promised in the pitch turn out not to exist	Medium	High	Section 38.2's per-system reality check is the single source of truth for what is live versus mocked; the pitch deck's integration slide should be built directly from that table, not from aspiration
A judge asks about a regime or provision this document flags as unverified	Medium	Medium	Section 51 lists these explicitly; the honest answer — “flagged as unverified in our own research, here is what we know and don't” — is a stronger answer than a confident guess
Scope creep back toward the twenty-regime framing from earlier sessions	Medium	Medium	Section 30's two-engine strategy and Section 48's exclusion list are the standing answer; any proposal to add a third bespoke engine should be tested against Section 30.3's configuration criteria first
PostGIS spatial query performance at scale is untested	Low	Medium	Index every geometry column from the start; test the 500-metre and district-aggregate queries against a synthetic dataset at the target parcel count before the demo, not during it
Field PWA offline sync consumes disproportionate build time relative to its priority	Medium	Low	P2 priority per Section 47; ship the responsive form first and treat offline sync as additive time allows, per Section 42

51. Open research gaps

This section exists because the source research for this document — an act-by-act corpus built and partially corrected across several sessions — contains specific, named errors that a later correction pass fixed in some places and not others. Presenting the unreconciled parts as settled would be worse than not researching them at all. Every item below should be independently verified against the gazette text before being asserted in a pitch or built into a compensation engine.

#	Issue	Status	Where it matters
1	Railways Act 1989 land acquisition sections	Corrected to Sections 20A–20P from an earlier claim of Sections 120–124 in the summary note, but the body text and stage headings of the underlying research file were not reconciled to match	Section 10 of this document uses the corrected sections; do not rebuild from the uncorrected source file without fixing the body text first
2	SEZ Act 2005 exemption status	The master comparative summary was corrected to show SEZ as not exempt from LARR (consistent with Section 12.1 of this document), but the standalone SEZ act-research file's body still describes a pre-2013-style notification-and-declaration workflow rather than the LARR SIA-and-consent process its own header now correctly states applies	Do not use the SEZ file's stage descriptions without rewriting them to match LARR's actual process
3	Petroleum & Minerals Pipelines Act compensation percentage	Three different figures appear in the same source file for the same calculation: 33–50% in the general description, 10% in one worked example and the comparison table, 40% in a second worked example. This document uses 10% (Section 11.3, Section 35.1) as the figure the correction pass identified as correct	The single highest-priority arithmetic defect in the corpus. If the compensation engine for this regime is ever built, verify 10% against the Act's text directly before trusting any source file
4	Fourth Schedule exempt-act count	Corrected from an earlier claim of 16 to the correct 13 acts, with SEZ and two miscategorised entries removed from the list. This document's Section 17 table reflects the corrected 13	Section 8.5 and Section 17 of this document
5	Atomic Energy Act 1962 section numbers	Recorded in the source corpus as Sections 9–12, correcting an earlier claim of Sections 16–18. Neither figure has been independently checked against the gazette text in producing this document	Section 11.5 states this explicitly as unverified. Do not cite a specific section number for this Act without checking the primary text first

#	Issue	Status	Where it matters
6	Requisitioning and Acquisition of Immovable Property Act 1952, key dates	The source file shows an enacted date of 14 March 1952 and an effective date of 7 March 1952 — effective before enacted, which reads as a data-entry swap rather than a genuine finding	Section 11.2 flags this. Do not build a validity calculation on either date without confirming both against the gazette
7	Post-2015 compensation basis for each Fourth Schedule act individually	The 2015 order under Section 105(3) is well documented as extending First, Second and Third Schedule provisions to the thirteen exempt acts. Whether each individual act's act-research file in the corpus reflects this post-2015 basis, rather than the pre-2015 bare compensation figures, has not been checked act by act	Section 9.2 and Section 35.1's coal_act_plus_2015_order selector are flagged as needing this verification before the compensation engine trusts any Fourth Schedule act's figures beyond the NH Act, which has been checked
8	Metro Railways (Construction of Works) Act 1978 — section references	Not independently verified in producing this document; described only structurally in Section 11.6	Do not cite specific section numbers for this Act without checking the primary text
9	Whether the Phase 2 items were reflected in an earlier scope-lock exercise	An earlier gap analysis (Section 24 legacy handling, Fifth Schedule/PESA consent, sector Acts beyond LARR and the NH Act, land pooling) was raised in a prior session as “Phase 2 roadmap” material; whether it was actually reflected consistently in the team's separately maintained in-scope and out-of-scope requirement documents was not confirmed at the time and should be checked directly against those documents	Worth confirming before final submission that Section 47 and Section 48 of this document and the team's own requirement tracker agree
10	Whether two correction passes over the ten-act research corpus fully agree in scope	One record from that work claims all eleven files received verification headers; another states only seven of eleven received substantive fixes, with four files — National Highways, Metro Railways, Electricity, Coal Bearing Areas — marked as having no errors found. Those four have not been individually spot-checked against that claim in producing this document	Worth a direct spot-check of those four files before treating “no errors found” as verified rather than merely asserted

Why this section belongs in the document rather than in a private note

A reference document that hides its own uncertain points is more dangerous than one that states them, because the confident version is the one that gets built from. Every entry above is a specific, checkable claim with a specific consequence if wrong — which is exactly the standard the rest of this document tries to meet. Treat this table as a verification checklist to close out before the compensation engine and the pitch deck are finalised, not as a list of caveats to mention once and forget.

52. Sources

Primary legal texts

- The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (Act 30 of 2013), gazette text
- The National Highways Act, 1956 (Act 48 of 1956), as amended, Sections 3A–3J
- The Railways Act, 1989, Chapter IVA, Sections 20A–20P
- The Land Acquisition Act, 1894 (repealed), for historical and Section 24 transitional reference only
- Supreme Court, *Indore Development Authority v. Manoharlal* (2020), Constitution Bench, on Section 24 lapse
- Supreme Court, *State of Gujarat v. Patel* (2023), on the same section

Official problem statement and platform documentation

- SIH26016 official problem statement PDF, Ministry of Rural Development, Department of Land Resources
- Bhoomi Rashi portal (Ministry of Road Transport & Highways / NHAI), public documentation and press material
- Digital India Land Records Modernization Programme (DILRMP), Department of Land Resources progress reporting
- SVAMITVA scheme, Ministry of Panchayati Raj, public reporting
- Kerala Ente Bhoomi / ILIMS, NIC Kerala public documentation

- Tamil Nadu Land Acquisition Management System and TNGLMS, Commissionerate of Land Administration
- Odisha LARRMS (Bhuarjan), public documentation
- National Generic Document Registration System (NGDRS)
- API Setu (MeitY / NeGD) and NAPIX (NIC), onboarding guidelines and public statistics

Internal research sessions synthesised into this document

- T200 AUV thruster specifications session — unrelated to this problem statement; not a source for this document
- “Land request workflow and stakeholder management system” session (3 September 2026) — stage stakeholder matrices, gap analyses, the 36-hour-versus-multi-phase framing correction, offline sync architecture, and the LARR Miro board decode
- SIH problem statement history research session — official PS deep dive, prior-year SIH comparison, international hackathon comparison, existing-system landscape, and the twelve scope items / eight deliverables decomposition
- The act-research corpus (ten sector acts plus market-research files) and its two correction passes, with the unreconciled points listed in Section 51
- The uploaded National Highways Act stakeholder document, including the 20-regime framework session synthesised into Part II of this document

A note on this document's own status

This is a synthesis and correction of prior research, not a first-principles legal opinion. Where Part II states a section number or a percentage without a flag in Section 51, it reflects the most consistent and best-supported figure across the sources reviewed. Where sources disagreed and the disagreement was not resolved, Section 51 says so explicitly. For any figure that will appear in a compensation engine actually computing amounts payable under law, verify against the gazette text directly before relying on this document alone.